

**THEORETICAL AND EXPERIMENTAL ASPECTS OF
ELECTROCATALYSIS OF THE OXYGEN REDUCTION REACTION AND
RELATED SYSTEMS**

By

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Table of Contents

Table of Contents	4
List of Tables	9
List of Figures	10
List of Abbreviations.....	27
Abstract	30
Chapter 1 Introduction	32
Chapter 2 Hydrogen Peroxide Reduction in Aqueous Electrolytes: Influence of a Heterogeneous Decomposition Step	37
Publication Details	38
Abstract.....	38
Introduction.....	38
Theoretical	39
Results and Discussion	41
Summary.....	45
Chapter 3 The Koutecky-Levich Plot for the Evaluation of Catalysts for the Oxygen Reduction Reaction: A Critical Analysis	46
Introduction.....	47
Theoretical	48

Results and Discussion	50
Suggested Methods for Finding n_{eff}	53
Conclusion	55
Chapter 4 A Critical Assessment of $X_{\text{H}_2\text{O}_2}$ As A Figure Of Merit For Oxygen Reduction	
Electrocatalysts In Aqueous Electrolytes	56
Publication Details	57
Contributors	57
Abstract.....	57
Introduction.....	58
Experimental.....	58
Theoretical Formalism	59
Chapter 5 A Rotating Ring-Disk Electrode Method for the Detection of Solution Phase	
Superoxide as a Reaction Intermediate of Oxygen Reduction in Neutral Aqueous Solutions	69
Publication Details	70
Contributors	70
Abstract.....	70
Introduction.....	71
Experimental Aspects.....	72
Results and Discussion	73
Theoretical Aspects	78

Chapter 6 Theoretical Analysis of Microelectrode Arrays Under Forced Convection.....	82
Abstract.....	83
Introduction.....	83
Theoretical Aspects	85
Results	88
Concluding Remarks	93
Chapter 7 Oxygen and Hydrogen Peroxide Reduction on Polycrystalline Platinum in Acid	
Electrolytes: Effects of Bromide Adsorption	95
Publication Details	96
Contributors	96
Abstract.....	96
Introduction.....	97
Experimental.....	99
Results and Discussion	100
Bromide Adsorption under Forced Convection	101
Oxygen Reduction	106
Hydrogen Peroxide Reduction	111
Theoretical	114
Conclusions.....	119

Chapter 8 Electrocatalytic Properties of Adsorbed Halides: The Reduction of Hexaaquairon(III) on Polycrystalline Gold in Aqueous Electrolytes	121
Contributors	122
Abstract.....	122
Introduction.....	123
Experimental.....	124
Results and Discussion	125
A. Bromide Adsorption Under Forced Convection: Chronoamperometric Studies	126
B. The Reduction of Hexaaquairon(III): Effects of Adsorbed Bromide	129
Conclusions.....	136
Chapter 9 Oxidation of Bromide on Platinum Electrodes in Aqueous Acidic Solutions: Electrochemical and In Situ Spectroscopic Studies	138
Contributors	139
Abstract.....	139
Introduction.....	140
Experimental.....	143
Results and Discussion	144
Conclusions.....	158
Chapter 10 New Advances in Ohmic Microscopy	160

Contributors	161
Abstract.....	161
Introduction.....	162
Experimental.....	162
Theoretical	165
Results	166
A. Current Pulses	166
B. Ring-Disk Electrode	169
C. Single-Crystal Experiments	173
Concluding Remarks	175
Chapter 11 The Electrocatalytic Properties of Adsorbed Hemin and its Nitroxyl Adduct on	
Glassy Carbon Surfaces Toward Hydroxylamine Reduction in Aqueous Neutral Electrolytes .177	
Contributors	178
Abstract.....	178
Introduction.....	178
Experimental.....	179
Results and Discussion	181
Catalytic and Electrocatalytic Properties of Hm GC Toward Hydroxylamine.....	186
Bibliography	201

List of Tables

Table 5.1. Boundaries and Boundary Conditions.....	79
Table 6.1. Governing Equations and Boundary Conditions Both in Dimensioned and Dimensionless Forms (see text for details).	86
Table 7.1. Governing Differential Equations and Boundary Conditions Involved in the Implementation of Levart's Formalism to Calculate Attenuation Factors for Partially Blocked Rotating Disk Electrodes.....	116
Table 7.2. Parameters Employed in the Simulations.....	117
Table 7.3. Boundary Conditions Associated with the Domain Shown in Figure 7.11 ...	117
Table 9.1. Steady State Governing Differential Equations and Boundary Conditions for the EC Mechanism in Scheme 2 in One-Dimension at the Diffusion Limited Current both in Dimensioned and Dimensionless Form.....	146
Table 10.1. Experimental and theoretical values $\Delta\phi_{\text{sol}}$ obtained in two different trials following application of two current pulses of different amplitude to a Pt disk electrode embedded in an insulting surface assuming a primary and uniform current distribution.	163
Table 11.1. Mass Balance Equations for the Major Species Involved	193
Table 11.2 Proposed Mechanism for the Electrocatalytic Reduction of NH_2OH Mediated by Hm GC	194
Table 11.3. Best Fit Parameters Based on the Model in & Scheme 11.1	194

List of Figures

Figure 2.1. Koutecky-Levich plots generated using Eq. 2.6 for various values of the rate constants for which $\gamma = 2.045$ (Panel A) and $\gamma = -2.045$ (Panel B), assuming $D_{O_2} = 1.3 \times 10^{-5}$ cm²/s and $D_{H_2O_2} = 1.9 \times 10^{-5}$ cm²/s and $\nu = 0.01$ cm²/s.43

Figure 2.2. Koutecky-Levich plots reproduced from the data reported by Amirfakhri *et al.* (see scattered symbols) and fits according to our theory (solid lines). The dotted lines are linear fits provided in the Ref. ². See text for additional details.44

Figure 3.1. Example plots for two (fictitious) systems of $\gamma = \pm 1.88$ with $n_{\text{eff, actual}} = 2.02$, $D_{O_2} = 1.3 \times 10^{-5}$ cm/s, and $D_{H_2O_2} = 1.9 \times 10^{-5}$ cm/s, that otherwise share all parameters except k_3 . Light lines are plots of the reciprocal current obtained from Eq. 3.10; light points are simulated data along those plots at square hundred rpm, in the range of 100-3600 rpm; and dark lines are the linear fits according to the data at those points, from which the respective $n_{\text{eff, meas}}$ is calculated. The black triangle points to the common intercept.51

Figure 4.1. Plots of I_d/I_r vs E_{disk} based on the data reported by Stamenkovic *et al.*(8) for a sputtered Pt₃Ni disk electrode of a Pt| Pt₃Ni RRDE collected at two different rotation rates as indicated.65

Figure 4.2. Panel A. Dynamic polarization curves recorded at a scan rate $\nu = 10$ mV/s with the Pt(poly) disk of a Pt(poly)|Pt(poly) RRDE in O₂-saturated 0.1 M HClO₄ solutions containing 10 μ M KBr at various values of ω , as indicated. Panel B. Ring currents collected while the disk data in Panel A was being collected, with the ring polarized at $E_{\text{ring}} = 1.4$ V vs

RHE. Panel C. Plots of I_d/I_r based on the data in Panels A and B in this figure, after correction for residual current contributions.66

Figure 4.3. Dynamic polarization curves recorded at $v = 10$ mV/s for the reduction of 1 mM H_2O_2 in 10 μ M KBr in Ar-purged 0.1 M $HClO_4$ on the Pt(poly) disk of the Pt|Pt RRDE at various rotation rates as indicated. The thick curve was obtained in the same solution devoid of bromide at $\omega = 400$ rpm.67

Figure 4.4. Plot of I_d/I_r vs. $\omega^{-1/2}$ based on the data in Panel C in Figure 4.2, for the specified values of E_{disk} , where the solid lines were obtained by a linear fit of the experimental data.68

Figure 5.1. Cyclic voltammetric curves ($v = 10$ mV/s) recorded with the Au disk of a Au|Au RRDE at $\omega = 900$ rpm in 0.1 M PB pH 7.4 (Left Panel) and pH 10 (Right Panel) in O_2 -saturated 0.1 M PB before (thick blue lines) and after adding 1mM H_2O_2 . The corresponding data collected after modifying the Au with a covalently bound MP layer are shown in expanded form (X100) in thin lines in these panels. Also shown in grey lines are voltammetric curves recorded with the bare Au in the corresponding neat deaerated solutions in expanded form. Inserts: Cyclic voltammetric curves ($v = 100$ mV/s) recorded with the bare Au disk of the Au|Au RRDE in the same neat deaerated solutions, where the cross represents the (0,0) point.74

Figure 5.2. **Top Panels.** Dynamic polarization curves (solid lines) recorded at $v = 10$ mV/s, with the glassy carbon (GC) disk of a Au|GC RRDE in O_2 -saturated PB pH 7.4 (Left Panels) and pH 10 (Right Panels) rotating at $\omega = 900$ rpm. **Middle Panels.** Ring currents acquired with the bare Au ring electrode polarized at the specified potentials while scanning the

disk. **Lower Panels.** Same as middle panels with the Au ring modified by a covalently linked MP monolayer. The dotted lines in all the panels were obtained in the same, thoroughly deaerated buffer solutions. The solid symbols represent the steady state currents measured at constant potential as specified (see text).75

Figure 5.3. Polarization curve of the MP-modified Au ring of a Au|GC RRDE recorded in O₂-saturated 0.1 PB at $\nu = 10$ mV/s while the GC disk was polarized at $E_{\text{disk}} = -0.4$ V in pH 7.4 (lower panel) and -0.41 V in pH 10 (upper panel) at various rotation rates as indicated. The curves in black were recorded in fully deaerated solutions.77

Figure 5.4. Schematic diagram of the two-dimensional domain employed in the simulations.79

Figure 5.5. Plot of $[O_2^-]^{\text{disk}}$ vs $I_{O_2^-}^{\text{ring}}$ for pH 7.4 and 10 for various values of ω as indicated, where the solid circles represent experimental values reported in an earlier communication. Insert: Data for pH 7.4 in an expanded scale.81

Figure 5.6. Plots of $[O_2^-]^{\text{disk}}$ vs $\omega^{1/2}$ determined from the ring currents, with the data at pH=7.4 expanded by 10x.81

Figure 6.1. **Left Panel:** Top-down view of a section of the surface of a RDE consisting of a hexagonal, closed-packed arrangement of disk microelectrodes of radius r_1 embedded in concentric inactive disks of radius r_2 . **Right Panel:** Domain employed in the theoretical simulations (see shaded area), where R_1 and R_2 are the corresponding dimensionless radii and H the dimensionless height of an individual cylinder in the array (see Table 6.1).85

Figure 6.2. **Panel A.** Plots of $\rho_{\text{lim}}^{\text{sim}}$ vs $1 - \theta$ for various values of R_1 as indicated in the legend, obtained from the simulations, where the region corresponding to $1 - \theta < 0.55$ has been expanded for clarity. For $1 - \theta > 0.55$, $\rho_{\text{lim}}^{\text{sim}}$ does not vary significantly. **Panel B.** Same as Panel A, but in log-log space. Also included in this panel in dashed lines are values of $\rho_{\text{lim}, R_1 < \Delta, 1 - \theta \rightarrow 0}$ determined from Eq. 6.2.....88

Figure 6.3. Plots of $\% \text{ error} = 100 \times (\rho_{\text{lim}}^{\text{Eq.3}} - \rho_{\text{lim}}^{\text{sim}}) / \rho_{\text{lim}}^{\text{sim}}$ vs $1 - \theta$, for various values of R_1 , where the symbol $\rho_{\text{lim}}^{\text{Eq.3}}$ represents the values of ρ_{lim} as determined from Eq. 6.3.....90

Figure 6.4. Plots of the absolute value of the $\% \text{ error} = 100 \left\{ 1 - \rho_K^{\text{sim}} \left(\frac{K_o / K}{(1 - \theta)} + \frac{1}{\rho_{\text{lim}}^{\text{Eq. x}}} \right) \right\}$ at $R_1 = 0.002$, as a function of K/K_o for values of $1 - \theta$ shown in the legend, where $\rho_{\text{lim}}^{\text{Eq. x}}$ represents the dimensionless limiting current specified for $x = 3$ and (Panel A) or $x = 5$ (Panel B).....91

Figure 7.1. Dynamic polarization curves (Panel B) recorded at a positive-going scan rate $v = 10 \text{ mV/s}$ (as indicated by the arrow in the lower left) with the Pt disk of the Pt|Pt RRDE, in O_2 -saturated 0.1 M HClO_4 at $\omega = 400 \text{ rpm}$, before (black) and after addition of either $10 \mu\text{M NaCl}$ (blue) or $10 \mu\text{M KBr}$ (red). The thin lines (Panel A) are the corresponding ring currents, i_{ring} , measured simultaneously with the ring polarized at $E_{\text{ring}} = 1.2 \text{ V}$ for the neat, and chloride-containing, and 1.4 V for bromide-containing solutions (see text for additional details).....98

Figure 7.2. Cyclic voltammograms recorded at $v = 20 \text{ mV/s}$ with the Pt disk of the Pt|Pt RRDE in Ar-purged 0.1 M HClO_4 before (black) and after addition of $20 \mu\text{M KBr}$ (2.2 ppm, blue).....101

Figure 7.3. Panel A. Chronoamperometric curves ($E_0 = 0.03$ V to E_{step} , as indicated) recorded with the Pt disk of a Pt|Pt RRDE in 10 μ M KBr in 0.1 M HClO₄ at $\omega = 400$ rpm. Panel B. $I_{\text{ring}} - I_{\text{ring}}^o$ vs t recorded simultaneously with the ring polarized at $E_{\text{ring}} = 1.4$ V vs RHE. The smooth solid lines in Panel A were calculated based on the shielded ring currents in Panel B. .102

Figure 7.4. Plots of Q_{Br} (left ordinate) and θ_{Br^-} (right ordinate) vs time based on the analysis of $I_{\text{disk}}^{\text{pr}}$ vs t recorded at $\omega = 400$ rpm (see Panel A, Figure 7.3) for different E_{step} , as indicated.105

Figure 7.5. Potential-dependent isotherm for bromide adsorption on Pt(poly) for 10 μ M KBr in a 0.1 M HClO₄ solution, at $\omega = 400$ rpm based on the data shown in Figure 7.4 (see text for details).....106

Figure 7.6. Chronoamperometric curves recorded with the Pt disk of a Pt|Pt RRDE at $\omega = 400$ rpm in O₂-saturated 0.1 M HClO₄ in the absence (Panel A) and in the presence of 10 μ M KBr (Panel B) in solution. Potential steps were applied from $E_0 = 0.03$ V vs RHE to various values of E_{step} as indicated in the legend. The bottom panels, i.e. a and b display the corresponding i_{ring} for $E_{\text{ring}} = 1.2$ V for the neat and $E_{\text{ring}} = 1.4$ V for the Br⁻-containing solution.108

Figure 7.7. Plots of i_{disk} vs t (left ordinate, black line) and θ_{Br^-} vs t (right ordinate, blue line) for $E_{\text{step}} = 0.6$ V at $\omega = 400$ rpm.....109

Figure 7.8. Plots of i_{disk} vs θ_{Br^-} for various E_{step} (Panel A) and i_{disk} vs E_{step} for various θ_{Br^-} based on the data shown in Panels B and b in Figure 7.6.110

Figure 7.9. Panel A. Plots of $I_{\text{ring}}/I_{\text{disk}}$ as a function of θ_{Br^-} in 0.1 M HClO_4 solutions for a Pt disk of a Pt|Pt RRDE at $\omega = 400$ rpm, based on chronoamperometric data acquired following a potential step from $E_0 = 0.03$ V to various values of E_{step} as indicated, and the corresponding i_{ring} of the type shown in Panel b, Figure 7.7. Panel B Plots of $I_{\text{ring}}/I_{\text{disk}}$ as a function of E_{step} various values of θ_{Br^-} , as indicated 111

Figure 7.10. Dynamic polarization curves ($v = 10$ mV/s) for the Pt disk of the Pt|Pt RRDE recorded in 0.1 M HClO_4 containing 1 mM H_2O_2 at a rotation rate, $\omega = 400$ rpm, before (black) and after the addition of 10 μM NaCl (blue) or 10 μM KBr (red). The arrow points in the direction in which the potential was scanned. 112

Figure 7.11. Chronoamperometric curves recorded at $\omega = 400$ rpm in 0.1 M HClO_4 containing 1 mM H_2O_2 in the absence (Panel A) and in the presence of 10 μM KBr at $\omega = 400$ rpm (Panel B) in solution starting from $E_0 = 0.05$ V (neat) or $E_0 = 0.03$ V (bromide) to various E_{step} values as indicated in the legends. 113

Figure 7.12. Comparison between $\Delta R/R = [R(0.05\text{V}) - R(0.7\text{V})]/R(0.05\text{V})$ transients collected for the Pt disk of the Pt|Pt RRDE at $\omega = 900$ rpm in Ar-saturated (black) and in H_2O_2 -saturated (magenta) 10 μM KBr in 0.1 M HClO_4 following a potential step from $E_0 = 0.05$ to $E_{\text{step}} = 0.7$ V applied at $t = 0.0$ s (blue, right ordinate, see text for details). 113

Figure 7.13. Panel A. Peroxide reduction currents, i_{disk} , as a function of θ_{Br^-} recorded in a 1 mM H_2O_2 solution containing 10 μM KBr in 0.1 M HClO_4 with the Pt disk of the Pt|Pt RRDE at $\omega = 200$ rpm, based on chronoamperometric data acquired following a potential step from $E_0 =$

0.03 V to various values of E_{step} , as specified Panel B. Plot of i_{disk} vs E_{step} for various values of θ_{Br} -as indicated.114

Figure 7.14. Schematic diagram of the equivalent array of active disks of radius r_s and alternating active and inactive rings of thickness a and b , respectively, surrounding a central disk of radius r_a described by Levart. In this figure, q is the radial period and b_{ext} is the width of the outermost inactive ring (see text and the original paper for details).115

Figure 7.15. Schematic diagram of the domain used in COMSOL for implementing the multi-ring rotating electrode model described by Levart.117

Figure 7.16. Comparison between plots of ρ vs $\log(r_e/\delta)$ reported by Levart (solid symbols) and obtained by solving the convective diffusion equation using COMSOL (solid line) for $\theta = 0.5$ and selected values of N_A assuming the same domain and boundary conditions as those employed by Levart. The black thick dotted line is based on ρ_{SN} values calculated from Eq. (7). The symbol A_{rde} refers to a completely unblocked disk electrode, i.e. $N_A = 0$118

Figure 7.17. Plots of ρ vs N_A , for various values of θ (Panel A), and of ρ vs θ , for various values of N_A (Panel B), for $\omega = 900$ rpm.119

Figure 8.1. Dynamic polarization curves collected with the Au(poly) disk of the Au|Au RRDE in the absence (red) and presence (black) of $10 \mu\text{M}$ KBr in deaerated 0.1 M HClO_4 aqueous solutions containing $1 \text{ mM Fe}(\text{ClO}_4)_3$ at a scan rate $v = 10 \text{ mV/s}$ and a rotation rate $\omega = 400 \text{ rpm}$, where the arrow specifies the direction of the scan. The blue scattered symbols in this figure (see right ordinate) represent values of the bromide coverage normalized by its value at E

$= 0.9 \text{ V}, \theta_{Br}^N$. The difference between the red and black lines is represented by the dash-dotted gray line.127

Figure 8.2. Chronoamperometric curves recorded with the Au ring of a Au|Au RRDE in 10 μM KBr in 0.1 M HClO_4 at $\omega = 400 \text{ rpm}$, from $E_0 = -0.15 \text{ V}$ to E_{step} , where the thin lines are the predicted disk currents based upon the ring currents shown in the bottom panel in this figure.

Bottom Panel. Plots of $i_{\text{ring}} - i_{\text{ring}}^0$ vs time recorded with the ring polarized at $E_{\text{ring}} = 1.4 \text{ V}$ vs RHE, during the acquisition of the chronoamperometric disk data.....128

Figure 8.3. Plots of Q_{Br} vs t (left ordinate) and the corresponding θ_{Br}^N vs t (right ordinate) for various values of E_{step} , based on the data in the bottom panel, Figure 8.2 (see text for details).129

Figure 8.4. Schematic diagram describing the graphical method employed to extract values of τ_{pl} and $\theta_{Br}^{N,cr}$ from chronoamperograms recorded with the Au(111) electrode under the conditions specified in Figure 8.5 using one such curve as an example (see solid black line), where the blue line represents its derivative.....130

Figure 8.5. Chronoamperometric curves recorded with the Au disk of a Au|Au RRDE in deaerated 0.1 M HClO_4 aqueous solutions containing both 10 μM KBr and 1 mM $\text{Fe}(\text{ClO}_4)_3$ for various values of ω as specified in each of the panels. For each experiment, the potential was stepped from $E_0 = -0.15 \text{ V}$ to different E_{step} , as indicated in the legend. Except for the jagged curves (raw data), collected for $E_{\text{step}} = 0.75 \text{ V}$, all others represent averages (see text for details).131

Figure 8.6. Plots of $\Delta R/R = [R(-0.15) - R(1.0)]/R(-0.15)$ vs time collected with the Au disk of the Au|Au RRDE at $\omega = 900$ rpm in Ar-saturated (black) 0.1 M HClO₄ containing 10 μ M KBr before (black) and after (blue) addition of 1 mM [Fe(H₂O)₆]³⁺ following a potential step from $E_0 = -0.15$ V to $E_{\text{step}} = 1.0$ V applied at $t = 0.0$ s (magenta, right ordinate, see text for details).132

Figure 8.7. Plot of the Au(poly) disk current associated with the reduction of [Fe(H₂O)₆]³⁺ (left ordinate; black) and θ_{Br}^N as a function of time for $E_{\text{step}} = 0.70$ V from Figure 8.5. The blue curve is the normalized coverage of bromide from Figure 8.3.134

Figure 8.8. Plot of the normalized critical bromide coverage, $\theta_{Br}^{N,cr}$ for the reduction of [Fe(H₂O)₆]³⁺ as a function of the step potential, E_{step} for different rotation rates, ω , calculated from fits of the data to Figure 8.5.134

Figure 8.9. Left Panel: experimental curves for the current as a function of potential at 400 rpm. Slices of the current were taken at times corresponding to the coverages as indicated in the legend.....135

Figure 8.10. Left panel: Plot of current as a function of time, with the time axis normalized to the corresponding coverage of bromide. Right panel: simulations of various R1 as a function of coverage. The current axis is again normalized such that the limiting current is 1.136

Figure 9.1. Plot of the dimensionless steady state limiting current, $I = \frac{Sc_{Br}^{1/3}}{0.621} \frac{dC_{Br}}{d\gamma} \Big|_{\gamma=0}$, as

a function of $\log_{10}\xi$, where $\xi = c_{Br}^{\infty} K$ is the dimensionless equilibrium constant for the homogeneous reaction associated with the EC mechanism in Scheme 9.3. The shaded area represents the range of values of κ involved in the experimental measurements reported herein.

.....147

Figure 9.2. Plot of the dimensionless current density, $I = \frac{Sc_{Br}^{1/3}}{0.621} \frac{dC_{Br}}{d\gamma} \Big|_{\gamma=0}$ as a function of

the dimensionless rate constant $\kappa = \frac{k_f}{K\omega}$ for selected values of $\xi = c_{Br}^{\infty} K$, namely, 1.7×10^{-2} (blue), 8.5×10^{-2} (red) and 0.17 (black), which correspond to values of the bulk concentration of bromide, $c_{Br}^{\infty} = 1, 5$, and 10 mM. The shaded area represents the range of values of κ involved in the experimental measurements reported herein.148

Figure 9.3. Plots for the dimensionless concentration C_j , $j = Br^-$ (solid line), Br_2 (dotted) Br_3^- (dashed, X 40) for various bulk concentrations of bulk bromide, $c_{Br}^{\infty} = 0.1$ (magenta), 1 (blue), 5 (red), and 10 mM (black) as a function of the dimensionless distance $\gamma = (\omega/\nu)^{1/2} x$.

.....149

Figure 9.4. Plot of the absorbance, due to solution phase tribromide, $A = \frac{2\epsilon_{Br_3}}{\sin \theta} \int_0^{\infty} c_{Br_3} dx$

(lower panel), and $\log_{10}A$ (upper panel), as a function of $\log_{10}\xi$ $\xi = c_{Br}^{\infty} K$150

Figure 9.5. Steady state polarization curves (scattered symbols) for the oxidation of bromide collected with the Pt(poly) disk of a Pt|Pt rotating ring disk electrode, RRDE, in a solution 5 mM KBr in 0.1M HClO₄, at various rotation rates, ω , i.e. 400 (magenta), 800 (blue), 1200 (green), 1600 (red) and 2000 rpm (black). Corresponding dynamic polarization curves recorded for the same Pt electrode at a scan rate $v = 10$ mV/s and $\omega = 400$ rpm in the same electrolyte and in 0.1 M HClO₄ devoid of bromide are shown in magenta and black solid lines, respectively, where the arrows in the former indicate the direction of the scan.151

Figure 9.6. Plots of i_{lim} vs. $\sqrt{\omega}$ for the oxidation of bromide recorded with a Pt RDE electrode in 0.1M HClO₄ containing 10 mM (black), 5 mM (red), 1 mM (blue), and 0.1 mM KBr (magenta). The open and half open circles represent data collected in two independent experiments. The solid symbols are the results of the theoretical simulations assuming $D_{Br^-} = 1.58 \times 10^{-5}$ cm²/s (see text). The straight lines represent the best linear fits to the experimental data. The error bars represent the average of four independent runs.152

Figure 9.7. Koutecky-Levich plots for the oxidation of bromide at a rotating Pt disk electrode based on the results shown in Figure 9.5. The solid and open circles represent data collected in two fully independent runs and the arrow points in the direction of decreasing potentials from 1.3 to 1.21 V in steps of 0.01 V.153

Figure 9.8. Plot of the kinetic currents, i_k , (left ordinate, solid symbols) and kinetic rate constants, k_f , (right ordinate, open symbols) based on the data in Figure 9.7.155

Figure 9.9. Potential difference reflection absorption spectra, $A(1.40 \text{ V}) - A(0.05 \text{ V})$, recorded at an angle of incidence of ca. 45° from a rotating Pt disk electrode in a 0.01M KBr

solution in 0.1M HClO₄ at various values of ω = 400 (black), 800 (red), 1200 (blue), 1600 (magenta) and 2000 rpm (olive). The prominent peak centered at λ = 266 nm is characteristic of tribromide ion. Each of the spectra represents the average of 15 acquisitions (see text). The smooth line is the best fitting of the data.....155

Figure 9.10. Potential difference absorbance, $A(1.4V) - A(0.05 V)$ vs $\omega^{-1/2}$, collected at λ = 266 nm from a Pt RDE in solutions 1 (blue), 5 (red) and 10 mM (black) KBr in 0.1M HClO₄ and at an angle of incidence, θ ca. 45°C (open symbols) during acquisition of the electrochemical data in Figure 9.6. The solid symbols are the results of the theoretical simulations (see text). The straight lines represent the best fits to the experimental data. The error bars represent the average of four or two independent measurements.....156

Figure 9.11. Potential difference absorbance, $A(1.4V) - A(0.05 V)$ vs $\omega^{-1/2}$ collected from a Pt RDE in a 5 mM KBr in 0.1M HClO₄ (open circles) at λ = 266 (blue), 390 (red), 633 nm (black). The data in solid circles was acquired in neat 0.1M HClO₄ under otherwise identical conditions. The smooth lines are the best fit of the data.....157

Figure 9.12. Upper panel: Dynamic polarization (v = 10 mV/s) curve collected with the Pt disk of the Pt | Pt RRDE in 0.1M HClO₄ containing 0 mM KBr(olive), 0.1 mM KBr(magenta), 1 mM KBr(blue), 5 mM KBr(red) and 10mM KBr(black) , ω = 900 rpm. Lower panel: Corresponding differential reflectance collected simultaneously while scanning the potential toward positive values, where the smooth solid lines represent best fits to the data in the range $0.6 \leq V \leq 1.4 V$158

Figure 10.1. **Panel A.** Schematic diagram of the cell in which the ohmic microscopy experiments involving for current pulse and ring-disk electrodes were carried out. WE, working electrode; CE, counter electrode; RE, commercial reference electrode; R1 and R2, μ -ref electrodes. The vertical red line represents a glass insulated Cu wire that connects the Au ring counter electrode to the outside. **Panel B.** Schematic diagram of the Au (yellow) disk|Pt-Ir ring (grey) electrode. **Panel C.** Scanning electron micrograph (5.0 kV, $\times 250$) of a faceted Pt single crystal (see text for details).....164

Figure 10.2. Schematic diagram and associated dimensions of the cell (in cm) used for the current pulse experiments, where the shadowed area represents the domain involved in the theoretical simulations, and the dotted line, the counter electrode loop.165

Figure 10.3. **Top Panel.** Plot of the $\Delta\phi_{\text{sol}}$ vs t (right ordinate, blue curve) recorded while applying a current pulse of 5.0 mA for 50 μs to the Pt disk (left ordinate, black curve) with the μ -Ag/AgCl electrodes placed at (1,3)_c as shown schematically in the insert. The spikes observed immediately before and after the pulse was applied are instrumental artifacts. **Bottom Panel.** Plot of the potential of the Pt disk working electrode vs a conventional Ag|AgCl reference electrode (right ordinate, magenta curve) during application of a similar current step (left ordinate, black curve).....167

Figure 10.4. **Right Panel.** Equipotential lines obtained from COMSOL simulations involving the shaded domain in Figure 10.2 (see text for details) assuming a current of 5 mA under primary current distribution. **Left Panel.** Expanded region of the area just above the

working electrode in the right panel in this figure assuming a primary (solid lines) and a *fictitious* uniform current distribution (dotted lines).168

Figure 10.5. Plots of $\Delta\phi_{\text{sol}}$ vs E for the Pt-Ir ring|Au disk electrode collected with individual microreference electrodes, along three different axis normal to the electrode surface as shown where the dots represent the position of the tips (see also text for details). The red curves represent the actual recorded data and the black curves are the results of the numerical simulations.....170

Figure 10.6. **Left Panel.** Cyclic voltammogram of the Pt-Ir|Au ring-disk assembly (red curve, $\nu = 10$ V/s), and of a best fit analysis using individual voltammetric curves recorded independently shown in the Right Panel in this figure. **Right Panel.** Individual voltammetric curves recorded at 10V/s for the Pt-Ir ring with the center cavity filled with epoxy (blue curve), and for a commercial Au disk encased in Teflon (black curve, see text for details).171

Figure 10.7. Plots of $\Delta\phi_{\text{sol}}$ vs E acquired with the double barrel capillary while the working electrode was scanned at 10 V/s, where each of the curves represents data collected with the tip placed along a line parallel to the surface of the ring disk intersecting the center of the disk ($X = 0$), at a fixed distance from the surface, i.e. 50 μm173

Figure 10.8. Cyclic voltammetry for the single crystal Pt faceted microsphere recorded at 10V/s in 0.1M H_2SO_4 solutions (Upper Panels), and $\Delta\phi_{\text{sol}}$ vs E (black curve lower Panels) acquired simultaneously with the tip of the double barrel capillary placed very close to the center of the Pt(111) (Left Panel) and Pt(100) microfacets (Right Panel). The red curves are the corresponding voltammetric curves for Pt(111) and Pt(100) faces collected using conventional

single crystal electrodes reported in the literature in deaerated 0.5 M H₂SO₄ solution at 50 mV/s.⁹⁷174

Figure 10.9. Cyclic voltammogram (right ordinate, red curve) and $\Delta\phi_{\text{sol}}$ vs E recorded simultaneously (left ordinate, blue curve) under the same conditions as specified in the caption Figure 10.8 with the tip of the capillary 0.3 mm away from the microfaceted single crystal sphere.175

Figure 11.1. Cyclic voltammograms of a bare GC (grey line) and a Hm|GC electrodes (black line) recorded in deaerated 0.5 M PB (pH 7) at a scan rate, $\nu = 0.5$ V/s under quiescent conditions. The dotted line was used as a background for the coulometric analysis.....182

Figure 11.2. Cyclic voltammograms ($\nu = 0.5$ V/s) of NOHm|GC (black line) recorded in deaerated PB under quiescent conditions. The scan was initiated at 0.0 V vs Ag/AgCl in the negative direction. The magenta line was obtained during the subsequent third cycle. Also shown for comparison is the voltammogram of Hm|GC in neat PB (dotted curve).....184

Figure 11.3. Series of cyclic voltammograms of NOHm|GC ($\nu = 0.05$ V/s) in 0.5 M PB (pH 7). The scan was initiated at 0.0 V in the positive direction reversed at the indicated limit, and then reversed once again at -0.9 V vs Ag/AgCl to end at 0.0 V. The curve in dashed-dot was recorded after scanning the potential three times over the entire potential range under otherwise identical conditions.185

Figure 11.4. Cyclic voltammograms of NOHm|GC ($\nu = 0.5$ V/s) (first cycle, black curve) initiated at 0.0 V in the negative direction in 0.5 M PB (pH 7) recorded under the same

conditions as those specified in the caption of Figure 11.2. The magenta and dash-dot line represent the second and fifth subsequent cycles.186

Figure 11.5. Cyclic voltammetry ($v = 0.5$ V/s) for a Hm|GC electrode in neat PBS which had been previously exposed to a 1mM NH_2OH in PBS at open circuit, ca. 0 V for 60 min. under rotation ($\omega = 400$ rpm), and subsequently rinsed with neat PB (black curve). The magenta curve was obtained without exposure to the NH_2OH solution. Unlike the data collected in nitrite solutions the nitrosyl adduct does not form readily on Hm under these conditions.187

Figure 11.6. Cyclic voltammograms ($v = 0.05$ V/s) of a GC RDE electrode recorded under quiescent conditions deaerated PB before (grey) and after addition of 1 mM NH_2OH (black). See text for details.188

Figure 11.7. Cyclic voltammogram ($v = 0.05$ V/s) of a GC (grey curve) and a HM|GC electrode recorded under quiescent conditions in deaerated PB (pH 7) after addition of 1 mM NH_2OH (black curve). See text for details.189

Figure 11.8. Dynamic polarization curves recorded at $v = 10$ mV/s with a Hm|GC RDE in PB buffer pH 7 as a function of ω as indicated in the legend for $c_{\text{NH}_2\text{OH}}^{x=\infty} = 2.5$ mM (Top Panel), and, as a function of $c_{\text{NH}_2\text{OH}}^{x=\infty}$ as indicated in the legend, at $\omega = 900$ rpm (Bottom Panel). The solid and dotted curves for each set of data correspond to scans toward negative and subsequent positive potentials, respectively.190

Figure 11.9. **Top Panel.** Plot of the experimental polarization curve collected at true steady state at $\omega = 900$ rpm (scattered symbols) and the simulation (solid line) based on Model in

Table I and Scheme A and the parameters specified in Table III. **Middle Panel.** Plots of the concentrations of the four relevant species as a function of the applied potential based on the best fit set of parameters. **Lower Panel.** Contributions to the current from the two electrochemical processes, where the contribution from R_6 , i.e. the reduction of the OHNHm adduct, has been multiplied by 1000.196

Figure 11.10. Simulated polarization data based on the parameters used to fit the steady state data in Figure 11.9 (see legends for details).197

List of Abbreviations

A	Area of the electrode
α	Electron transfer coefficient
Ag/AgCl	Silver/silver chloride reference electrode
c	concentration
δ	diffusion boundary layer thickness
D	Diffusion coefficient
E	Electrical Potential
f	shorthand for F/RT
F	Faraday's Constant, i.e. 96485 C/mol
i	current density
I	current
k	rate constant, usually cm/s
λ	wavelength, nm
n	number of electrons
N	Collection efficiency

Pt(poly)	Polycrystalline Platinum
Q	Charge
R	Ideal gas constant, i.e. 8.314 J/mol K , also reaction rate
RHE	Reversible hydrogen electrode
R(R)DE	Rotating (ring) disk electrode
SHE	Standard hydrogen electrode
T	Temperature
ν	Kinematic viscosity, cm ² /s, also scan rate, V/s
ω	rotation rate

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“Isn’t it funny that, day by day, nothing changes, but when you look back, everything is different” –C.S. Lewis

Theoretical and Experimental Aspects of Electrocatalysis of the Oxygen Reduction Reaction and Related Systems

Abstract

By

NICHOLAS STEFAN GEORGESCU

This thesis addresses theoretical aspects of systems of relevance to energy conversion, energy storage and the nitrogen cycle, as studied by rotating disk and ring-disk electrode techniques. In particular, the non-linear character of Koutecky-Levich plots often reported in the literature for the oxygen reduction reaction, ORR, and the hydrogen peroxide reduction, HPRR, have been ascribed to the interplay between mass transport of intermediates away from the electrode surface and electrode kinetics. Also assessed was the validity of $X_{\text{H}_2\text{O}_2}$, a popular figure of merit used to rank the efficacy of electrocatalysts towards the ORR. In another set of studies, numerical simulations were performed to determine the amount of superoxide produced at the disk of a rotating ring-disk electrode as detected by a functionalized concentric ring electrode.

Also examined were systems involving partially-active surfaces involving in one case bromide adsorbed on Pt as a poisoning species for the ORR and HPR and bromide adsorption on Au as an electrocatalyst for the reduction of hexaaquairon(III) ion in aqueous electrolytes. In both cases, very small effects were found for bromide at small coverages. In related studies involving nanoparticles of electrocatalysts dispersed on otherwise inert supports, a full

theoretical analysis for thin films of such materials bonded to a rotating disk electrode yielded results which support the use of a modified Koutecky-Levich equation for the determination of rate constants for first order reactions.

Finally, two further reactions of interest were studied, namely the bromide-bromine-tribromide reaction couple, where it was accurately predicted that negligible effects on the current would result from the formation of tribromide, as well as the reduction of hydroxylamine on adsorbed hemin on glassy carbon, for which a model was proposed that could account semiquantitatively for data collected with a RDE. Lastly, theoretical aspects of ohmic microscopy were investigated that served to validate interpretation of experimental data in terms of a primary current distribution.

Chapter 1 Introduction

The explosive growth of the human population over the past century is beginning to stress our resources in terms of fuel availability, as well as our ability to produce enough food and energy to meet societal needs, creating complex environmental problems which, unless properly addressed, will impact adversely life on earth as we know it today. Two factors may be regarded as particularly threatening. First, the continuous burning of fossil fuels and natural gas has led to an increase in the concentration of carbon dioxide in the atmosphere, a prime contributor to climate change. Second, the over-fertilization of agricultural lands has generated imbalances in the nitrogen cycle leading to the accumulation of nitrate in lakes and rivers leading to the development of hypoxic dead zones. Electrochemistry affords, in principle, a viable means for meeting some of these challenges. Particularly attractive is the use of hydrogen derived from the electrolysis of water powered by a photovoltaic device, which can then be used in a conventional fuel cell to generate electricity producing water in the process.

This thesis focuses on theoretical and experimental aspects of electrocatalysis as it relates to the oxygen reduction reaction, ORR, in aqueous electrolytes, as well as selected model redox reactions involving atomic and molecular species that can profoundly affect its kinetics and mechanisms. Also contained are other reactions of interest to electrochemical energy storage and the nitrogen cycle.

Chapter 2 presents a mechanistic model for the reduction of $\text{H}_2\text{O}_2(\text{aq})$, a solution phase intermediate in the ORR in aqueous electrolytes on a rotating disk electrode, RDE, that includes a possible heterogeneous dismutation of $\text{H}_2\text{O}_2(\text{aq})$. As will be shown, a theoretical analysis of this model predicts, in agreement with the behavior found experimentally for a variety of electrode materials, Koutecky-Levich plots that deviate from linearity at high rotation rates. In a

similar fashion, Chapter 3 seeks to demonstrate that the Koutecky-Levich plots for the ORR itself are an unreliable means of extracting information, even neglecting the possibility of heterogeneous dismutation.

In related studies (Chapter 4), the assumptions underlying the use of $X_{\text{H}_2\text{O}_2}$, the fraction of the disk current that generates $\text{H}_2\text{O}_2(\text{aq})$, as a figure of merit for ranking the efficacy of ORR electrocatalysts, were critically assessed. The results obtained indicated the classical mechanism proposed by Damjanovic et al. (*J. Chem. Phys.* **45**, 4057 (1966)), which neglects hydrogen peroxide disproportionation, predicts that at a fixed potential, a plot of the ratio of the disk to the ring currents for the ORR vs $\omega^{-1/2}$, where ω is the rotation rate of the disk, should be linear. According to this analysis, the slope of this line is proportional to $k_3 / X_{\text{H}_2\text{O}_2}$, where k_3 is the rate constant for the reduction of solution phase hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$, and $X_{\text{H}_2\text{O}_2}$ can be determined from the line intercept. Illustrations of this theoretical framework are provided for ORR data reported in the literature for experiments involving Pt-based electrodes and collected in our laboratories for Pt in aqueous acidic solutions.

Chapter 5 presents a mathematical model of a method for detection of $\text{O}_2^-(\text{aq})$ reported in our laboratories that allows for the concentration of this intermediate in the ORR next to the disk of a rotating ring disk electrode, RRDE to be determined during the ORR based on the magnitude of the current collected with a Au ring specifically functionalized to detect exclusively $\text{O}_2^-(\text{aq})$, using data reported earlier in our laboratories for a glassy carbon disk electrode as a model system.

Chapter 6 provides analytical expressions that reproduce rather accurately mass transport to an array of hexagonally-packed active disks embedded in an otherwise inert rotating disk electrode, which can serve as an accurate model system for electrocatalytic nanoparticles dispersed on an otherwise inert support. Conditions were found under which one of such expressions reduces to that proposed earlier by Zoski et al., (*J. Electroanal. Chem.*, **651**, 80 (2011)) for an microelectrode array under linear convection, which in the absence of convection reduces to a Koutecky-Levich type equation recently proposed by Kim et al. (*Anal. Chem.*, **88**, 1742 (2016).) for a first-order redox reaction on a microelectrode under stagnant conditions.

Chapter 7 discusses changes in the electrocatalytic activity of Pt for the ORR as well as for the hydrogen peroxide reduction reaction, HPOR, in an aqueous acidic electrolyte induced by the adsorption of bromide. To this end, experiments were carried out which could determine the current at the electrode surface as a function of both coverage and applied potential. As evidenced from the data collected, significant drops in the diffusion-limited currents for O₂ reduction induced by the presence of 10 μ M KBr could only be observed for very large coverages regardless of E_{step}. Similar measurements involving the same Pt(poly) rotating disk electrode performed in 1 mM H₂O₂ in deaerated 0.1 M HClO₄ devoid of O₂ displayed a similar trend. This behavior was found to be consistent with a simple blockage of surface active sites by adsorbed bromide, as predicted by the attenuation model proposed by Levart.

The work in Chapter 8 employs a similar tactic, wherein the coverage of bromide is correlated with an enhancement in the rates of reduction of hexaaquairon(III), [Fe(H₂O)₆]³⁺, in acid media. Rather surprisingly, no evidence for increased electrocatalytic activity was found for $\theta_{Br}^N < 0.15$. At higher coverages, the results obtained were in qualitative agreement with theories

put forward in years past that correlate the activities of arrays of active microdisks arranged in a hexagonal pattern, as discussed in a previous chapter.

Chapter 9 involves a series of electrochemical measurements of the potential drop in solution near the surface of the electrode induced by the passage of current, which were then simulated in Comsol. Excellent quantitative agreement was found between the experimental values of this potential drop, $\Delta\phi_{\text{sol}}$, and the predictions based upon a model involving uniform current distribution.

Chapter 10 focuses on the oxidation of bromide, which forms bromine, a heterogeneous product that consumes the reactant in a homogeneous reaction, thereby affecting the measured current. Numerical simulations of the electrochemical-chemical (EC) mechanism that take into account the dynamics of formation and dissociation of tribromide, yielded good agreement with both the optical and electrochemical data collected in this work under strict diffusion control.

In Chapter 11, Cyclic voltammetry and dynamic polarization at a rotating disk electrode, RDE, have been employed to elucidate the electrocatalytic properties of hemin, Hm, adsorbed on glassy carbon, GC, surfaces, Hm|GC, toward hydroxylamine, NH_2OH , in neutral aqueous solutions. The results obtained have provided ample evidence that Hm|GC promotes both the reduction and oxidation of NH_2OH , as well as its disproportionation. Also presented in this work are mathematical simulations based on a simple mechanism for the adsorbed Hm-mediated reduction of NH_2OH that accounts quantitatively for experimental data collected with a RDE as a function of rotation rate.

Chapter 2 Hydrogen Peroxide Reduction in Aqueous Electrolytes: Influence of a Heterogeneous Decomposition Step

Publication Details

The contents of this chapter are published in the Journal of the Electrochemical Society.¹
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Abstract

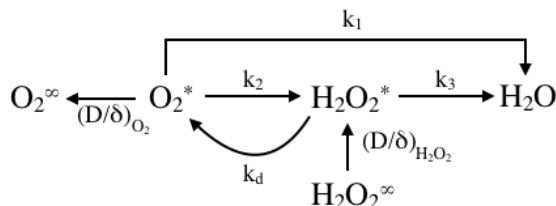
A mechanistic model is herein presented for the reduction of $\text{H}_2\text{O}_2(\text{aq})$ in aqueous electrolytes on a rotating disk electrode that considers the possible heterogeneous dismutation of $\text{H}_2\text{O}_2(\text{aq})$. A theoretical analysis of this model predicts Koutecky-Levich plots that deviate from linearity at high rotation rates depending on the relative values of the rates constants of the processes involved, in agreement with the behavior found for a variety of electrode materials.

Introduction

The search for inexpensive, non-noble-metal-based electrocatalysts for the oxygen reduction reaction (ORR) has ignited renewed interest in the study of the redox properties of hydrogen peroxide, H_2O_2 , in aqueous electrolytes. As is well known, solution-phase H_2O_2 is often found to be an intermediate in the ORR, leading to losses in the operating voltage of hydrogen/oxygen fuel cells and also to the progressive degradation of proton-exchange membranes. Despite considerable efforts, certain aspects of the reduction of $\text{H}_2\text{O}_2(\text{aq})$ have not as yet been fully elucidated. In particular, no satisfactory explanation has as yet been given for the non-linear behavior of Koutecky-Levich (KL) plots found for this reaction²⁻⁸, although the possible involvement of a heterogeneous dismutation of H_2O_2 has indeed been raised⁹. This brief communication presents a theoretical model that considers explicitly such possibility, and

predicts that the direction and extent of the bending depends critically on the values of the rate constants associated with all the processes involved.

Theoretical



Scheme 2.1. Reaction pathways for the reduction of $\text{H}_2\text{O}_2(\text{aq})$ considered in this work, where k_1 is rate constant for the reduction of oxygen to water, k_2 and k_3 are the rate constants for the reduction of solution phase oxygen, $\text{O}_2(\text{aq})$, to solution phase peroxide, $\text{H}_2\text{O}_2(\text{aq})$, and water, respectively, k_d is the first-order rate constant for the catalytic disproportionation of $\text{H}_2\text{O}_2(\text{aq})$, and δ_j is the diffusion boundary layer thickness for species $j = \text{H}_2\text{O}_2(\text{aq})$ and $\text{O}_2(\text{aq})$. The symbols * and ∞ represent the region immediately adjacent to the electrode surface and bulk, respectively.

The mechanistic pathways for $\text{H}_2\text{O}_2(\text{aq})$ reduction to be considered herein (see Scheme 2.1)^{10, 11} assumes that none of the species involved displays any affinity for the electrode surface, i.e. the possible role of adsorbed reaction intermediates is fully neglected. As shown in the scheme, $\text{H}_2\text{O}_2(\text{aq})$ present in a region immediately adjacent to the electrode surface, denoted as $\text{H}_2\text{O}_2^*(\text{aq})$, can be reduced via a two-electron process to yield water, i.e. $2\text{e}^- + \text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O}$. In addition, $\text{H}_2\text{O}_2^*(\text{aq})$ can undergo a surface-catalyzed disproportionation generating water and $\text{O}_2(\text{aq})$, i.e. $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$, which, in turn, can be reduced, regenerating $\text{H}_2\text{O}_2^*(\text{aq})$ via a two-electron process, i.e. $2\text{e}^- + \text{O}_2 \rightarrow \text{H}_2\text{O}_2$, or producing water via a four-electron reaction, i.e. $4\text{e}^- + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$. Also specified in the scheme are the rate constants associated with each of processes involved, as well as the mass transport contributions, denoted as $(D/\delta)_j$, $j = \text{H}_2\text{O}_2(\text{aq})$ and $\text{O}_2(\text{aq})$, where D_j is the diffusion coefficient of species j , $\delta = 1.61(D^2\omega^{-3}\nu)^{1/6}$, the diffusion

boundary layer thickness, ν , the kinematic viscosity of the solution, and the superscript ∞ refers to the bulk concentration. On this basis, and assuming all of the reactions are first order in the reactant, as assumed originally by Bagotskii et al.,⁹ the mass balances for $O_2^*(aq)$ and $H_2O_2^*(aq)$ can be shown to be given by:

$$k_d [H_2O_2(aq)]^* = [O_2(aq)]^* (k_1 + k_2 + (D/\delta)_{O_2}) \quad [2.1]$$

and

$$k_2 [O_2(aq)]^* + ([H_2O_2(aq)]^\infty - [H_2O_2(aq)]^*) (D/\delta)_{H_2O_2} = (k_3 + 2k_d) [H_2O_2(aq)]^* \quad [2.2]$$

respectively, from which explicit expressions for $O_2^*(aq)$ and $H_2O_2^*(aq)$, can be derived, namely,

$$[O_2(aq)]^* = \frac{k_d (D/\delta)_{H_2O_2} [H_2O_2(aq)]^\infty}{(k_1 + k_2 + (D/\delta)_{O_2}) (k_3 + 2k_d + (D/\delta)_{H_2O_2}) - k_2 k_d} \quad [2.3]$$

$$[H_2O_2(aq)]^* = \frac{(k_1 + k_2 + (D/\delta)_{O_2}) (D/\delta)_{H_2O_2} [H_2O_2(aq)]^\infty}{(k_1 + k_2 + (D/\delta)_{O_2}) (k_3 + 2k_d + (D/\delta)_{H_2O_2}) - k_2 k_d} \quad [2.4]$$

The current density, i , will be given by the sum of the three electrochemical processes in Scheme I, i.e.

$$i = F \left\{ 4k_1 [O_2(aq)]^* + 2k_2 [O_2(aq)]^* + 2k_3 [H_2O_2(aq)]^* \right\} \quad [2.5]$$

where F is Faraday's constant. Substituting Eqs. 2.3 and 2.4 into Eq. 2.5, the reciprocal current density can be expressed as:

$$\frac{1}{i} = \frac{1}{2F [H_2O_2(aq)]^\infty} \left\{ \frac{1}{(D/\delta)_{H_2O_2}} + \frac{2(D_{O_2}/D_{H_2O_2})^{2/3} + (k_1 + k_2 + (D/\delta)_{O_2})/k_d}{(2k_1 + k_2) + k_3(k_1 + k_2 + (D/\delta)_{O_2})/k_d} \right\} \quad [2.6]$$

As expected, for $k_d = 0$, i.e. no $H_2O_2^*(aq)$ dismutation, Eq. 2.6 will reduce to a conventional Koutecky-Levich (KL) expression for the two-electron reduction of $H_2O_2(aq)$ proceeding at a rate proportional to k_3 :

$$\frac{1}{i_{k_d \rightarrow 0}} = \frac{1}{2F [H_2O_2(aq)]^\infty (D/\delta)_{H_2O_2}} + \frac{1}{2F [H_2O_2(aq)]^\infty k_3} \quad [2.7]$$

Furthermore, as k_d reaches values much larger than $k_3(k_1 + k_2 + (D/\delta)_{O_2})/(2k_1 + k_2)$, Eq. 2.6 can be expressed as:

$$\frac{1}{i_{k_d \rightarrow \infty}} = \frac{1}{2F [H_2O_2(aq)]^\infty (D/\delta)_{H_2O_2}} + \frac{(D_{O_2}/D_{H_2O_2})^{2/3}}{F [H_2O_2(aq)]^\infty (2k_1 + k_2)} \quad [2.8]$$

In this specific case, a KL plot will also be linear with exactly the same slope as that in Eq. 2.7 and an intercept inversely proportional to the sum of the rates constants for $O_2(aq)$ reduction to peroxide and water, i.e. k_1 and k_2 , respectively.

Results and Discussion

Insight into the effect of k_d on the shape of the KL plots was obtained using, as a means of illustration, values for the electrochemical rate constants similar to those reported by

Muthukrishnan *et al.*¹² for a pyrolyzed Fe-containing carbonaceous material derived from mixtures of fine particles of polyimide and tris(acetylacetonato) iron(III) recorded at a potential $E \approx 0.6\text{V}$ vs RHE (as specified in the legend in Panel A, Figure 2.1), as well as those by Uehara *et al.* for a tantalum-based catalyst dispersed on multi-walled carbon nanotubes¹³ at $E \approx 0.45\text{V}$ vs RHE (cf. Figure 3 in Ref. ¹³). Shown in Panel A, Figure 2.1 are a series of $1/i$ vs $\omega^{-1/2}$ (KL) plots for various k_d based on the parameters provided in the legend, and for values of D_{O_2} , $D_{H_2O_2}$ and v listed in the caption. As clearly evident from these results, the curves obtained for intermediate values of k_d were found to bend upward as ω increased approaching the limit predicted by Eq. 2.7 for $k_d \rightarrow 0$ (see grey dashed line).

A more detailed mathematical analysis of Eq. 2.6 reveals that the bending direction of the KL plots or, equivalently, the sign of their first derivative with respect to $\omega^{-1/2}$ in the limit as $\omega \rightarrow \infty$, is governed by the difference between the limiting values of the reciprocal current densities in Eqs. 2.7 and 2.8, which can be conveniently expressed in terms of a dimensionless parameter γ , defined as:

$$\gamma = \ln \left\{ \left(\frac{1}{k_3} \right) \middle/ \left(\frac{2(D_{O_2} / D_{H_2O_2})^{2/3}}{(2k_1 + k_2)} \right) \right\} = \ln \left\{ \left(\frac{D_{H_2O_2}}{D_{O_2}} \right)^{2/3} \frac{(2k_1 + k_2)}{2k_3} \right\} \quad [2.9]$$

In particular, $\gamma > 0$ if $(2k_1 + k_2)/D_{O_2}^{2/3} > 2k_3/D_{H_2O_2}^{2/3}$, whereas $\gamma < 0$ if $(2k_1 + k_2)/D_{O_2}^{2/3} < 2k_3/D_{H_2O_2}^{2/3}$.

In other words (and ignoring the rather small differences in the values of $D_{H_2O_2}$ and D_{O_2}), if the rates of oxygen reduction, regardless of the mechanism involved, are larger than the corresponding rates of peroxide reduction, γ would be positive, whereas should the opposite be the case, γ would be negative. The latter is illustrated in Panel B, Figure 2.1, for the set of

(fictitious) parameters specified in the legend for which γ is the same as in Panel A, but of opposite sign.

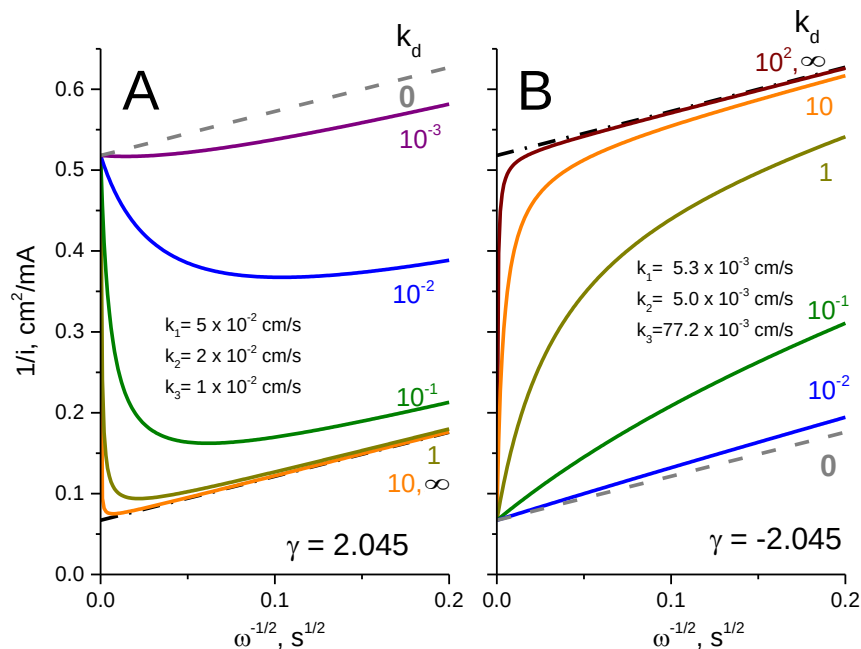


Figure 2.1. Koutecky-Levich plots generated using Eq. 2.6 for various values of the rate constants for which $\gamma = 2.045$ (Panel A) and $\gamma = -2.045$ (Panel B), assuming $D_{\text{O}_2} = 1.3 \times 10^{-5} \text{ cm}^2/\text{s}$ and $D_{\text{H}_2\text{O}_2} = 1.9 \times 10^{-5} \text{ cm}^2/\text{s}$ and $\nu = 0.01 \text{ cm}^2/\text{s}$.

In yet another paper, Amirfakhri *et al.* examined the reduction of $\text{H}_2\text{O}_2(\text{aq})$ on metal-free nitrogen-doped graphene nanoflake electrodes in unbuffered 0.1 M Na_2SO_4 aqueous solutions and found KL plots bending downward² (see scattered symbols in Figure 2.2 below), where the solid lines are linear fits to the data in the original paper). As indicated, the experimental data displays a curvature which according to our theory, is consistent with a rate for peroxide reduction that exceeds that of oxygen reduction, i.e. $\gamma < 0$. Assuming, for simplicity, a value of $k_1 = 0$, i.e. total neglect of a four-electron reduction of oxygen to water, a potential independent two-electron reduction of oxygen to peroxide, and a constant value of k_d , the best fits to the data

based on our model (see Eq. 2.6; solid lines in Figure 2.2), yielded values of $k_2 = 0.0016$ cm/s, and $k_d = 0.25$ cm/s, which reproduce the curvature observed in the experimental results for all rotation rates, except perhaps the lowest one for the most negative voltage. It should be emphasized that for large ω our model provides a much better fit than their linear analogue (dotted lines in Figure 2.2).

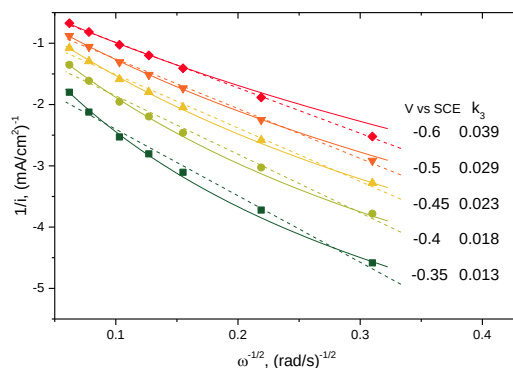


Figure 2.2. Koutecky-Levich plots reproduced from the data reported by Amirfakhri *et al.* (see scattered symbols) and fits according to our theory (solid lines). The dotted lines are linear fits provided in the Ref. ². See text for additional details.

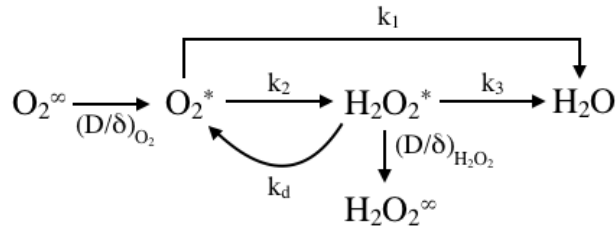
Also included in this report² was a thorough theoretical analysis of mechanistic aspects of $\text{H}_2\text{O}_2(\text{aq})$ reduction using a similar approach to that employed in this work yielding expressions (cf. Eqs. 13 and 14 in the Ref. ²) equivalent to those in Eq. 2.6. However, the model considered therein assumed that the O_2 generated from $\text{H}_2\text{O}_2(\text{aq})$ dismutation remained adsorbed on the surface undergoing a subsequent irreversible desorption, which led them to conclude that KL plots would always be linear, a fact that is not supported by their own measurements. These contradictory results stem from their neglect of the mass transport contribution associated with the $\text{O}_2(\text{aq})$ and thus its dependence on ω .

Summary

The theoretical analysis herein presented has shown that the curvature of Koutecky-Levich plots observed for the reduction of $\text{H}_2\text{O}_2(\text{aq})$ in aqueous electrolytes reported in the literature can be accounted for by including the heterogeneous dismutation of $\text{H}_2\text{O}_2(\text{aq})$, whereby the direction of the bending depends critically on the values of the rates of the processes involved. On this basis, caution must be exercised when forcing a linear fit to such plots, as the slope and intercept could not be interpreted in the conventional way, and thus yield unreliable values for the rate constants for $\text{H}_2\text{O}_2(\text{aq})$ reduction. One possible means of avoiding this problem is by measuring the rate constant for the heterogeneous dismutation of $\text{H}_2\text{O}_2(\text{aq})$ under the same identical conditions as those employed in the electrochemical experiments and include those in the mechanistic analysis.

Chapter 3 The Koutecky-Levich Plot for the Evaluation of Catalysts for the Oxygen Reduction Reaction: A Critical Analysis

Introduction



Scheme 3.1

Improving the electrocatalytic efficiency of the Oxygen Reduction Reaction, ORR, is a common target for advancement in the electrochemical sciences. Of particular concern when developing novel catalysts is the effective number of electrons transferred per oxygen molecule reacted at the catalyst, n_{eff} , which typically ranges from 2 to 4, as a measure of how much oxygen goes through the two- and four-electron pathways respectively (See Scheme 3.1). This quantity can be written in terms of the rates k_1 and k_2 as seen in the Scheme, representing the quantity of oxygen reacting to form oxygen and peroxide respectively, i.e.:

$$n_{eff} = \frac{4}{1 + (1 + 2k_1 / k_2)^{-1}} \quad [3.1]$$

It is common practice to extract this information by measuring the current at a variety of rotation rates and constructing a Koutecky-Levich, KL, plot, which can be written for a current density at the electrode, i :

$$\frac{1}{i} = \frac{1}{n_{eff} F [O_2(aq)]^\infty} \left\{ \frac{1}{0.62 D^{2/3} \nu^{-1/6} \omega^{-1/2}} + \frac{1}{k_{eff}} \right\} \quad [3.2]$$

where k_{eff} is some effective first-order rate constant, and other symbols have their usual meanings (*vide infra*). An n_{eff} can be extracted from the slope of a linear fit of the data via:

$$n_{\text{eff}, \text{meas}} = \frac{1.612 D^{-2/3} \nu^{1/6}}{F [O_2(aq)]^\infty} \left/ \left(\frac{d[1/i]}{d\omega^{-1/2}} \right) \right. \quad [3.3]$$

Foremost amongst the assumptions embedded in this formalism is that there is a single species reacting at the electrode with no subsequent electroactive products generated. For many less complicated reactions, this assumption is met to a reasonable degree. However, it is well known for the ORR that electroactive side products such as hydrogen peroxide are generated (See Scheme 3.1). In this work we show that the use of Eq. 3.3 applied to a fit of a KL plot, even of a seemingly straight line, does not yield an effective measure of the electrocatalytic properties of catalysts and therefore caution is advised when using it as a tool to rank electrocatalysts.

Theoretical

The mechanism to be considered here (see Scheme 3.1) assumes that none of the species have any affinity for the surface. As shown in the Scheme oxygen can undergo a two-, or four-electron reduction to form peroxide or water, respectively. In addition, peroxide can undergo a two-electron reduction to form water, or undergo a first-order disproportionation to form water and oxygen, i.e. $2H_2O_2(aq)^* \rightarrow O_2(aq)^* + 2H_2O$. As in the previous chapter, the resulting mass balances for oxygen and peroxide, respectively, are:

$$(D_{O_2} / \delta) ([O_2(aq)]^\infty - [O_2(aq)]^*) + [H_2O_2(aq)]^* k_d = [O_2(aq)]^* (k_1 + k_2) \quad [3.4]$$

and

$$-(D_{O_2} / \delta) \xi [H_2O_2(aq)]^* + [O_2(aq)]^* k_2 = [H_2O_2(aq)]^* (k_3 + 2k_d) \quad [3.5]$$

where $\xi = D_{H_2O_2}^{2/3} / D_{O_2}^{2/3}$ and the diffusion boundary layer thickness, δ , can be shown to be given

by $\delta = 1.612 (\omega^{-3} D^2 V)^{1/6}$. Given these two equations, the respective concentrations at the

electrode surface can be solved for, yielding:

$$[O_2(aq)]^* = \frac{[O_2(aq)]^\infty (D_{O_2} / \delta) (\xi D_{O_2} / \delta + k_3 + 2k_d)}{(D_{O_2} / \delta + k_1 + k_2) (\xi D_{O_2} / \delta + k_3 + 2k_d) - k_2 k_d} \quad [3.6]$$

And

$$[H_2O_2(aq)]^* = \frac{[O_2(aq)]^\infty (D_{O_2} / \delta) k_2}{(D_{O_2} / \delta + k_1 + k_2) (\xi D_{O_2} / \delta + k_3 + 2k_d) - k_2 k_d} \quad [3.7]$$

respectively. These equations can be substituted into the expression for the current density, i , namely:

$$i = F \left((4k_1 + 2k_2) [O_2(aq)]^* + 2k_3 [H_2O_2(aq)]^* \right) \quad [3.8]$$

to yield:

$$i = \frac{2F [O_2(aq)]^\infty (D_{O_2} / \delta) \{ (2k_1 + k_2) (\xi D_{O_2} / \delta + k_3 + 2k_d) + k_2 k_3 \}}{(D_{O_2} / \delta + k_1 + k_2) (\xi D_{O_2} / \delta + k_3 + 2k_d) - k_2 k_d} \quad [3.9]$$

By substituting in the definition for δ , $1/i$ can be expressed as follows:

$$\frac{1}{i} = \frac{1.612D_{O_2}^{-2/3}\nu^{1/6}\omega^{-1/2}}{4F[O_2(aq)]^\infty} + \frac{1}{4F[O_2(aq)]^\infty} \times \frac{2\xi + (\xi k_2 + 2k_3 + 4k_d)1.612D_{O_2}^{-2/3}\nu^{1/6}\omega^{-1/2}}{\xi(2k_1 + k_2) + 2(k_3(k_1 + k_2) + k_d(2k_1 + k_2))1.612D_{O_2}^{-2/3}\nu^{1/6}\omega^{-1/2}} \quad [3.10]$$

Results and Discussion

On this basis, in the limit of infinitely slow rotation rates for $k_3 + k_d \gg 0$, Eq. 3.10 reduces an apparent linear KL plot:

$$\frac{1}{i_{\omega^{-1/2} \rightarrow \infty}} = \frac{1.612D^{-2/3}\nu^{1/6}\omega^{-1/2}}{4F[O_2(aq)]^\infty} + \frac{\xi k_2 + 2k_3 + 4k_d}{8F[O_2(aq)]^\infty((k_1 + k_2)k_3 + (2k_1 + k_2)k_d)} \quad [3.11]$$

which has an intercept that convolutes information about all of the rate constants, as well as a measured number of effective electrons transferred, n_{eff} , always equal to 4. This differs from the actual limit of the KL as the intercept is approached, namely:

$$\frac{1}{i_{\omega^{-1/2} \rightarrow 0}} = \frac{1}{2F[O_2(aq)]^\infty(2k_1 + k_2)} \quad [3.12]$$

indicating that there will be a region of curvature in between the linear region shown in Eq. 3.11 and the intercept. The ratio of the difference between the intercept from the two intercepts is given by

$$\left. \frac{i_{\omega^{-1/2} \rightarrow \infty}}{i} \right|_{\omega^{-1/2} \rightarrow 0} = \frac{4(1 - (2 + k_2/k_1)^{-1})k_3 + 4k_d}{\xi k_2 + 2k_3 + 4k_d} \quad [3.13]$$

It can be shown that the figure of merit describes whether this function is greater than or less than unity is precisely the figure defined previously for the hydrogen peroxide reduction reaction, namely:

$$\gamma = \ln \left[\frac{(2k_1 + k_2)\xi}{2k_3} \right] = \ln \left[\frac{(2k_1 + k_2)D_{H_2O_2}^{2/3}}{2k_3D_{O_2}^{2/3}} \right] \quad [3.14]$$

Based upon this parameter, one can predict the sign of the curvature. However, it behaves in the inverse fashion to that predicted in our previous publication. That is, the direction of curvature for the ORR will be share the sign with $-\gamma$ (see Figure 3.1). This is exemplified by the changing of sign in the curvature between the ORR and HPORR in publications that contain both¹⁴.

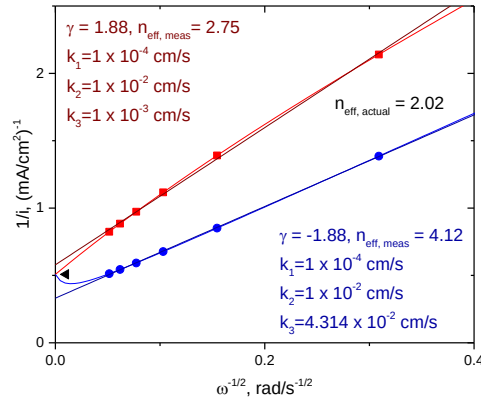


Figure 3.1. Example plots for two (fictitious) systems of $\gamma = \pm 1.88$ with $n_{\text{eff, actual}} = 2.02$, $D_{O_2} = 1.3 \times 10^{-5}$ cm/s, and $D_{H_2O_2} = 1.9 \times 10^{-5}$ cm/s, that otherwise share all parameters except k_3 . Light lines are plots of the reciprocal current obtained from Eq. 3.10; light points are simulated data along those plots at square hundred rpm, in the range of 100-3600 rpm; and dark lines are the linear fits according to the data at those points, from which the respective $n_{\text{eff, meas}}$ is calculated. The black triangle points to the common intercept.

Most importantly, an analysis of the expression in Eq. 3.10 using Eq. 3.3 to attempt to extract n_{eff} shows that the measure number of electrons convolutes information about the entire system:

$$n_{\text{eff}, \text{meas}} = \frac{4}{1 + \left(1 + \frac{2k_1}{k_2}\right)^{-1} \left\{ \left(1 - e^{-\gamma}\right) \left(1 + \frac{e^{-\gamma} (k_1 + k_2) + 2k_d / \xi}{1 / (1.612 D_{\text{O}_2}^{-2/3} \nu^{1/6} \omega^{-1/2})}\right)^{-2} \right\}} \quad [3.15]$$

Which means that the number of effective electrons measured will vary between a value of $n_{\text{eff}, \text{meas}} = 4$ at very far from the intercept and $n_{\text{eff}, \text{meas}} = 4 / \left\{ 1 + (1 - e^{-\gamma}) (1 + 2k_1 / k_2)^{-1} \right\}$ at the intercept, the latter of which could be larger than 4 or even negative. In fact, there have been publications which report an n_{eff} for oxygen greater than 4¹⁵ and n_{eff} less than 2¹⁶. This type of result is one of the predictions of this model, and is otherwise not understandable.

It is also possible that γ will change sign over the scan of the potential, simply by one set of rates overtaking the other. For these systems γ is typically remains very close to 0. Eq. 3.15 predicts that one will measure from the slope an n_{eff} of 4, even if it proceeds strictly through a two-electron pathway. This has also been seen in the literature¹⁷.

For most of the reported systems, it is impossible to determine from the limited measurements what the actual rates are precisely because radically different systems will generate otherwise identical KL plots. So, to illustrate the problems more fully, let us consider the pair of $\gamma = \pm 1.88$ plots shown in Figure 3.1, ignoring, for simplicity, the possibility of disproportionation via k_d . Although both systems have identical k_1 and k_2 , and thus will have identical performance at high ω , one material is only slightly overestimated in the specificity for

the four-electron path, while the other has an apparent n_{eff} much greater than 4. In reality, both materials take almost exclusively the two-electron path. Also interesting is that the plot with the less accurate result showed less curvature, demonstrating that a linear plot is no guarantee of a reliable measurement. In a similar fashion the same pair of $\gamma=\pm 2.044$ systems as in the former publication (see the previous chapter for more information),. The actual number of electrons transferred is 3.43 and 3.03 for the positive- and negative- γ systems respectively. However, if one were to take measurements at every square hundred rpm up to 3600, the results would be 3.93 and 4.03 electrons transferred respectively even though the plots appear nearly linear. That is to say, not only are the specificities for the four-electron pathway for these systems overestimated by a significant factor, but the material with a worse specificity might be favored by the methodology, even though it may significantly underperform by comparison.

These examples highlight the inability to effectively evaluate catalysts with simply a Koutecky-Levich plot alone, even if the results appear to be linear. It causes unrealizable and contradictory results owing to a misunderstanding of the use of the technique, and the results hardly corroborate the actual behavior of the systems in question.

Suggested Methods for Finding n_{eff}

A much more effective methodology for collecting n_{eff} involves measuring these rates with the use of a ring in the presence of bulk oxygen. First, the ring is coated with a material for which all of the rates in Scheme 3.1 except for k_2 are zero, such as selenium or sulfur-coated Pt.¹⁸ The potential of the disk can then be scanned in the region of interest. The difference between the unshielded ring current (i.e. when $i_{\text{disk}}=0$) and the shielded ring current at a given potential is given by the flux of oxygen into the disk multiplied by the collection efficiency, N :

$$\begin{aligned}
i_{ring,O_2}^0 - i_{ring,O_2} &= i_{ring,O_2}^{shielding} \\
&= 2NFA_{disk} (D_{O_2} / \delta) \left([O_2(aq)]^\infty - [O_2(aq)]^* \right)
\end{aligned}
\tag{3.16}$$

Next, the ring can then be pushed positive to the desorption or oxidation of the species on the ring, to a potential where peroxide can oxidize, but water cannot, and the disk can be scanned over the same potentials. The current due to the collection of peroxide at these potentials is given by:

$$i_{ring,H_2O_2} = 2NFA_{disk} \xi (D_{O_2} / \delta) [H_2O_2(aq)]^* \tag{3.17}$$

By substituting Eqs. 3.6 and 3.7 into 3.16 and 3.17 respectively, it can be shown that a ratio of these two quantities is predicted to be linearly dependent under this model:

$$\frac{i_{ring,H_2O_2}}{i_{ring,O_2}^{shielding}} = \left[\left(\frac{k_1 + k_2}{k_2} \right) + \left\{ \frac{(k_1 + k_2)k_3 + (2k_1 + k_2)k_d}{\xi k_2 / (1.612 D_{O_2}^{-2/3} \nu^{1/6})} \right\} \omega^{-1/2} \right]^{-1}
\tag{3.18}$$

and, therefore, n_{eff} can be extracted from the intercept of this ratio:

$$n_{eff} = 4 - 2 \left(\frac{i_{ring,H_2O_2}}{i_{ring,O_2}^{shielding}} \right)_{\omega \rightarrow \infty} \tag{3.19}$$

When the shielding current is the same as the collection current, i.e. as the ratio approaches unity, the amount of oxygen in is the same as peroxide out, and exclusively the two-electron pathway is taken. However, if no collection is noted at high rotation rates, then no peroxide is being flushed from the disk and the number of effective electrons is 4. This

methodology avoids any of the problems noted with the Koutecky-Levich analysis, and does not rely on the ring current.

Conclusion

It has been shown that the presence of intermediates, which can interfere with the measurement of the specificity, not only in giving nonsensical results (i.e. greater than four electrons per oxygen), but also in creating inversions between the actual and measured specificity, could potentially result in objectively better materials getting discarded because of the problematic nature of this analysis . Given that even the apparent linearity of a Koutecky-Levich plot cannot guarantee the reliability of the measurement, researchers are encouraged to employ a ring to measure n_{eff} using a shielding and collection experiment.

**Chapter 4 A Critical Assessment of $X_{\text{H}_2\text{O}_2}$ As A Figure Of Merit For Oxygen
Reduction Electrocatalysts In Aqueous Electrolytes**

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Abstract

Certain aspects of the interpretation of rotating ring-disk electrode, RRDE, data for the oxygen reduction reaction, ORR, in aqueous electrolytes have been examined. In particular, the classical mechanism proposed by Damjanovic et al. (*J. Chem. Phys.* **45**, 4057 (1966)) predicts that at a fixed potential, a plot of the ratio of the disk to the ring currents for the ORR vs $\omega^{-1/2}$, where ω is the rotation rate of the disk, should be linear. According to this analysis, the slope of this line is proportional to $k_3 / X_{\text{H}_2\text{O}_2}$, where k_3 is the rate constant for the reduction of solution phase hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$, and $X_{\text{H}_2\text{O}_2}$ is the fraction of the disk current that generates $\text{H}_2\text{O}_2(\text{aq})$, the magnitude of which can be determined from the line intercept. Illustrations of this theoretical framework are provided for ORR data reported in the literature for experiments involving Pt-based electrodes and collected in our laboratories for Pt in aqueous acidic solutions. Also critically discussed is the use of $X_{\text{H}_2\text{O}_2}$ as a figure of merit for ranking the efficacy of ORR electrocatalysts.

Introduction

Continuing efforts are being made in laboratories worldwide toward the search for inexpensive and highly active electrocatalysts that can replace Pt, and other noble metal-based materials in technical applications involving energy conversion, such as fuel cells and electrolyzers. A commonly used figure of merit to rank the performance of electrocatalysts for the oxygen reduction reaction, ORR, is a parameter derived from data collected with a rotating ring-disk electrode, RRDE²⁰⁻³¹, denoted as $X_{\text{H}_2\text{O}_2}$, which affords a measure of the fraction of the disk current, I_d , associated with the generation of solution phase hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$. Most often, $X_{\text{H}_2\text{O}_2}$ is calculated via an expression that assumes the disk electrode material to be inactive toward the reduction of $\text{H}_2\text{O}_2(\text{aq})$, namely

$$X_{\text{H}_2\text{O}_2} = \frac{2}{I_d N / I_r + 1} \quad [4.1]$$

where I_r is the ring current as measured with the ring polarized at a potential positive enough for $\text{H}_2\text{O}_2(\text{aq})$ to be oxidized under strict diffusion limited control, and N is the collection efficiency of the RRDE assembly. It thus follows from Eq. 4.1 that if $X_{\text{H}_2\text{O}_2} = 0$, the reaction proceeds strictly by a four electron pathway, i.e. no $\text{H}_2\text{O}_2(\text{aq})$ being detected by the ring, and if $X_{\text{H}_2\text{O}_2} = 1$, the ORR generates exclusively $\text{H}_2\text{O}_2(\text{aq})$.

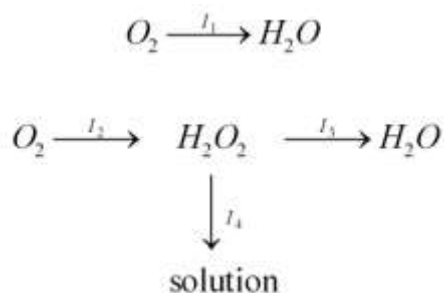
Experimental

Measurements were performed in an all-glass, three-compartment cell, filled with O_2 -saturated (Air Gas) 0.1 M HClO_4 solutions (75 mL) prepared from ultrapure perchloric acid (Ultrapure, EMD), and ultrapure, UHP water generated by a Barnstead system (18.3 M Ω cm,

EASYpureUV) containing 10 μM KBr (Fisher Scientific, IR grade, > 99%). Experiments were carried out using a polycrystalline Pt(poly)|Pt(poly) ring-disk electrode (Pine Instruments, disk diameter: 0.457 cm; ring inner diameter 0.493 cm; gap, $G = 180\ \mu\text{m}$ and ring outer diameter: 0.538 cm, or, correspondingly, disk and ring areas of $0.164\ \text{cm}^2$ and $0.037\ \text{cm}^2$, respectively, and a collection efficiency, $N = 0.22$) and a Pt wire and a reversible hydrogen electrode (RHE) as counter and reference electrodes, respectively. The disk and ring potentials, E_{disk} and E_{ring} , respectively, were controlled with a bipotentiostat (Pine Instruments, Model AFCBP1) and the rotation rate of the RRDE, ω , adjusted with a commercial rotator (Pine Instruments, Model AFMSRX). All the data were recorded using a National Instruments acquisition card (USB-6009) programmed in Labview.

Theoretical Formalism

The analysis presented in this section assumes, as originally proposed by Damjanovic *et al.*³², that the ORR proceeds either via a 4-electron, or a 2-electron pathway generating $\text{H}_2\text{O}_2(\text{aq})$, a species that can subsequently escape into the bulk electrolyte, or be further reduced via a 2-electron process to yield water as the final product (see Scheme 4.1).



Scheme 4.1

where I_i are the rates of the corresponding processes expressed as currents. On this basis, the molar fraction of O_2 that generates $H_2O_2(aq)$, denoted as $X_{H_2O_2}$, may be written as follows^{20, 25}:

$$X_{H_2O_2} = \frac{\left(\frac{I_2}{2}\right)}{\left(\frac{I_1}{4} + \frac{I_2}{2}\right)} \quad [4.2]$$

The symbols I_1 and I_2 in this equation represent the currents as specified in Scheme I. In particular, if the disk electrode displays no activity for $H_2O_2(aq)$ reduction, i.e. $I_3 = 0$ in Scheme I, $X_{H_2O_2}$ reduces to Eq. 4.1, where $I_d = I_1 + I_2$ and $I_r = I_2N$.

Yet another parameter often used to assess the activity of a given electrocatalyst for the ORR is n_e , (also known as the selectivity) defined as

$$n_e = \frac{4}{I_r / (I_d N) + 1} \quad [4.3]$$

which may be regarded as the effective number of electrons transferred per O_2 molecule, e.g. 4 or 2, depending on whether the process generates exclusively water or $H_2O_2(aq)$, respectively.³³

Rather surprisingly, however, hardly any attempts have been made to determine the validity of the assumptions upon which Eq. 4.1 is based, casting doubts regarding the reliability of

$X_{H_2O_2}$ calculated on such basis. This brief communication reexamines theoretical aspects of Scheme I to shed light into some of these issues.

As reported by Damjanovic *et al.*³², the currents in Scheme I are related by the following set of equations^{32, 34}:

$$I_d = I_1 + I_2 + I_3 \quad [4.4]$$

$$I_2 = I_3 + I_4 \quad [4.5]$$

$$I_2 = I_r / N \left(1 + 1.61 k_3 D_{H_2O_2}^{-2/3} \nu^{1/6} \omega^{-1/2} \right) \quad [4.6]$$

$$I_4 = I_r / N \quad [4.7]$$

where k_3 is the rate constant for the reduction of $H_2O_2(aq)$ at the disk, assumed to be first order process, $D_{H_2O_2}$ is the diffusion coefficient of $H_2O_2(aq)$, and ν the kinematic viscosity of the solution. These relationships allow a more general form of $X_{H_2O_2}$ to be derived, namely,

$$X_{H_2O_2} = \left(1 + 1.61 k_3 D_{H_2O_2}^{-2/3} \nu^{1/6} \omega^{-1/2} \right) \frac{2}{I_d N / I_r + 1} \quad [4.8]$$

which, as expected, reduces to Eq. 4.1 when $k_3 = 0$. This expression bears striking similarity to that proposed recently by Ke *et al.*²⁴ to account for the ORR behavior of thin films of catalyst nanoparticles homogeneously distributed on the surface of a glassy carbon electrode for the ORR in 0.1 M $HClO_4$, namely,

$$X_{H_2O_2} = \frac{1}{1 - P_{re}} \times \frac{2}{I_d N / I_r + 1} \quad [4.9]$$

As described by these authors, P_{re} is an empirical parameter that relates to the percent of desorbed peroxide that can readsorb and react on the disk to yield water as the final product, so

that if there is no affinity of $H_2O_2(aq)$ for sites of the electrode surface, Eq. 4.9 reduces to Eq. 4.1

1. It thus follows from Eqs. 4.8 and 4.9 that, in fact,

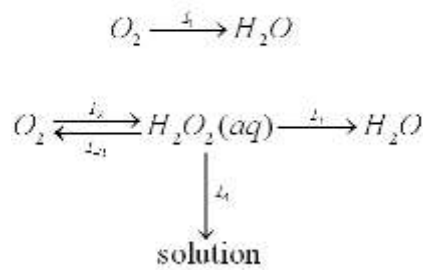
$$P_{re} = 1 - \frac{1}{1 + 1.61k_3D_{H_2O_2}^{-2/3}V^{1/6}\omega^{-1/2}} \quad [4.10]$$

A more useful relationship between the actual observables can be obtained by rearranging Eq. 4.8, i.e.

$$\frac{I_d}{I_r} = \left(\frac{3.22k_3D_{H_2O_2}^{-2/3}V^{1/6}}{X_{H_2O_2}N} \right) \omega^{-1/2} + \left(\frac{2}{X_{H_2O_2}N} - \frac{1}{N} \right) \quad [4.11]$$

In particular, and provided the ORR follows strictly the mechanism in Scheme I, this equation predicts that a plot of I_d/I_r vs $\omega^{-1/2}$ would be linear, and that $X_{H_2O_2}$ and k_3 can, in principle, be determined from the intercept and the slope, S , of data collected at a fixed potential, E . In particular, the slope would be zero, and thus independent of ω if $k_3 = 0$, and positive otherwise.

Recently, L. Zhang et al. attempted to address the same issues as those in this work by including explicitly the oxidation of $H_2O_2(aq)$ in Scheme I³⁵, i.e.



Scheme 4.2

In their analysis, Zhang et al. assumed correctly at first that I_d has four contributions as specified in Scheme 4.2 above; later in their work, however, they mistakenly rewrote I_d as originating solely from the 4e-, and 2e-, pathways, namely,

$$\frac{4I_d}{n_e} = 4F\pi r^2(k_1 + k_2)C_{O_2}^s \quad [4.12]$$

where n_e (denoted in their work simply as n) is the selectivity, which led to mathematical expressions of highly questionable validity. It must be stressed that Eq. 4.12 should not be confused with that reported by Antoine and Durand,³³ namely,

$$\frac{4I_d}{n_e} = I_1 + 2I_2 \quad [4.13]$$

As these authors correctly stated in their paper, this expression only applies provided the series pathway is neglected, i.e. $k_3 = 0$, or is included in the 4-e⁻ disk current³³. Furthermore, a closer examination of Zhang et al.'s contribution revealed yet another error (cf. Eq. 15 in this reference) which, when corrected, predicts $k_3 = k_2'$, an equality that would only be valid if both quantities were zero at all potentials.

In fact, a rigorous analysis of Scheme II may be shown to yield the following expressions for $X_{H_2O_2}$ and I_d/I_r :

$$X_{H_2O_2} = \frac{2I_2}{I_1 + 2I_2} = \frac{2 \left[1 + 1.61(k_2 + k_3) D_{H_2O_2}^{-2/3} \nu^{1/6} \omega^{-1/2} \right]}{I_d N / I_r + 1 + 3.22 k_2 D_{H_2O_2}^{-2/3} \nu^{1/6} \omega^{-1/2}} \quad [4.14]$$

and

$$\frac{I_d}{I_r} = \frac{3.22 D_{H_2O_2}^{-2/3} \nu^{1/6} [k_{-2} (1 - X_{H_2O_2}) + k_3]}{X_{H_2O_2} N} \omega^{-1/2} + \frac{1}{N} \left(\frac{2}{X_{H_2O_2}} - 1 \right) \quad [4.15]$$

where k_{-2} represents the rate constant for the reduction of $H_2O_2(aq)$, assumed to be a first order process. Hence, for high overpotentials for $H_2O_2(aq)$ reduction, $k_3 \gg k_{-2}$, and Eqs. 4.14 and 4.15 correctly reduce to Eqs. 4.8 and 4.11 above.

From a general perspective, the vast majority of papers published over the past two decades have reported values of $X_{H_2O_2}$ ^{20-24, 29, 30} and n_e ^{29, 30, 36, 37} as defined in Eqs. 4.1 and 4.3, respectively, either at a single, or at multiple rotations rates, but failed to perform the required tests to assess the validity of the assumptions upon which their mathematical expressions are based.^{26-28, 31}

Consider, for example, the dynamic polarization curves reported by Stamenkovic et al.²⁷ (see Figure 4.1) for the ORR on a sputtered Pt_3Ni disk electrode of a $Pt|Pt_3Ni$ RRDE ($N = 0.2$) in 0.1 M $HClO_4$. These data were recorded for several values of ω , while the disk potential, E_{disk} , was being scanned at a rate $\nu = 50$ mV/s toward positive values and the ring was polarized at a potential, E_{ring} , positive enough for the oxidation of $H_2O_2(aq)$ to proceed under diffusion-limited conditions. Shown in Figure 4.1 is a plot of I_d/I_r vs E_{disk} using I_r vs E_{disk} results for the two values of ω reported by these authors, i.e. 1600 and 2500 rpm. Although I_d/I_r over the range $0.075 \text{ V} \leq E_{disk} \leq 0.23 \text{ V}$ were found to be quite similar, the corresponding values positive to this range were quite different. This latter behavior indicates that the main assumption upon which their

reported values of $X_{\text{H}_2\text{O}_2}$ were calculated (also provided in Fig. 2 in their work) i.e. no reduction of solution phase $\text{H}_2\text{O}_2(\text{aq})$, was not fulfilled. In fact, the values of $X_{\text{H}_2\text{O}_2}$ determined from their I_d and I_r data using Eq. 4.1 appear to be inconsistent with those in their figure.

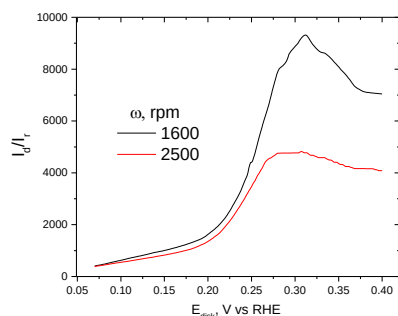


Figure 4.1. Plots of I_d/I_r vs E_{disk} based on the data reported by Stamenkovic et al.(8) for a sputtered Pt_3Ni disk electrode of a $\text{Pt}|\text{Pt}_3\text{Ni}$ RRDE collected at two different rotation rates as indicated.

Yet another illustration of the formalism herein developed is provided by dynamic polarization data recorded in this laboratory at a scan rate $\nu = 10$ mV/s for the $\text{Pt}(\text{poly})$ disk of a $\text{Pt}|\text{Pt}$ RRDE in O_2 -saturated 0.1 M HClO_4 containing 10 μM KBr at various rotation rates (see Figure 4.2). Also included in this figure are the corresponding ring currents recorded simultaneously with the ring polarized at $E_{\text{ring}} = 1.4$ V vs RHE (Panel B). The data in Panels A and B can then be used to generate plots of I_d/I_r vs E_{disk} which, as shown in Panel C in this figure, where in this case the contributions to I_r derived from the oxidation of bromide and other sources were subtracted (see Appendix A for details). As evidenced from these data, I_d/I_r was found to be virtually independent of E_{disk} over the range $0.37 < E_{\text{disk}} < 0.63$ V vs RHE. Hence, according to the model herein developed, the bromide-modified surface in this potential region is not active for the direct reduction of $\text{H}_2\text{O}_2(\text{aq})$, Strong support for this conclusion was obtained from dynamic polarization curves recorded in Ar-purged 1 mM H_2O_2 solutions in the same

electrolyte for the Pt(poly) disk of the RRDE assembly (see Figure 4.3), for which the very small reduction currents observed in the range $0.4 < E_{\text{disk}} < 0.8$ V vs RHE, were found to be independent of ω , i.e. strict kinetic control. Also included in this figure for comparison (see thick line, right ordinate) are data collected at $\omega = 400$ rpm in the same solution devoid of bromide to underscore the marked reduction in the electrocatalytic activity of Pt(poly) induced by the adsorbed halide.

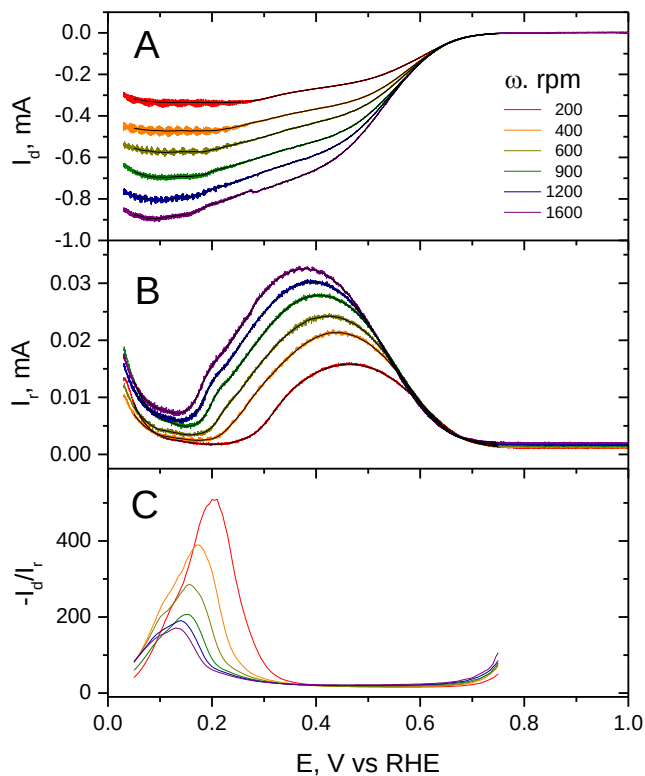


Figure 4.2. Panel A. Dynamic polarization curves recorded at a scan rate $v = 10$ mV/s with the Pt(poly) disk of a Pt(poly)|Pt(poly) RRDE in O_2 -saturated 0.1 M $HClO_4$ solutions containing 10 μ M KBr at various values of ω , as indicated. Panel B. Ring currents collected while the disk data in Panel A was being collected, with the ring polarized at $E_{\text{ring}} = 1.4$ V vs RHE. Panel C. Plots of I_d/I_r based on the data in Panels A and B in this figure, after correction for residual current contributions.

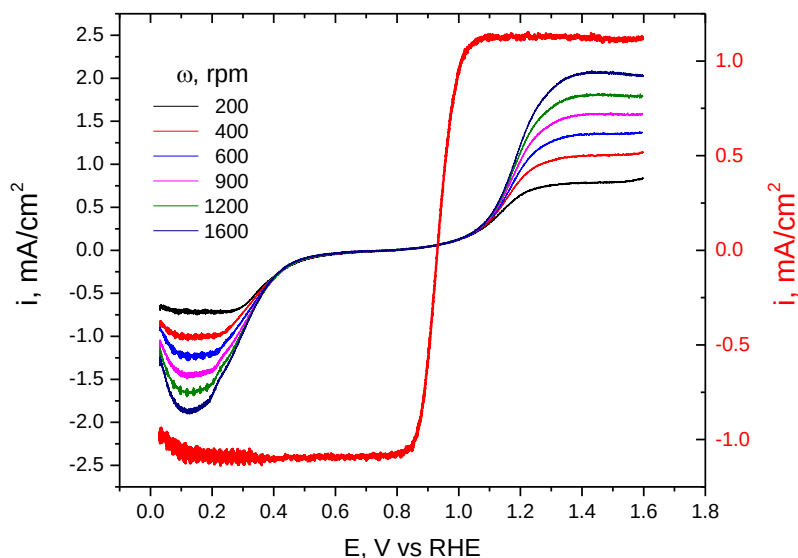


Figure 4.3. Dynamic polarization curves recorded at $v = 10$ mV/s for the reduction of 1 mM H_2O_2 in 10 μM KBr in Ar-purged 0.1 M HClO_4 on the Pt(poly) disk of the Pt|Pt RRDE at various rotation rates as indicated. The thick curve was obtained in the same solution devoid of bromide at $\omega = 400$ rpm.

Plots of I_d/I_r vs $\omega^{-1/2}$ over the same range (see Figure 4.4) were found to be linear with an average intercept of ca. 19.5, which based on Eq. 4.8 would yield a value of $X_{\text{H}_2\text{O}_2}$ ca. 0.38.

However, no clear correlations were found between I_d/I_r and $\omega^{-1/2}$ for $E_{\text{disk}} < 0.4$ V vs RHE affording strong evidence that Scheme I is not operative in this potential range. This may not be surprising, as it is precisely in this region where bromide desorbs and hydrogen adsorbs introducing complexities in the overall mechanism of the ORR.

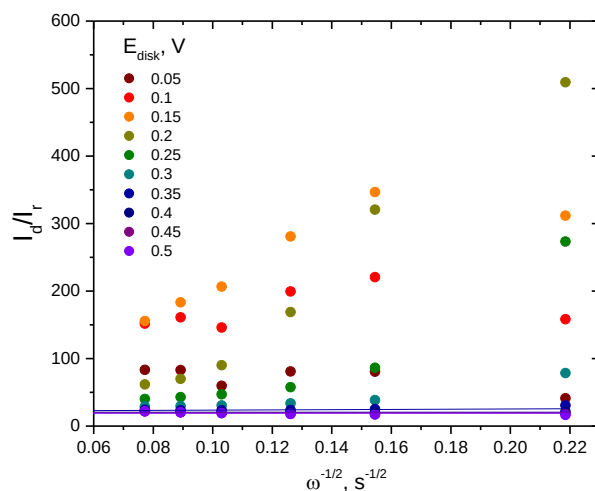


Figure 4.4. Plot of I_d/I_r vs. $\omega^{-1/2}$ based on the data in Panel C in Figure 4.2, for the specified values of E_{disk} , where the solid lines were obtained by a linear fit of the experimental data.

In conclusion, the classical mechanism proposed decades ago by Damjanovic *et al.* for the ORR predicts a linear relationship between the ratio of the disk and ring currents collected with a RRDE assembly, and the reciprocal of the square root of the rotation rate, ω . The formalism presented in this work makes it possible to extract from the intercept of these lines, the fraction of the disk current associated with the generation of $\text{H}_2\text{O}_2(\text{aq})$, a figure of merit defined as $X_{\text{H}_2\text{O}_2}$, which can then be used to determine the rate constant for the reduction of $\text{H}_2\text{O}_2(\text{aq})$ on the disk. On this basis, reliable values of $X_{\text{H}_2\text{O}_2}$ cannot be obtained by performing measurements at a single rotation rate, a common practice in research published in the literature.

**Chapter 5 A Rotating Ring-Disk Electrode Method for the Detection of
Solution Phase Superoxide as a Reaction Intermediate of Oxygen Reduction
in Neutral Aqueous Solutions**

Publication Details

The contents of this chapter are published, in large part, in *Analytical Chemistry*,³⁸.
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Contributors

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Abstract

A method is herein described that allows for solution phase superoxide generated via the reduction of dioxygen in neutral aqueous solutions at a rotating disk electrode to be oxidized at a concentric Au ring electrode bearing a covalently linked monolayer of 3-mercapto-1-propanol, a modified surface that blocks the oxidation of solution phase of hydrogen peroxide. Experiments in which the potential of a glassy carbon disk electrode was linearly scanned in the oxygen reduction region, and the ring voltage was poised at a value at which superoxide oxidation ensued yielded bell-shaped ring currents. This behavior is consistent with changes in the relative rates constant for the processes involved in the mechanism of oxygen reduction on this carbonaceous material induced by the applied potential which so far had remained undetected using other techniques. Also included is a mathematical model that allows for the concentration of $O_2^-(aq)$ next to the disk of a RRDE to be determined during the ORR based on the magnitude of the current collected with a Au ring specifically functionalized to detect exclusively

$O_2^-(aq)$, using data reported earlier in our laboratories for a glassy carbon disk electrode as a model system.

Introduction

Despite decades of research, certain aspects of the mechanism of dioxygen, O_2 , reduction on solid electrodes in aqueous electrolytes still remain poorly understood.³⁹⁻⁴² Particularly elusive has been the role of solution phase superoxide, $O_2^-(sol)$, a relatively short lived species, as a reaction intermediate. Considerable insight into this issue was gained by Yang and McCreery for carbon electrodes,⁴³ who found that chemically functionalization of the surface by a covalently bonded methylphenyl monolayer promotes O_2 reduction via a one-electron step to yield $O_2^-(sol)$ which undergoes facile electrochemical oxidation at potentials far negative to the onset of peroxide oxidation. More recently, Zhang et al. employed an ingenious arrangement involving a micropipette filled with an organic solvent immiscible with an external aqueous solution with its tip placed at a very short distance from a Pt microelectrode.⁴⁴ This tactic allowed for oxygen dissolved in the organic phase to diffuse into the aqueous phase reach the Pt electrode where it could be reduced to form $O_2^-(sol)$, and subsequently migrate back into the organic phase where it could be detected using scanning electrochemical microscopy. Although not pointed out in that publication, the organic phase employed displays finite solubility in water and, as shown in preliminary experiments performed in our laboratories, can adsorb on the Pt surface blocking hydrogen adsorption and reduce the activity of the metal toward oxygen reduction. This brief communication describes a method that makes it possible to detect $O_2^-(sol)$, on a general class of electrodes during the reduction of O_2 in neutral and mildly

alkaline electrolytes, $7.4 \leq \text{pH} \leq 10$ without introducing foreign species into the media. This strategy relies on the use of a gold ring of a rotating ring-disk electrode (RRDE) assembly modified by a covalently bonded layer of 3-mercaptopropyl-1-propanol (MP), a short chain alkyl thiol bearing an OH terminus. As shown by Chen et al. in phosphate buffer (PB) aqueous solutions of pH 7.4,⁴⁵ MP-modified Au electrodes promote the oxidation of $\text{O}_2^-(\text{sol})$, while blocking that of solution phase hydrogen peroxide, H_2O_2 , up to concentrations as high as 0.1 mM, when polarized at +0.2 V vs Ag/AgCl (1 M KCl).

Experimental Aspects

All measurements were performed at room temperature in a conventional three electrode glass cell using either a gold rotating disk electrode (RDE, Pine Instruments, area: 0.164 cm^2), Au ring- glassy carbon (GC) disk RRDE (Pine Instruments: AFE7R9GGCAU, disk area: 0.2475 cm^2 ; ring area 0.1866 cm^2 , gap $320 \text{ }\mu\text{m}$; collection efficiency, $N = 0.37$) as working electrodes. A Au wire and a Ag/AgCl in 3.5M KCl contained within a Pt cracked seal capillary served as counter and reference electrodes, respectively. Experiments were carried out in 0.1M PB of pH 7.4 and pH 10 prepared, respectively, by mixing Na_2HPO_4 (J.T. Baker Ultrapure Bioreagent 99.7%), and NaH_2PO_4 (J.T. Baker Ultrapure Bioreagent 99.9%), and only Na_2HPO_4 , and then adjusting the pH in each case with NaOH (Sigma Aldrich, semiconductor pure, 99.99%) using ultrapure water (UPW, $18.3 \text{ M}\Omega \text{ cm}$, EASYpure UV system, Barnstead). Oxygen (Airgas, 99.999% research grade), Ar (Airgas, PP300, 99.998%), as well as hydrogen peroxide (Fisher scientific, certified ACS, 31.5%) were used as received. Shown as inserts in Figure 5.1 are typical cyclic voltammetric curves recorded with the Au electrodes employed in this study in deaerated 0.1 PB solutions of pH 7.4 (left panel) and 10 (right panel) displaying characteristic

features associated with the clean surfaces. Such Au electrodes were chemically modified by immersing the RDE or RRDE in a solution 10 mM 3-mercaptopropyl-1-propanol (MP, Sigma Aldrich, 95%) in ethyl alcohol (AAPER, 200 proof) for 20 h, as described by Gobi et al.⁵ Electrode potentials were controlled with a commercial bipotentiostat (Pine Instrument Company, Model AFCBP1) and the data collected with an acquisition card (National Instruments, NI USB-6009) installed in a PC and programmed in Labview (2012).

Results and Discussion

Shown in thick lines in Figure 5.1 are dynamic polarization curves recorded with the Au RDE at a scan rate, $v = 10$ mV/s and a rotation rate, $\omega = 900$ rpm, in O_2 -saturated 0.1 M PB aqueous solutions, pH 7.4 (left panel) and pH 10 (right panel), before (blue) and after (black) adding 1mM H_2O_2 . The corresponding curves acquired following modification of the Au disk with a covalently bound MP layer under otherwise identical conditions are displayed in thin lines (X100) in these panels. As expected, the interfacial capacity of the bare Au (see grey line amplified by a factor of 100) decreases by about half an order of magnitude upon MP adsorption. Cursory inspection of these results clearly indicates that the MP-modified Au electrode blocks the reduction of O_2 and both the oxidation and reduction of H_2O_2 over the specified potential range.

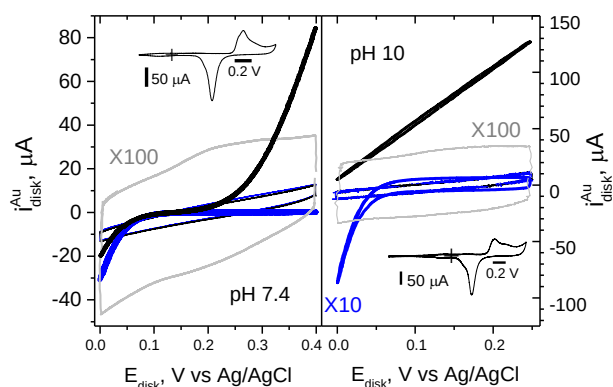


Figure 5.1. Cyclic voltammetric curves ($v = 10$ mV/s) recorded with the Au disk of a Au|Au RRDE at $\omega = 900$ rpm in 0.1 M PB pH 7.4 (Left Panel) and pH 10 (Right Panel) in O_2 -saturated 0.1 M PB before (thick blue lines) and after adding 1mM H_2O_2 . The corresponding data collected after modifying the Au with a covalently bound MP layer are shown in expanded form (X100) in thin lines in these panels. Also shown in grey lines are voltammetric curves recorded with the bare Au in the corresponding neat deaerated solutions in expanded form. Inserts: Cyclic voltammetric curves ($v = 100$ mV/s) recorded with the bare Au disk of the Au|Au RRDE in the same neat deaerated solutions, where the cross represents the (0,0) point.

Dynamic polarization curves were then recorded at $v = 10$ mV/s, for the glassy carbon (GC) disk electrode of the Au|GC RRDE in O_2 -saturated PB aqueous solutions of pH 7.4 and pH 10 at $\omega = 900$ rpm. These are shown in solid lines in the top left and right panels in Figure 5.2, respectively, where the arrows indicate the direction of the scan. Also displayed in this figure are the ring currents, i_{ring} , collected while scanning the disk with the ring polarized at various fixed potentials, E_{ring} , in the range 0 to 0.4 V vs Ag|AgCl, as specified in the legends, using either the bare Au ring (middle panels) or modified by a covalently linked MP monolayer. The solid symbols in these panels represent data obtained at fixed disk potentials, E_{disk} , by stepping E_{disk} from 0.0 V to -0.3 V and then in increments of 0.1 V in a staircase fashion down to -0.7 V, allowing about 25 s following each step for the current to achieve steady state. A number of interesting observations can be gleaned from these data. In particular, clearly defined ring currents, i_{ring} , were observed for E_{disk} in the range $-0.7 < E_{disk} < -0.1$ V (pH 7.4) and $-0.8 < E_{disk} <$

-0.2 V (pH 10) for E_{ring} in the range $0.1 < E_{\text{ring}} < 0.4$ V (pH 7.4) and $0.0 < E_{\text{ring}} < 0.25$ V (pH 10) both for the bare and MP-modified Au ring. The disk and ring currents became undetectable after deaerating the buffer solutions by bubbling Ar gas (see dotted lines in all panels, Figure 5.2) providing unambiguous evidence that the finite current observed are indeed derived from the presence of O_2 in the solutions. As has been discussed elsewhere,⁴⁶ i_{ring} for the bare Au ring (see middle panels) can be ascribed primarily to the oxidation of H_2O_2 generated at the GC disk approaching limiting values as E_{disk} reaches the negative end of the scan, which are far better defined for the solution of pH 10 than for that of pH 7.4.

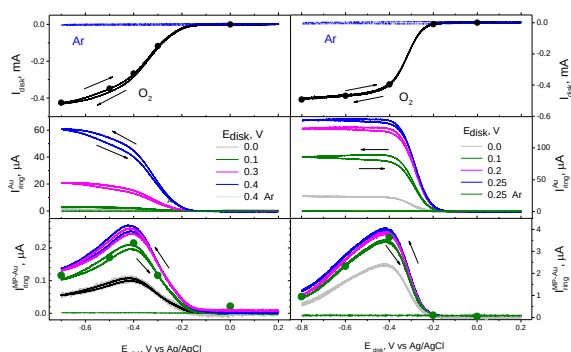


Figure 5.2. **Top Panels.** Dynamic polarization curves (solid lines) recorded at $v = 10$ mV/s, with the glassy carbon (GC) disk of a Au|GC RRDE in O_2 -saturated PB pH 7.4 (Left Panels) and pH 10 (Right Panels) rotating at $\omega = 900$ rpm. **Middle Panels.** Ring currents acquired with the bare Au ring electrode polarized at the specified potentials while scanning the disk. **Lower Panels.** Same as middle panels with the Au ring modified by a covalently linked MP monolayer. The dotted lines in all the panels were obtained in the same, thoroughly deaerated buffer solutions. The solid symbols represent the steady state currents measured at constant potential as specified (see text).

The results obtained in virtually identical experiments in which the Au ring was modified by a covalently linked MP monolayer also yielded an increase in i_{ring} at the onset potential for O_2 reduction on the disk. In marked contrast with the behavior found for the bare Au ring, however, i_{ring} in this case, displayed a bell shaped curve with a maximum at E_{disk} ca. -0.4 V independent of

the pH of the solution, for which the currents were ca. two orders of magnitude smaller than those found for the bare Au ring. In agreement with the data shown in Figure 5.1, the increase in i_{ring} for the MP-modified Au induced by adding H_2O_2 to the buffered solution to a concentration of 1 mM amounted to ca. 3 nA for pH 7.4 at 0.4 V and 10 nA for pH 10 at 0.25 V, i.e. less than an order of magnitude smaller than those found in lower panels, Figure 5.2. It may be concluded on this basis that the species being oxidized by the MP-modified Au ring is indeed $\text{O}_2^-(\text{sol})$ with a very minor contribution due to solution phase peroxide, which appears to reach a maximum at $E_{\text{disk}} = -0.4$ and thus consistent with its further reduction as E_{disk} becomes more negative. Although somewhat speculative, this might explain the continued increase in i_{disk} as the E_{disk} was polarized toward more negative values.

In another series of experiments, the GC disk was polarized at a potential close to the current plateau for O_2 reduction in both electrolytes, i.e. $E_{\text{disk}} = -0.4$ V (pH 7.4) and $E_{\text{disk}} = -0.41$ V (pH 10) and E_{ring} scanned at $v = 10$ mV/s over the ranges $-0.2 < E_{\text{ring}} < 0.4$ V (pH 7.4) and $-0.2 < E_{\text{ring}} < 0.25$ V (pH 10) with the electrode rotating at various values of ω . As shown in Figure 5.3, i_{ring} began to increase at ca. 0.0 V for pH 7.4 (see bottom panel) and ca. -0.2 V for pH 10 (upper panel) approaching for both solutions a plateau as E_{ring} was scanned toward more positive potentials. Also shown in these panels are data collected at $\omega = 900$ rpm in the same solutions following deaeration, which served as a basis for subtracting contributions to the polarization curves derived from the interfacial capacity. Attempts to extend the negative limit beyond -0.2 V to study the reduction of solution phase $\text{O}_2^-(\text{sol})$ were unsuccessful as the MP layer failed to fully block O_2 reduction in that potential region.

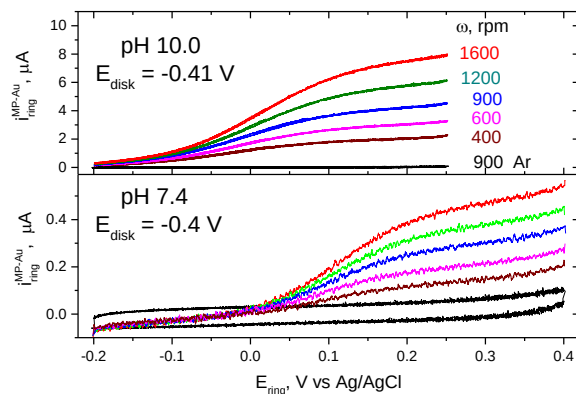


Figure 5.3. Polarization curve of the MP-modified Au ring of a Au|GC RRDE recorded in O_2 -saturated 0.1 PB at $v = 10$ mV/s while the GC disk was polarized at $E_{\text{disk}} = -0.4$ V in pH 7.4 (lower panel) and -0.41 V in pH 10 (upper panel) at various rotation rates as indicated. The curves in black were recorded in fully deaerated solutions.

Several factors complicate a quantitative analysis of the data shown in Figure 5.3. In particular, the $O_2^-(\text{sol})$ oxidation currents do not reach well-defined plateaus suggesting the involvement of contributions other than those derived from a simple diffusion limited process, such as the heterogeneous dismutation of $O_2^-(\text{sol})$ on the MP-modified Au surface. In fact, the corresponding process for the case of solution phase peroxide has been well documented in the literature for a number of surfaces. In addition, the mechanism of O_2 reduction on GC as well as the values of the rate constants of some of the elementary processes known to be involved have not been determined with certainty. Nevertheless, the strategy herein described for the specific detection of $O_2^-(\text{sol})$ affords a critical piece of information toward a better understanding of processes of relevance to both fundamental and applied research without the problems associated with the presence of foreign species in the media which could well be responsible for the generation of $O_2^-(\text{sol})$ detected recently by other workers.

Theoretical Aspects

The primary objective of this section is to determine quantitative correlations between the concentration of O_2^- in the immediate vicinity of the disk and the magnitude of the ring current measured at steady state with a RRDE as a function of pH. The problem so stated involves finding solutions to the time-independent convective diffusion equation for O_2^- (aq) in Eq. 5.1 below,

$$0 = D_{O_2^-} \left\{ \frac{\partial^2 [O_2^-]}{\partial z^2} + \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial [O_2^-]}{\partial r} \right) \right\} - u_r \frac{\partial [O_2^-]}{\partial r} - u_z \frac{\partial [O_2^-]}{\partial z} - 2R \quad [5.1]$$

where $[O_2^-]$ and $D_{O_2^-}$ represent the concentration and diffusion coefficient of O_2^- , respectively, u_r and u_z are the fluid velocities along the radial, r , and axial, z , coordinates, and R is the pH-dependent rate of the second-order dismutation of O_2^- given by⁷

$$R = k_{dis} [O_2^-]^2 = \left\{ \frac{7.61 \times 10^8 + 1.58 \times 10^{pH+6}}{(1 + 1.78 \times 10^{pH-5})^2} \right\} [O_2^-]^2 \quad [5.2]$$

where the rate constant k_{dis} is given in units of $\text{cm}^3 \text{mol}^{-1} \text{s}^{-1}$. Numerical solutions of Eq. 5.1, subject to the boundary conditions specified in Table 5.1 were obtained using COMSOL, assuming the system could be accurately represented by the finite 2-dimensional axisymmetric domain shown schematically in Figure 5.4, and explicit expressions for the fluid velocities as prescribed by the Levich formalism⁷, i.e.:

$$u_r = 0.51023 \cdot (\omega^3 / \nu)^{1/2} r z \quad [5.3]$$

$$u_z = -0.51023 \cdot (\omega^3 / \nu)^{1/2} z^2 \quad [5.4]$$

where ω is the rotation rate of the electrode in rad/s, and ν is the kinematic viscosity of the solution, i.e. ca. $0.01 \text{ cm}^2/\text{s}$.

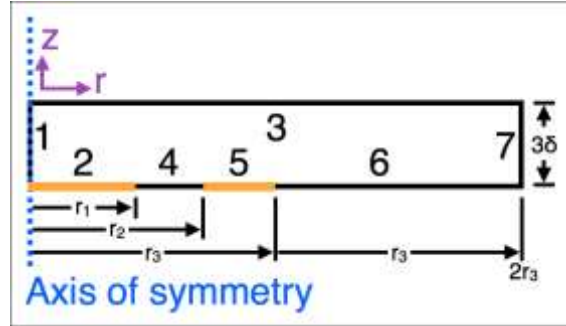


Figure 5.4. Schematic diagram of the two-dimensional domain employed in the simulations.

Table 5.1. Boundaries and Boundary Conditions

Boundary	Description	Boundary Condition
1	Center Axis	Axis of Symmetry
2	Disk Electrode	$[O_2^-] = [O_2^-]_{\text{disk}}$
3	Bulk	Inflow; $[O_2^-]^{\text{bulk}} = 0$
4	Insulator	No Flux
5	Ring	$[O_2^-] = 0$
6	Insulator	No Flux
7	Outflow	Outflow

Such calculations were performed for a series of rotation rates, in the range of 400 to 1600 rpm and for pH 7.4 and 10.0. As indicated in Figure 5.4, the height of the domain was assumed to be three times longer the thickness of the diffusion boundary layer, δ , given by

$$\delta = 1.612 D_{O_2}^{1/3} \nu^{1/6} \omega^{-1/2} \quad [5.5]$$

where $D_{O_2} = 6.7 \times 10^{-6} \text{ cm}^2/\text{s}$ ^{1,8}. For the simulations, the concentration of O_2^- immediately adjacent to the disk electrode (boundary 2), $[O_2^-]^{\text{disk}}$, was assumed to be fixed over a range of

values chosen such that the flux did not exceed the flux of oxygen to the surface given the

solubility of oxygen in the bulk solution of $[O_2]^{bulk} = 1 \text{ mM}$, i.e. $\left(\frac{1.3 \times 10^{-5} \text{ cm}^2 / \text{s}}{6.7 \times 10^{-6} \text{ cm}^2 / \text{s}} \right)^{2/3} [O_2]^{bulk}$,

and independent of the radial distance along the disk, r_1 . The simulated current for a single-electron process flowing across the ring (boundary 5) is given by:

$$I_{O_2}^{ring} = FD_{O_2} \int_5 \frac{\partial [O_2^-]}{\partial z} \Big|_{z=0} 2\pi r dr \quad [5.6]$$

where F is Faraday's constant, and boundary 5 represents the thickness of the ring electrode.

Figure 5.5 displays a plot of $[O_2^-]^{disk}$ vs $I_{O_2}^{ring}$ for various values of ω determined from the simulations, where the solid circles represent experimental values reported in our earlier publication involving a glassy carbon (GC) disk and a modified Au ring of a RRDE specific for the detection of solution phase O_2^- .³ As indicated, the magnitudes of $[O_2^-]^{disk}$ are indeed very reasonable and their dependence on pH in line with the relative rates of disproportionation reported in the literature.² Also interesting is the dependence of $[O_2^-]^{disk}$ on $\omega^{1/2}$ shown in Figure 5.6, which provide valuable information to elucidate the overall mechanism of the ORR in the disk.

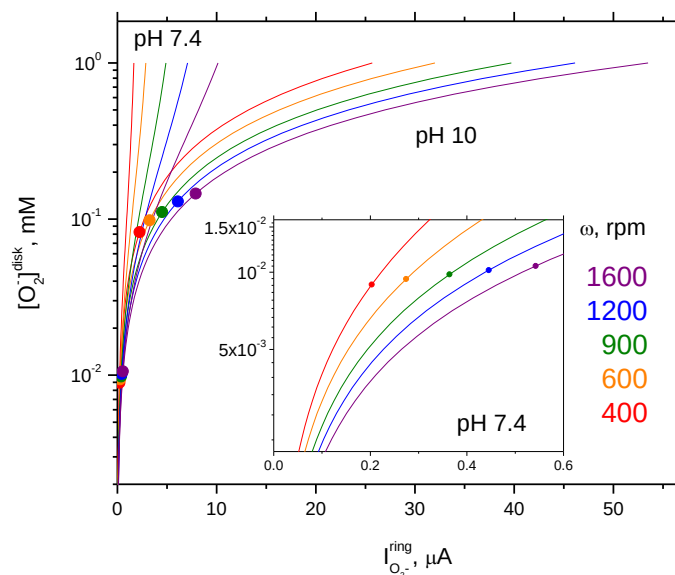


Figure 5.5. Plot of $[O_2^-]_{\text{disk}}$ vs $I_{O_2^-}^{\text{ring}}$ for pH 7.4 and 10 for various values of ω as indicated, where the solid circles represent experimental values reported in an earlier communication. Insert: Data for pH 7.4 in an expanded scale.

Unfortunately, the actual mechanism of the ORR on many surfaces including carbon is not known with certainty and the rates of the elementary processes involved have not as yet been determined with sufficient accuracy. This scarcity of key information makes it very difficult to gain additional insights regarding the ORR based strictly on the data analyzed in this work.

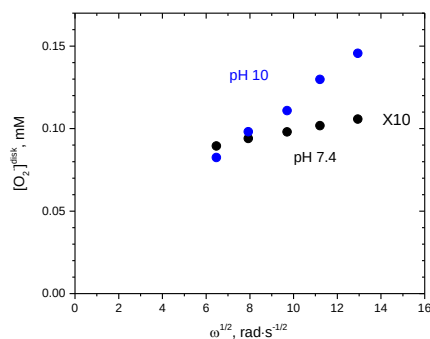


Figure 5.6. Plots of $[O_2^-]_{\text{disk}}$ vs $\omega^{1/2}$ determined from the ring currents, with the data at pH=7.4 expanded by 10x.

Chapter 6 Theoretical Analysis of Microelectrode Arrays Under Forced Convection

Abstract

Numerical solutions to the equation that governs steady-state mass transport to a hexagonal array of active small disks embedded in an otherwise inert rotating disk electrode under both diffusion limited and partial kinetic control were obtained using COMSOL Multiphysics. Analytical expressions were found which accurately reproduced the simulations yielding a limiting behavior in agreement with that predicted in the literature. Application of this formalism to dispersions of nanoparticles in inert supports at loadings within the range of relevance to technical gas-fed electrodes made it possible to demonstrate the validity of a modified Koutecky-Levich-like approach for determining rate constants of first order redox processes as proposed by Zoski et al. (*Journal of Electroanalytical Chemistry*, **651**, 80 (2011)).

Introduction

Technical electrodes for low-temperature fuel cells and a few other applications are composed of electrocatalytic particles of nm in size dispersed in an otherwise inert matrix, most commonly high-area carbon, a tactic that makes it possible to optimize utilization of often expensive and scarce materials⁴⁷. Over the last two to three decades, strategies have been developed and implemented to reduce and hopefully replace such materials by more abundant and cheaper elements. Some schemes involve the use of submonolayer amounts of noble metals adsorbed on suitable inert core substrates⁴⁸, whereas others seek to modify carbon with functional groups, including common transition metals, e.g. iron and cobalt, yielding discrete microscopic active sites, the structure of which continues to be the subject of heated debate⁴⁹. Among the most common techniques for assessing the potency of these types of electrocatalysts is the rotating disk electrode, RDE, whereby a very thin layer of the material is spread on the

surface of an inert disk made of an inactive substrate, such as glassy carbon, using a binder, commonly Nafion, to impart the layer mechanical integrity^{50, 51}. It has become common practice among researchers in the area to analyze the results obtained from such measurements by using the Koutecky-Levich, KL, formalism, i.e. plots of the reciprocal of the observed current flowing through the RDE vs the reciprocal of the square root of the rotation rates for various values of the applied potential⁵¹⁻⁵³. This approach often yields straight lines from which kinetic parameters can be extracted based on the values of the slope and intercepts provided the system fulfills the strict requirements upon which the aforementioned formalism is based. Considerable care must be exercised, however, to properly account for mass transport effects toward such an ensemble of individual active sites. In particular, as the density of nanoparticles or active sites is decreased, each site behaves as an independent ultramicroelectrode, a factor that is not considered in the KL formalism. Although significant progress has been made toward gaining a theoretical understanding of arrays of electrodes incorporating active materials in the form of spheres or disks either attached onto or embedded into otherwise inert flat substrates⁵⁴⁻⁵⁸, a strict and explicit analysis of data collected for these systems under forced convection has not, as of yet, been reported in the literature.

This contribution presents a rigorous analysis of data collected with a RDE incorporating arrays of disks embedded in a hexagonally close-packed arrangement on its surface under steady-state conditions based on numerical simulations using COMSOL. This arrangement affords an accurate representation of ensembles of spherical particles of the same diameter⁵⁵. Also to be discussed are analytical expressions which, provided certain conditions are met, closely reproduce the results of the simulations.

Theoretical Aspects

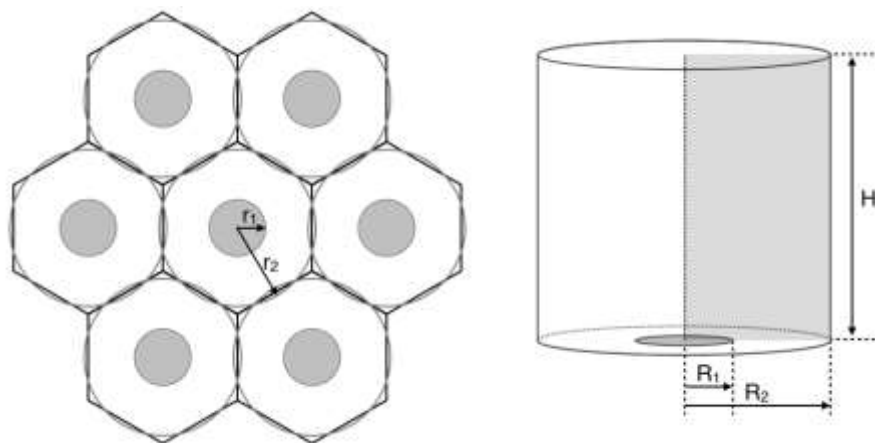


Figure 6.1. **Left Panel:** Top-down view of a section of the surface of a RDE consisting of a hexagonal, closed-packed arrangement of disk microelectrodes of radius r_1 embedded in concentric inactive disks of radius r_2 . **Right Panel:** Domain employed in the theoretical simulations (see shaded area), where R_1 and R_2 are the corresponding dimensionless radii and H the dimensionless height of an individual cylinder in the array (see Table 6.1).

Of concern here is to find solutions of the steady state convective-diffusion equation that governs mass transport for a rotating disk electrode (RDE) of radius r_e consisting of an array of a number, N_s , of hexagonally packed, electrochemically active disk electrodes of radius r_1 , individually surrounded by a concentric insulating ring, of radius, r_2 , as depicted schematically in the Left Panel, Figure 6.1. The approach to be employed regards the response of the system to be given by the sum of contributions due to identical individual disk-ring electrodes assuming the axial fluid velocity to be given by the leading term in the Cochran expansion for a RDE⁵⁹, and a total neglect of radial convection (see “Governing Equation”, Table 6.1). This tactic reduces the problem to solving the governing equation for a single cylinder or, equivalently, within the axisymmetric domain specified in the Right Panel, Figure 6.1, subject to the appropriate boundary conditions (see Table 6.1). The latter include no flux across the insulating segments, i.e. the curved wall and the ring, and constant concentration of reactant at the flat surface

opposite to that of the ring disk. In this analysis, the redox reaction responsible for the current was assumed to be first order in the reactant. Also defined in Table 6.1 are the dimensionless variables and dimensionless parameters expressed in terms of their dimensioned counterparts.

Table 6.1. Governing Equations and Boundary Conditions Both in Dimensioned and Dimensionless Forms (see text for details).

	Dimensioned	Dimensionless
Governing Equation	$D \left\{ \frac{\partial^2 c}{\partial z^2} + \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial c}{\partial r} \right) \right\} = -a \left(\frac{\omega^3}{\nu} \right)^{1/2} z^2 \frac{\partial c}{\partial z}$	$\frac{1}{Sc} \left\{ \frac{\partial^2 C}{\partial Z^2} + \frac{1}{R} \frac{d}{dR} \left(R \frac{\partial C}{\partial R} \right) \right\} = -a Z^2 \frac{\partial C}{\partial Z}$
Geometric Parameters	$r_2 = r_1 / \sqrt{1 - \theta}$ $h = 3\delta$ $\delta = 1.612(\omega^{-3} D^2 \nu)^{1/6}$	$R_2 = R_1 / \sqrt{1 - \theta}$ $H = 3\Delta$ $\Delta = 1.612 Sc^{-1/3}$
Kinetics	$\psi_n^t _{z=0} = k c_{z=0}$ $k_o = D / \delta$	$\Psi_n^t _{Z=0} = K C_{Z=0}$ $K_o = 1 / (Sc \cdot \Delta)$
Limiting Current	$c_{z=0, r \leq r_1} = 0$	$C_{Z=0, R \leq R_1} = 0$
Other Boundaries	$\left. \frac{\partial c}{\partial z} \right _{z=0, r > r_1} = \left. \frac{\partial c}{\partial r} \right _{r=0} = \left. \frac{\partial c}{\partial r} \right _{r=r_2} = 0$ $c_{z=h} = c_{bulk}$	$\left. \frac{\partial C}{\partial Z} \right _{Z=0, R > R_1} = \left. \frac{\partial C}{\partial R} \right _{R=0} = \left. \frac{\partial C}{\partial R} \right _{R=R_2} = 0$ $C_{Z=H} = 1$
Current	$\psi_n^t _{z=0} = D \left. \frac{\partial c}{\partial z} \right _{z=0, r \leq r_1}$ $I^{UME} = \frac{2nF(1-\theta)}{r_1^2} \int_0^{r_1} \psi_n^t _{z=0} r dr$	$\Psi_n^t _{Z=0} = \frac{1}{Sc} \left. \frac{\partial C}{\partial Z} \right _{Z=0, R \leq R_1}$ $\rho = \frac{2\Delta Sc(1-\theta)}{R_1^2} \int_0^{R_1} \Psi_n^t _{Z=0} R dR$
Definitions	$Z = (\omega/\nu)^{1/2} z, R_j = (\omega/\nu)^{1/2} r_j, \theta = (r_2^2 - r_1^2)/r_2^2, C = c / c_{bulk},$ $I^{array} = I^{UME} N_s, N_s = r_e^2 / r_2^2, K = k(\nu\omega)^{-1/2}, a = 0.51023, Sc = \nu / D$ $\rho = \frac{I^{UME} \cdot \delta(1-\theta)}{nF(r_1^2 \pi) c_{bulk} D}, \Psi_n^t _{Z=0} = \frac{\psi_n^t _{z=0} (\omega\nu)^{1/2}}{c_{bulk}}$	

Calculations were performed using COMSOL Multiphysics v5.2. by selecting the height of the domain in terms of dimensionless variables, H , (see right panel, Figure 6.1) to be three times the dimensionless thickness of the diffusion boundary layer, $\Delta = 1.612Sc^{-1/3}$, where $Sc = \nu / D$, the Schmidt number, ν is the kinematic viscosity of the solution and D , the diffusion coefficient of the redox active species. No changes in the results were observed by increasing H beyond this value. Furthermore, the dimensionless radius of the domain, R_2 , in terms of the corresponding radius of the microelectrode, R_1 , is given by $R_2 = R_1 / \sqrt{1-\theta}$, where $1 - \theta$ is the coverage of the active area, Lastly, the results will be expressed in terms of a dimensionless current, ρ , defined in Eq. 6.1 as the ratio of the actual current normalized by the limiting current for a fully active disk surface, as predicted by the Levich classical formalism, a quantity referred to in the literature as the attenuation^{58, 60}. As indicated in Eq. 6.1, the integral in the numerator is over the surface of the active disk, R_1 , whereas the normalizing factor, πR_2^2 , represents the dimensionless area of the disk-ring,

$$\rho = \left(\frac{1}{\pi R_2^2} \int_0^{R_1} 2\pi R \Psi_n^t \Big|_{Z=0} dR \right) / \left(\frac{1}{\Delta \cdot Sc} \right) \quad [6.1]$$

and $\Psi_n^t \Big|_{Z=0} = \frac{1}{Sc} \frac{\partial C}{\partial Z} \Big|_{Z=0, R \leq R_1}$ represents the total (convective and diffusive), normal outward flux

across the boundary at $Z = 0$, as implemented in COMSOL. The domain was meshed with a free-triangular mesh of size $M = (10/\Delta + 1/R_1)^{-1}$, with an electrode boundary mesh 100 times smaller than M , and a mesh surrounding $Z=0, R=R_1$ 200 times smaller than the mesh of the domain. Lastly, the Codina crosswind stabilization was used, because it correctly predicted the behavior of a macroscopic fully homogeneous electrode.

For the type of applications of interest herein, r_1 ranges from 2 to 40 nm, and the rotation rates ω can vary from 100 to 2500 rpm (the maximum ω achievable with a popular commercial RDE brand); hence, $2 \times 10^{-5} \leq R_1 \leq 5 \times 10^{-3}$. Furthermore, $\nu = 0.01 \text{ cm}^2/\text{s}$, and D , for common redox species, is in the range $10^{-6} < D < 5 \times 10^{-5} \text{ cm}^2/\text{s}$. On this basis, Δ would fall in the range 0.075 to 0.276. Without loss of generality, and unless otherwise noted, all the calculations presented herein will assume $D = 10^{-5} \text{ cm}^2/\text{s}$, i.e. $\Delta = 0.16$, and $Sc = 1000$, which represent values commonly found in actual experiments.

Results

A. Diffusion Limited Current for Arrays of Active Disk-Insulating Ring Electrodes.

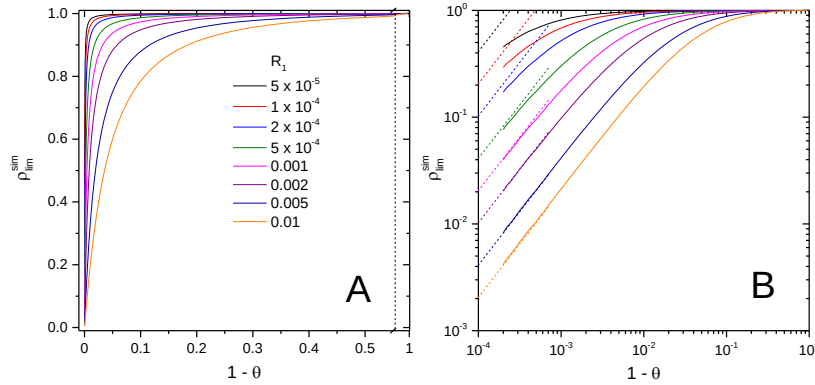


Figure 6.2. **Panel A.** Plots of $\rho_{\text{lim}}^{\text{sim}}$ vs $1 - \theta$ for various values of R_1 as indicated in the legend, obtained from the simulations, where the region corresponding to $1 - \theta < 0.55$ has been expanded for clarity. For $1 - \theta > 0.55$, $\rho_{\text{lim}}^{\text{sim}}$ does not vary significantly. **Panel B.** Same as Panel A, but in log-log space. Also included in this panel in dashed lines are values of $\rho_{\text{lim}, R_1 < \Delta, 1-\theta \rightarrow 0}$ determined from Eq. 6.2.

Shown in Panel A, Figure 6.2, are plots of $\rho_{\text{lim}}^{\text{sim}}$, the value of ρ under diffusion limited conditions, i.e. $c_{z=0, r \leq r_1} = 0$ (see Table 6.1), obtained from the simulations, as a function of $1 - \theta$, i.e. the coverage of the active area of the entire electrode. As indicated in the figure, in the limit

as $1 - \theta \rightarrow 1$, $\rho_{\text{lim}}^{\text{sim}} \rightarrow 1$, i.e. the current approaches that corresponding to a fully active surface, and, as $1 - \theta \rightarrow 0$, $\rho_{\text{lim}}^{\text{sim}} \rightarrow 0$, i.e. the electrode becomes fully blocked. A more detailed representation of these data for small values of $1 - \theta$ is given in log-log form in Panel B, Figure 6.2. In this region, the microelectrodes in the array become separated by distances large enough so that their diffusion boundary layers are not expected to overlap significantly, and r_1 (or R_1) is much smaller than the thickness of the diffusion boundary layer of the RDE, δ (or Δ). Under such conditions, each disk behaves as a regular microelectrode, μdisk , for which the steady state diffusion limited current is indeed well-known^{61, 62}, i.e. $I_{\text{ss}}^{\mu\text{disk}} = 4nFAc_{\text{bulk}}D / \pi r_o$, where r_o is the radius of the μdisk . On this basis, the diffusion limited current for the entire array can be expressed in terms of the dimensionless quantities defined herein as follows:

$$\rho_{\text{lim}, R_1 < \Delta, 1-\theta \rightarrow 0} = (1-\theta) \left(\frac{4\Delta}{\pi R_1} \right) \quad [6.2]$$

i.e. $\rho_{\text{lim}, R_1 < \Delta, 1-\theta \rightarrow 0}$ for a fixed value of Δ/R_1 , becomes proportional to $1 - \theta$, the total active area of the array, which correctly predicts the diminishing role of convection when the disks are small and well separated. In related work, Amatore *et al.*⁵⁷ derived a theoretical expression for the limiting current flowing through an active disk embedded in the flat end of a cylinder under quiescent conditions, assuming the flux of the reactive species was constant over the entire opposing wall. In fact, equating Eqs. (8a) and (9) in Amatore *et al.*⁵⁷ and substituting the result into Eq. (54) in Treimer *et al.*, yields exactly Eq. 6.2 above⁵⁶.

An empirical equation that fits the data across all values of $1 - \theta$ and R_1 may be shown to be given by:

$$\frac{1}{\rho_{\lim}} = \frac{1}{(1-\theta)\left(\frac{4\Delta}{R_1\pi} + 1\right)} + \frac{1}{\frac{R_1\pi}{4\Delta} \left\{ \lambda \left(\frac{1}{\sqrt{1-\theta}} - 1 \right) + 1 \right\} + 1} \quad [6.3]$$

where for Sc ranging from 800 to 1400, $\lambda \approx 1.6(9R_1 + 1)$.

Shown in Figure 6.3 are plots of the $\% \text{ error} = 100 \times (\rho_{\lim}^{Eq.3} - \rho_{\lim}^{sim}) / \rho_{\lim}^{sim}$ as a function of $1 - \theta$ for various values of R_1 (see legend), where the symbol $\rho_{\lim}^{Eq.3}$ represents the values of $\rho_{\lim}$ as determined from Eq. 6.3. As clearly evidenced from these data, Eq. 6.3 reproduces within less than 1% the results obtained from our simulations.

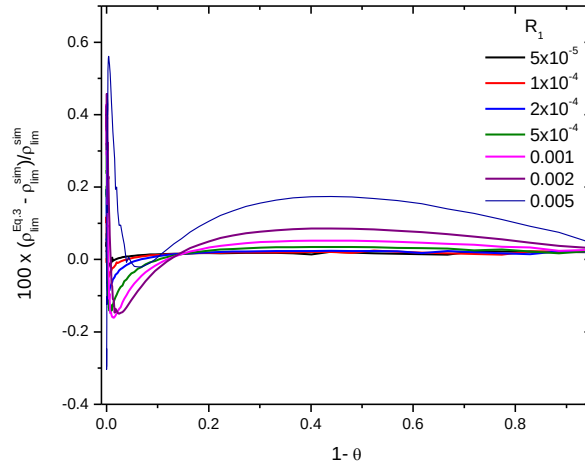


Figure 6.3. Plots of $\% \text{ error} = 100 \times (\rho_{\lim}^{Eq.3} - \rho_{\lim}^{sim}) / \rho_{\lim}^{sim}$ vs $1 - \theta$, for various values of R_1 , where the symbol $\rho_{\lim}^{Eq.3}$ represents the values of $\rho_{\lim}$ as determined from Eq. 6.3.

B. Active Disk Electrode Below Limiting Current

Additional calculations were performed, wherein the total normal flux of reactant to the disk electrode was assumed to be controlled by the kinetics of a first-order reaction in the

reactant, i.e. $\Psi_n^t = KC_{Z=0}$, for a range of dimensionless rate constants, $K = k(\nu\omega)^{-1/2}$, where k is

the dimensioned potential-dependent rate constant of a redox process. The results obtained could be represented to a very good degree of accuracy by the following analytical expression:

$$\frac{1}{\rho_K^{sim}} = \frac{1}{(1-\theta)K / K_o} + \frac{1}{\rho_{lim}^{Eq.3}} \quad [6.4]$$

In fact, for $0.01 < 1 - \theta < 0.99$, $0.0005 < R_1 < 0.002$, and $10^{-4} < K/K_o < 10^5$, where K_o is defined as the rate of mass transport to a fully-active electrode surface under diffusion limited conditions (see Table I), the agreement was better than 2% at a value of $R_1=0.002$ (see Figure 6.4), and even smaller for decreasing values of R_1 (data not shown).

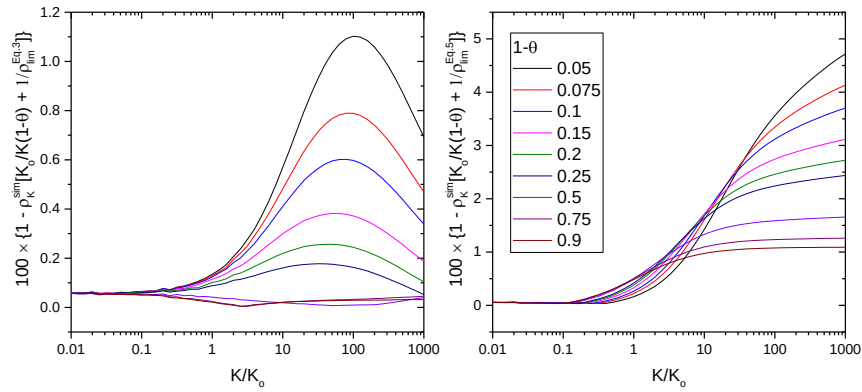


Figure 6.4. Plots of the absolute value of the % error = $100 \left\{ 1 - \rho_K^{sim} \left(\frac{K_o / K}{(1-\theta)} + \frac{1}{\rho_{lim}^{Eq. x}} \right) \right\}$ at $R_1 = 0.002$, as a function of K/K_o for values of $1 - \theta$ shown in the legend, where $\rho_{lim}^{Eq. x}$ represents the dimensionless limiting current specified for $x = 3$ and (Panel A) or $x = 5$ (Panel B).

For very small values of K/K_o , the first term in the right hand side of Eq. 6.4 will be larger than the second term, and, as expected, the system will be under kinetic control. Furthermore, for R_1 much smaller than the thickness of the diffusion layer of the RDE (as would be the case within the range of parameters of interest herein), $4\Delta / \pi R_1 \gg 1$, and, therefore, Eq. 6.3 simplifies to

$$\frac{1}{\rho_{\text{lim}, R_1 \ll \Delta}} = \frac{\pi R_1}{4\Delta(1-\theta)} + 1 \quad [6.5]$$

This expression can then be substituted into Eq. 6.4 to yield

$$\frac{1}{\rho_K} = \frac{K_o}{(1-\theta)K} + \frac{\pi R_1}{4\Delta(1-\theta)} + 1 \quad [6.6]$$

which, as shown in Figure 6.4, reproduces the numerical simulations to within 5% even up to a value of R_1 as large as 0.002 and a $1-\theta$ as low as 0.05. Rather unexpectedly, Eq. 6.6 is identical to that proposed by Zoski et al.⁵⁴ (cf. Eq. 10 in this reference) once the dimensionless quantities are expressed in terms of their dimensioned counterparts, i.e.

$$\frac{1}{I^{\text{array}}} = \frac{1}{nFA_{\text{disk}}(1-\theta)c_{\text{bulk}}} \left\{ \frac{\pi r_1}{4D} + \frac{\delta}{D}(1-\theta) + \frac{1}{k} \right\} \quad [6.7]$$

where I^{array} is the current of the whole array and A_{disk} is the cross-sectional area of the entire disk electrode. As noted by these authors, the first two terms in parenthesis are attributed to mass transport to the individual nanoparticles under quiescent conditions and to convection, respectively, and the last term accounts for the kinetics. Moreover, in the absence of convection, Eq. 6.7 reduces to that proposed recently by Kim and Bard⁶³ (See Eqs. 2 and 3 in that reference) in their analysis of Koutecky-Levich plots for microelectrodes. On the other hand, when the transport due to convection becomes much larger than diffusion toward the small particles, or equivalently, $\frac{4\delta}{\pi r_1}(1-\theta) \gg 1$, Eq. 6.7 yields:

$$\frac{1}{I^{array}} = \frac{1}{nFA_{disk}c_{bulk}} \left\{ \frac{\delta}{D} + \frac{1}{k(1-\theta)} \right\} \quad [6.8]$$

Eq. 8 is a Koutecky-Levich expression, where the kinetic term becomes proportional to the active area of the electrode, $1-\theta$, and the convective contribution to the cross-sectional area of the electrode, A_{disk} . Since the range of rotation rates of RDE measurements involving dispersed electrocatalysts rarely exceeds 2500 rpm and $r_1 < 30$ nm, even for a 1% coverage, the first term in parenthesis in the right hand of Eq. 6.7 would represent a contribution of 3% of the mass transport limitation. In fact, for four surveyed papers^{50, 64-66} the ratio $4\delta(1-\theta)/\pi r_1$ was on the order of 10^3 . A similar criterion has been suggested by Masa et al.⁵³ and Davies et al.⁶⁷ who use the relationship between the diffusion boundary layer thickness and the separation between the particles, i.e. $\delta/2r_2$, or equivalently, $\delta\sqrt{1-\theta}/2r_1 \gg 1$. Hence, contributions to mass transport to the nanoparticles other than those derived from convection are indeed negligible and Eq. 6.8 can be used with confidence.

Concluding Remarks

Analytical expressions have been herein proposed that reproduce rather accurately mass transport to an array of hexagonally-packed active disks embedded in an otherwise inert rotating disk electrode. Conditions were found under which one such expression reduces to that proposed earlier by Zoski et al., for a microelectrode array under linear convection. Moreover, in the absence of convection, this expression reduces to a Koutecky-Levich type equation recently proposed by Bard et al. for a first-order redox reaction on a microelectrode under stagnant

conditions. It followed from the analysis presented herein that for particles of nm size and loadings most commonly used in the literature, a Koutecky-Levich analysis is indeed warranted.

**Chapter 7 Oxygen and Hydrogen Peroxide Reduction on Polycrystalline
Platinum in Acid Electrolytes: Effects of Bromide Adsorption**

Publication Details

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Abstract

Changes in the electrocatalytic activity of Pt for the oxygen and hydrogen peroxide reduction reactions (ORR, and HPRR, respectively) in an aqueous acidic electrolyte induced by the adsorption of bromide, as a model impurity, have been investigated using chronoamperometric techniques under forced convection. Experiments were carried out using a polycrystalline Pt|Pt rotating ring-disk electrode in O₂-saturated 0.1 M HClO₄ containing 10 μM KBr (2.2 ppm). Potential steps were applied to the Pt disk from E_o, at which Br⁻ is fully desorbed, to more positive values, E_{step}, at which Br⁻ undergoes adsorption. During this period, the currents both at the disk, i_{disk}, and at the ring, i_{ring}, were monitored, with the ring polarized at a potential at which H₂O₂(aq) oxidation proceeds under diffusion-limited conditions. The results obtained, assuming the ORR does not interfere with Br⁻ adsorption, made it possible to correlate i_{disk} and i_{ring} with the coverage of adsorbed Br⁻, θ_{Br⁻}, and E_{step}. As evidenced from the data collected, significant drops in the diffusion-limited currents for O₂ reduction induced by the

presence of 10 μM KBr could only be observed for $\theta_{\text{Br}^-} > 0.25$ regardless of E_{step} . Similar measurements involving the same Pt(poly) rotating disk electrode performed in 1 mM H_2O_2 in deaerated 0.1 M HClO_4 devoid of O_2 displayed a similar trend. This behavior was found to be consistent with a simple blockage of surface active sites by adsorbed bromide, as predicted by the attenuation model proposed by Levart.

Introduction

The influence of adsorbed species on the kinetics and mechanism of the oxygen reduction reaction, ORR, in aqueous electrolytes has been recognized for decades.¹ Considerable insight into the factors that govern these effects has been gained from studies performed with single crystal Pt electrodes.²⁻⁹ In particular, the ORR on Pt(111) in aqueous perchloric acid has been found to generate solution phase hydrogen peroxide, $\text{H}_2\text{O}_2(\text{aq})$, over the potential region where hydrogen adsorbs, to yield, upon full hydrogen desorption, water as the only detectable product.²⁻⁵ A similar phenomenon has been reported upon adsorption of halides on Pt in acidic electrolytes regardless of their surface microtopography.⁶⁻⁹ This is illustrated by dynamic polarization data in Figure 7.1, recorded with a polycrystalline Pt, Pt(poly), disk, of a Pt(poly)|Pt(poly) rotating ring|disk electrode, RRDE, in O_2 -saturated 0.1 M HClO_4 before (black) and after addition of either 10 μM NaCl (blue) or 10 μM KBr (red) (bottom panel). Also displayed in the top panel in this figure are ring currents, i_{ring} , measured simultaneously with the ring polarized at potentials, E_{ring} , positive enough for the oxidation of $\text{H}_2\text{O}_2(\text{aq})$ to proceed under strict diffusion control. As clearly evidenced by these results, the presence of the halides is responsible for two prominent effects: a shift in the onset potential for the ORR toward more negative values, and the

generation of significant amounts of $\text{H}_2\text{O}_2(\text{aq})$, the extent of which being far more pronounced for the bromide-, compared to chloride-containing solutions.

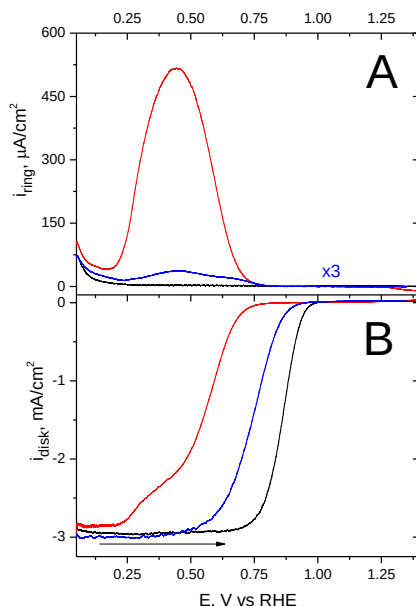


Figure 7.1. Dynamic polarization curves (Panel B) recorded at a positive-going scan rate $v = 10 \text{ mV/s}$ (as indicated by the arrow in the lower left) with the Pt disk of the Pt|Pt RRDE, in O_2 -saturated 0.1 M HClO_4 at $\omega = 400 \text{ rpm}$, before (black) and after addition of either $10 \text{ }\mu\text{M NaCl}$ (blue) or $10 \text{ }\mu\text{M KBr}$ (red). The thin lines (Panel A) are the corresponding ring currents, i_{ring} , measured simultaneously with the ring polarized at $E_{\text{ring}} = 1.2 \text{ V}$ for the neat, and chloride-containing, and 1.4 V for bromide-containing solutions (see text for additional details).

This contribution examines various aspects of the ORR and hydrogen peroxide reduction reaction, HPRR, on Pt(poly) electrodes in 0.1 M HClO_4 solutions containing $10 \text{ }\mu\text{M KBr}$, as a function of the bromide coverage, θ_{Br^-} , and the applied potential, E_{step} . Two factors make this electrolyte ideally suited for investigating the role of a simple adsorbate on the rates and mechanism of this important redox reaction. First, perchlorate displays no affinity for Pt surfaces¹⁰ and, second, the adsorption characteristics of bromide on this metal are well understood.^{6,7} The unique chronoamperometric strategy employed in these studies involves

application of a potential step to the disk of a RRDE from a value negative enough to induce the full desorption of the halide, E_o , to a potential E_{step} , at which its adsorption is expected to ensue. The coverage of bromide, θ_{Br^-} , on the Pt disk was determined quantitatively by measuring the shielded i_{ring} using the method reported originally by Markovic et al.^{6,69} This information made it possible to correlate the disk current, i_{disk} , and i_{ring} , associated, respectively, with the ORR, and the oxidation of $H_2O_2(aq)$ under diffusion-limited conditions, with θ_{Br^-} in O_2 -saturated solutions. Also to be presented in this work are the results of analogous experiments involving deaerated $H_2O_2(aq)$ solutions in 0.1 M $HClO_4$, from which insight was gained into mechanistic changes induced by the adsorption of bromide on the ORR. Evidence that the halide adsorption process is not affected by the HPRR was obtained by monitoring θ_{Br^-} using normalized differential reflectance, $\Delta R/R$,¹¹ while acquiring chronoamperometric data. In addition, theoretical simulations were performed aimed at providing a more quantitative account of the decrease in the electrocatalytic activity for the HPRR derived from the blockage of Pt sites by the halide.

Experimental

All measurements were performed in an all-glass, three-compartment cell, filled with Ar-purged, (UHP, 99.999%, Airgas) or O_2 -saturated (Air Gas) 0.1 M $HClO_4$ solutions (75 mL) prepared from perchloric acid (Ultrapure, EMD), and ultrapure water generated by a Barnstead system (18.3 M Ω cm, EASYpureUV), either neat, or following addition of 10 μ M KBr (Fisher Scientific, IR grade, > 99%) i.e. 800 ppb. Experiments were carried out using a Pt(poly)|Pt(poly) RRDE (Pine Instruments, disk diameter: 0.457 cm; ring inner diameter 0.493 cm; gap, $G = 180$ μ m and ring outer diameter: 0.538 cm, or, correspondingly, disk and ring areas of 0.164 cm² and 0.037 cm², respectively, yielding a theoretical collection efficiency, $N = 0.22$) cleaned via

polishing with a series of alumina (Al_2O_3) powders, followed by rinsing with ultra-pure water, and finally cycling between the onsets of hydrogen and oxygen evolution until the voltammogram displayed features characteristic of those attributed to a clean polycrystalline platinum surface, and a carbon rod and a reversible hydrogen electrode (RHE) as counter and reference electrodes, respectively. The disk and ring potentials, E_{disk} and E_{ring} , respectively, were controlled with a bipotentiostat (Pine Instruments, Model AFCBP1) and the rotation rate of the RRDE, ω , was adjusted with a commercial rotator (Pine Instruments, Model AFMSRX). All the data were recorded using a personal computer equipped with a National Instruments acquisition card (USB-6009) programmed in Labview.

Similar experiments involving H_2O_2 (ULTREX, Ultrapure reagent, JT Baker) were carried out in Ar-purged 0.1 M HClO_4 solutions either neat, or containing 10 μM KBr. Measurements aimed at monitoring the magnitude of the normalized optical differential reflectance, $\Delta R/R = [R(E_{\text{ref}}) - R(E_{\text{step}})]/R(E_{\text{ref}})$, where $R(E_{\text{ref}})$ and $R(E_{\text{step}})$ represent the intensity of the light collected at the detector following reflection from the rotating Pt disk electrode polarized at a reference, E_{ref} , and an arbitrary sampling potential, E_{step} , were performed using a blue diode laser (Polaris 700, $\lambda = 447$ nm, 900 mW, Laserglow), while collecting chronoamperometric data. Details regarding the instrumentation and methods involved in these measurements can be found in the original literature¹². Theoretical calculations were performed within the COMSOL platform (version 4.3a) running on a desktop computer.

Results and Discussion

Cyclic voltammetric curves recorded at a scan rate, $v = 20$ mV/s, with the Pt(poly) disk of the Pt|Pt RRDE in Ar-purged 0.1 M HClO_4 (see Figure 7.2) before (black) and after addition of

20 μM KBr (blue), displayed characteristic features associated with these solutions. In particular, the shift in the onset of oxide formation toward positive potentials, as well as the decrease in the area under the oxide reduction peak centered at about 0.75 V to 0.8 V for the halide-containing media compared to the neat electrolyte, are consistent with the presence of adsorbed bromide blocking sites on the Pt surface. Evidence that the large increase in the current found for $E_{\text{disk}} > 1.4$ V in the scan toward positive potentials is due in part to the oxidation of bromide is provided by the presence of a small peak pointing in the negative direction centered at ca. 1.3 V (blue curve), ascribed to the reduction of bromine.

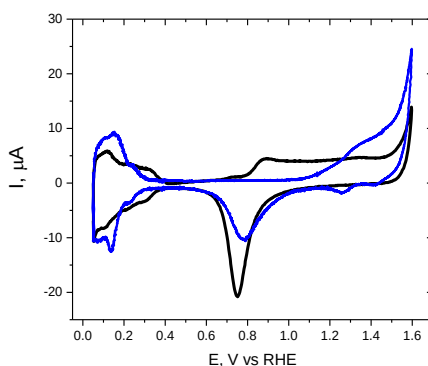


Figure 7.2. Cyclic voltammograms recorded at $v = 20$ mV/s with the Pt disk of the Pt|Pt RRDE in Ar-purged 0.1 M HClO_4 before (black) and after addition of 20 μM KBr (2.2 ppm, blue).

Bromide Adsorption under Forced Convection

Shown in Panel A, Figure 7.3 (see jagged curves) are a series of chronoamperometric curves recorded in 10 μM KBr in 0.1 M HClO_4 at $\omega = 400$ rpm, for potential steps from $E_o = 0.03$ V to different values of E_{step} . As clearly indicated, the curves displayed, upon application of the step, a response characteristic of an RC circuit, i.e. an almost instantaneous rise in the current followed by an exponential decay, where in this case, R is attributed to the electrolyte resistance, and C to the (pseudo) capacitance associated with the desorption of adsorbed hydrogen and the

adsorption of bromide. At longer times, i_{disk} either continued to decrease monotonically, or turned upwards, reaching a maximum to finally drop down to negligible values. As shown in the figure, the total time elapsed for i_{disk} to vanish was longer as E_{step} increased, consistent with an expected increase in the bromide coverage, θ_{Br^-} , with potential. This overall behavior resembles in many regards that reported earlier for the same type of experiments involving Pt(111)⁶ and Pt(100)⁶⁹ in 80 μM KBr in 0.1 M HClO_4 , which the authors explained in terms of the displacement of adsorbed hydrogen (for low E_{step}) or hydroxyl (for high E_{step}) by the halide.

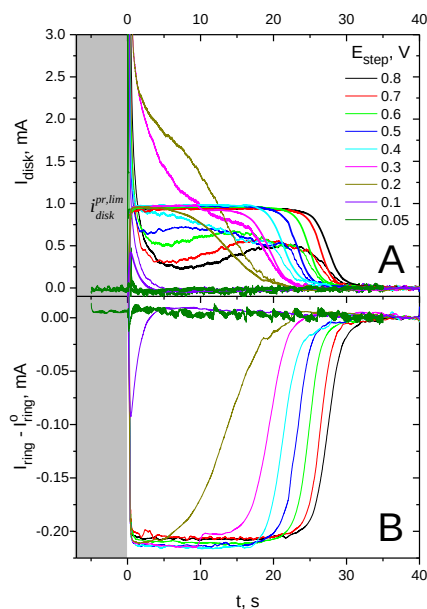


Figure 7.3. Panel A. Chronoamperometric curves ($E_o = 0.03$ V to E_{step} , as indicated) recorded with the Pt disk of a Pt|Pt RRDE in 10 μM KBr in 0.1 M HClO_4 at $\omega = 400$ rpm. Panel B. $I_{\text{ring}} - I_{\text{ring}}^o$ vs t recorded simultaneously with the ring polarized at $E_{\text{ring}} = 1.4$ V vs RHE. The smooth solid lines in Panel A were calculated based on the shielded ring currents in Panel B.

Yet another factor that may contribute to the “dip” found in the chronoamperograms in the region $0.6 \leq E_{\text{step}} \leq 0.8$ V (see Panel A, Figure 7.3), is a “shorting” of the double layer induced by bromide adsorption. More specifically, this voltage range is positive to the potential

of zero charge (pzc) of Pt.^{13,14} Hence, immediately following the step, adsorbed hydrogen will oxidatively desorb, and the surface will acquire a net positive charge, which will be at least partially neutralized by the adsorption of bromide, i.e.



This process would involve no net current flowing through the external circuit, even though bromide adsorption may continue taking place for some time. However, as the surface charge is depleted the adsorption will proceed predominantly through an oxidative process and the current will then increase (see e.g. the curves for $E_{\text{step}} \geq 0.6$ V in Panel A, Figure 7.3). Assuming the electrosorption valency of bromide is unity, the dominant process on the disk generating current can be attributed to:



in agreement with data reported elsewhere^{6,7}.

Markovic et al.⁶ developed a more elegant and indeed highly quantitative method to determine the time evolution of θ_{Br^-} . Their strategy relies on the use of the ring of a RRDE polarized at $E_{\text{ring}} = 1.08$ V vs SCE, i.e. positive enough to oxidize Br^- under strict diffusion control:



For conditions under which the disk is inactive, I_{ring} will be given by the classical Levich equation,¹⁵ i.e.

$$I_{ring}^o = 0.62nF \left[\pi (r_3^3 - r_2^3)^{2/3} \right] c_{bulk} D^{2/3} \omega^{1/2} \nu^{-1/6} \quad [7.4]$$

also known in the context of this work as the *unshielded* ring current, I_{ring}^o . However, if Br^- adsorbs on the disk electrode, I_{ring} will decrease by a corresponding amount, i.e.

$$I_{ring} - I_{ring}^o = -N \times I_{disk}^{pr} \quad [7.5]$$

from which the adsorption rate at the disk, expressed as a current, denoted as I_{disk}^{pr} , can be predicted, provided the electrosorption of the halide is unity, $n = 1$. Shown in Panel B, Figure 7.3, is a plot $I_{ring} - I_{ring}^o$ vs t recorded during the acquisition of I_{disk} in Panel A. As indicated therein, for $E_{disk} = 0.03$ V, i.e. before the potential step is applied (see shaded region), no adsorption of Br^- is expected to occur and, therefore, $I_{ring} = I_{ring}^o$. In stark contrast, for data collected for $E_{step} \geq 0.2$ V, i.e. positive enough for the adsorption to attain diffusion-limited values, I_{ring} will reach a plateau lasting from 5 s for $E_{step} = 0.2$ V, to 25 s for $E_{step} = 0.8$ V. It thus follows from Eq. (5), that the flux of bromide to the disk, and, thereby θ_{Br^-} (vide infra), can be calculated, even when the actual process might produce a non-stoichiometric flow of current through the external circuit, such as, the displacement reactions proposed by Markovic et al.,^{6,69} or the shorting of the double layer (see Eq. 7.1 above). The results of such an analysis are shown in solid lines in Panel A, Figure 7.3, where the symbol $I_{disk}^{pr.lim}$ refers to the limiting current for the adsorption of bromide on the disk predicted from I_{ring} .

The I_{disk}^{pr} vs time curves, shown in smooth solid lines in Panel A, Figure 7.3 enables the “equivalent” charge associated with Br^- adsorption on the disk, Q_{Br^-} , to be determined as a function of time via the equation:

$$Q_{\text{Br}^-} = \int_0^t I_{disk}^{pr} d\tau \quad [7.6]$$

yielding plots such as those shown in Figure 7.4 (left ordinate) for various E_{step} . These values can be converted into transient coverages, by dividing Q_{Br^-} by the charge associated with hydrogen adsorption for the same Pt electrode in a solution devoid of Br^- (see right ordinate, Figure 7.1). Based on the data in Figure 7.2, the latter was found to be ca. $360 \mu\text{C}/\text{cm}^2$, which is consistent with a roughness factor of ca. 1.5. Moreover, the steady state values of θ_{Br^-} make it possible to construct the corresponding adsorption isotherm, as depicted in Figure 7.5.

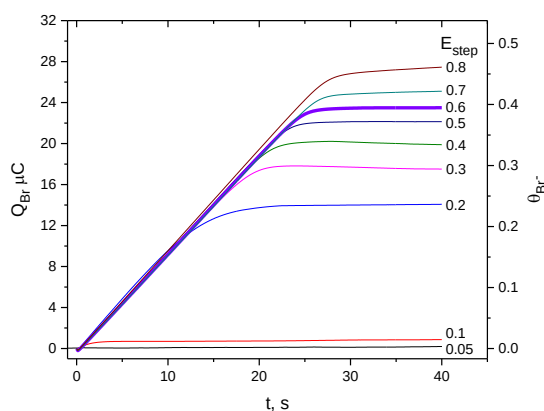


Figure 7.4. Plots of Q_{Br} (left ordinate) and θ_{Br^-} (right ordinate) vs time based on the analysis of I_{disk}^{pr} vs t recorded at $\omega = 400$ rpm (see Panel A, Figure 7.3) for different E_{step} , as indicated.

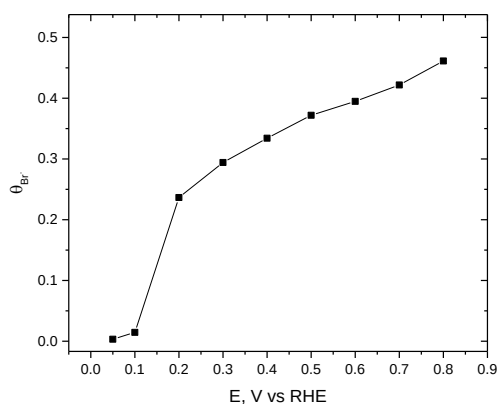


Figure 7.5. Potential-dependent isotherm for bromide adsorption on Pt(poly) for 10 μM KBr in a 0.1 M HClO_4 solution, at $\omega = 400$ rpm based on the data shown in Figure 7.4 (see text for details).

Oxygen Reduction

As clearly illustrated by the polarization curves in Figure 7.1, halide adsorption depresses the electrocatalytic activity for the ORR of the otherwise bare Pt surface, and modifies markedly its overall mechanism. Insight into the factors associated with these effects was obtained from the analysis of chronoamperometric data. For these experiments, the disk potential was stepped from $E_o = 0.03$ V vs RHE, to various values of E_{step} , with the RRDE rotating at $\omega = 400$ rpm, while the ring was polarized at either $E_{\text{ring}} = 1.2$ V or 1.4 V, as specified in the figure, to detect $\text{H}_2\text{O}_2(\text{aq})$ generated at the disk. Data collected in 0.1 M HClO_4 immediately following the step, both in the absence and in the presence of 10 μM KBr, shown in the left and right panels in Figure 7.6, respectively, displayed a contrasting behavior. Unlike the chronoamperograms recorded in the neat O_2 -saturated electrolyte (see left panels), which, except for $E_{\text{step}} = 0.65$ V, yielded both constant ring and disk currents immediately following the potential step, both the disk and ring currents were found to change in response to the gradual increase in the amount of adsorbed bromide (see right panels).

Whereas the onset for the ring current increase was a monotonic function of the applied potential, from close to zero for 0.2 V, up to 15 s for 0.65 V, the disk currents were found to hardly change up to about 15 s for all potentials. The limiting disk currents observed at longer times decreased monotonically with potential; whereas the ring currents reached a maximum at 0.4 V and then decreased thereafter. These trends are in line with those found in the polarization curves in Figure 7.1.

Assuming that the ORR does not interfere with bromide adsorption, as was found to be the case using differential reflectance spectroscopy for chloride-containing solution in virtually identical experiments, this strategy makes it possible to correlate i_{disk} , i_{ring} , E_{step} , with θ_{Br^-} .

Careful inspection of the curves in Figure 7.6 revealed a number of interesting features. In particular, following application of the step, i_{disk} achieved diffusion-limited steady state values for E_{step} within the range selected in neat O₂-saturated 0.1 M HClO₄ (Panel A). The same exact values were also found in the bromide-containing media (Panel B) for times shorter than ca. 10 s; however, at longer times, i_{disk} decreased achieving steady state values, the magnitude of which became smaller as E_{step} increased. This latter trend is in line with that found in the polarization curves in Figure 7.1, and may be ascribed to the progressive blockage of surface sites by the adsorbed halide. Not shown in the figure are data collected for $E_{\text{step}} < 0.2$ V for which i_{disk} was hardly affected by the presence of bromide in solution in agreement with the adsorption isotherm in Figure 7.5.

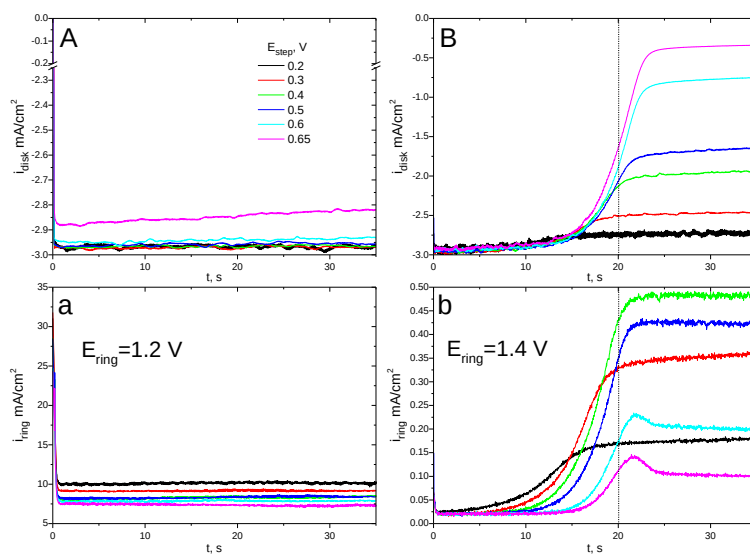


Figure 7.6. Chronoamperometric curves recorded with the Pt disk of a Pt|Pt RRDE at $\omega = 400$ rpm in O_2 -saturated 0.1 M $HClO_4$ in the absence (Panel A) and in the presence of 10 μM KBr (Panel B) in solution. Potential steps were applied from $E_0 = 0.03$ V vs RHE to various values of E_{step} as indicated in the legend. The bottom panels, i.e. a and b display the corresponding i_{ring} for $E_{ring} = 1.2$ V for the neat and $E_{ring} = 1.4$ V for the Br^- -containing solution.

The ring chronoamperograms recorded for $E_{ring} = 1.2$ V in the neat O_2 -saturated electrolyte (see Panel a, Figure 7.6) were found to yield very small and constant i_{ring} , ca. 20 $\mu A/cm^2$, and virtually independent of E_{step} over the range of values examined. Corresponding data collected in the bromide-containing solution for $E_{ring} = 1.4$ V displayed a very different behavior (see Panel b, Figure 7.6). Specifically, the time elapsed between the application of the step and the rise in i_{ring} increased monotonically with E_{step} , from close to zero for 0.2 V, up to 15 s for 0.65 V. The steady state i_{ring} , on the other hand, increased with E_{step} reaching a maximum at ca. 0.4 V, and then decreased, a trend mirroring that found in the dynamic polarization results in Figure 7.1.

The data in Panel B in Figure 7.6 can be combined with that shown in Figure 7.4, to yield plots of i_{disk} vs θ_{Br^-} for various E_{step} , and of i_{disk} vs. E_{step} for various θ_{Br^-} . As a first step, one may select an arbitrary time, or equivalently, θ_{Br^-} , and draw a vertical line, which will intercept various values of E_{step} , e.g. the vertical dotted line in the right hand panels in Figure 7.6 at $t = 20$ s, would correspond to θ_{Br^-} ca. 0.3, from which correlations can be drawn between i_{disk} and E_{step} for that specific θ_{Br^-} . The actual procedure involved is illustrated in Figure 7.7, which displays i_{disk} vs t (black, left ordinate) for data collected at 400 rpm for $E_{\text{step}} = 0.60$ V, and θ_{Br^-} vs t from Figure 7.4 (see thicker line), allowing a direct and continuous correlation to be made between i_{disk} and θ_{Br^-} for that specific E_{step} .

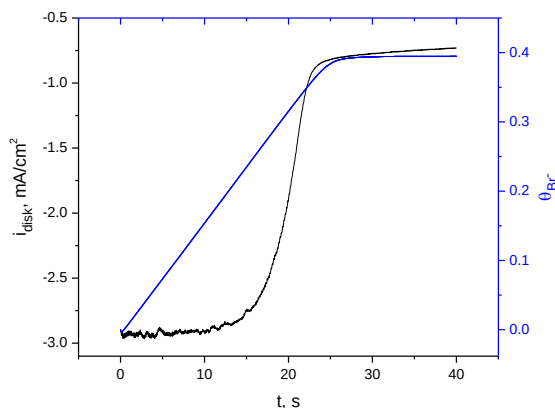


Figure 7.7. Plots of i_{disk} vs t (left ordinate, black line) and θ_{Br^-} vs t (right ordinate, blue line) for $E_{\text{step}} = 0.6$ V at $\omega = 400$ rpm.

Shown in Figure 7.8, are plots of i_{disk} vs θ_{Br^-} for various E_{step} (Panel A) and of i_{disk} vs E_{step} for different θ_{Br^-} (Panel B), induced by the analysis of the data in Figure 7.6, where the two sets of data extend only up to the equilibrium values of θ_{Br^-} , as prescribed by the adsorption isotherm in Figure 7.5. cursory inspection of these curves indicates that for $\theta_{\text{Br}^-} < 0.2$, i_{disk} is insensitive to

E_{step} . In contrast, for $\theta_{\text{Br}^-} > \text{ca. } 0.25$, i_{disk} vs θ_{Br^-} plots are sigmoidal, for which i_{disk} decreases monotonically with E_{step} at sufficiently high θ_{Br^-} (see Panel A).

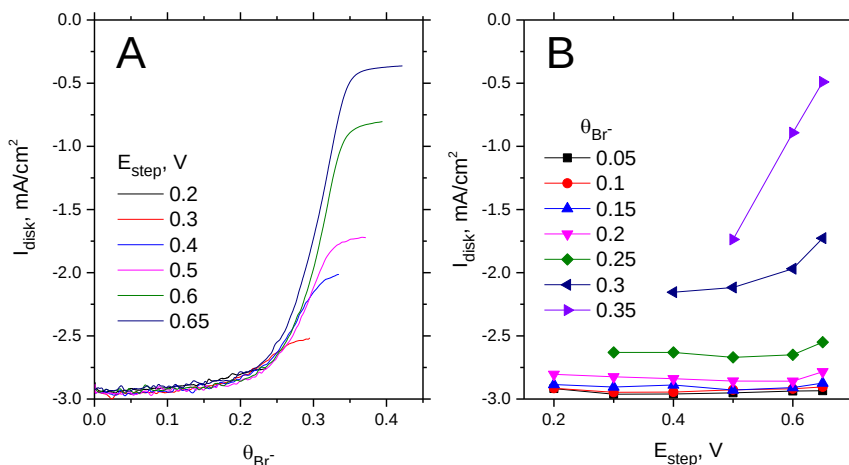


Figure 7.8. Plots of i_{disk} vs θ_{Br^-} for various E_{step} (Panel A) and i_{disk} vs E_{step} for various θ_{Br^-} based on the data shown in Panels B and b in Figure 7.6.

As may be inferred from the data in Panel b, Figure 7.6, i_{ring} , associated with the oxidation of $\text{H}_2\text{O}_2(\text{aq})$ generated at the disk, also exhibits a strong dependence on time, and thus on θ_{Br^-} . These effects can be better illustrated using the ratio $i_{\text{ring}}/i_{\text{disk}}$ as a figure of merit. Shown in Figure 7.9 are plots of $i_{\text{ring}}/i_{\text{disk}}$ as a function of θ_{Br^-} , for fixed values of E_{step} (Panel A), and as a function of E_{step} for fixed values of θ_{Br^-} (Panel B), which revealed, for coverages far below equilibrium, a monotonic increase in $i_{\text{ring}}/i_{\text{disk}}$ with E_{step} at constant θ_{Br^-} , and also with θ_{Br^-} at constant E_{step} .

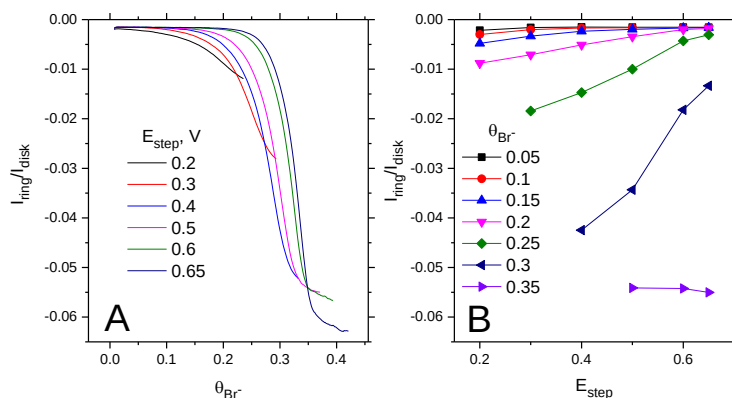


Figure 7.9. Panel A. Plots of $I_{\text{ring}}/I_{\text{disk}}$ as a function of θ_{Br^-} in 0.1 M HClO_4 solutions for a Pt disk of a Pt|Pt RRDE at $\omega = 400$ rpm, based on chronoamperometric data acquired following a potential step from $E_0 = 0.03$ V to various values of E_{step} as indicated, and the corresponding i_{ring} of the type shown in Panel b, Figure 7.7. Panel B Plots of $I_{\text{ring}}/I_{\text{disk}}$ as a function of E_{step} various values of θ_{Br^-} , as indicated

Additional insights into certain aspects of the results presented in this section was obtained from data collected in virtually identical experiments involving the reduction of $\text{H}_2\text{O}_2(\text{aq})$ using the same RRDE to be described in the next section.

Hydrogen Peroxide Reduction

Shown in Figure 7.10, are dynamic polarization curves obtained in 1 mM H_2O_2 in deaerated 0.1 M HClO_4 with the Pt disk of the Pt|Pt RRDE rotating at $\omega = 400$ rpm in the absence (black) and presence of either 10 μM NaCl (blue), or 10 μM KBr (red). Unlike chloride, for which the effect on the rates of both $\text{H}_2\text{O}_2(\text{aq})$ reduction and oxidation was only slight, bromide induced very large shifts in the onset of both processes. Addition of 10 μM KBr reduced the activity of Pt significantly in the range $0.6 < E < 0.9$ V; however, at very high overpotentials, the limiting currents were found to be very close to the values observed in the neat electrolyte.

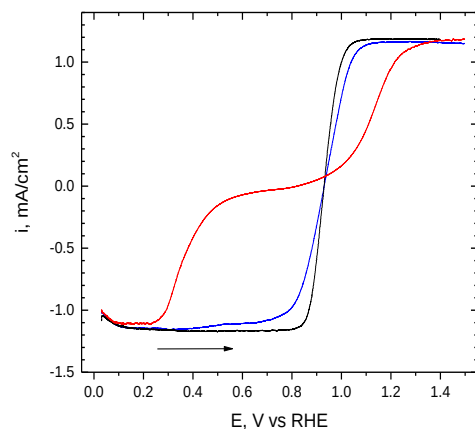


Figure 7.10. Dynamic polarization curves ($v = 10$ mV/s) for the Pt disk of the Pt|Pt RRDE recorded in 0.1 M HClO_4 containing 1 mM H_2O_2 at a rotation rate, $\omega = 400$ rpm, before (black) and after the addition of 10 μM NaCl (blue) or 10 μM KBr (red). The arrow points in the direction in which the potential was scanned.

Chronoamperometric curves recorded at $\omega = 400$ rpm under otherwise the same conditions both in the absence and in the presence of 10 μM KBr are displayed in Panels A and B, Figure 7.11, respectively. For these measurements, the potential was stepped from E_0 to different E_{step} values as specified. As evidenced from these data, no changes in activity were observed for short times, $t < 10$ s, i.e. $\theta_{\text{Br}^-} < 0.2$. This behavior is consistent with that predicted by theory as will be discussed in the next section. For longer times, i.e. $\theta_{\text{Br}^-} > 0.25$, the effects became quite noticeable. In particular, for E_{step} ca. 0.7 V, the current dropped by several orders of magnitude with respect to that observed for the halide-free surface under otherwise the same conditions. Differential reflectance spectroscopy confirmed $\text{H}_2\text{O}_2(\text{aq})$ reduction did not interfere with bromide adsorption, as evidenced by the fact that, within the noise, the two $\Delta R/R$ vs t curves in Figure 7.12 overlap.

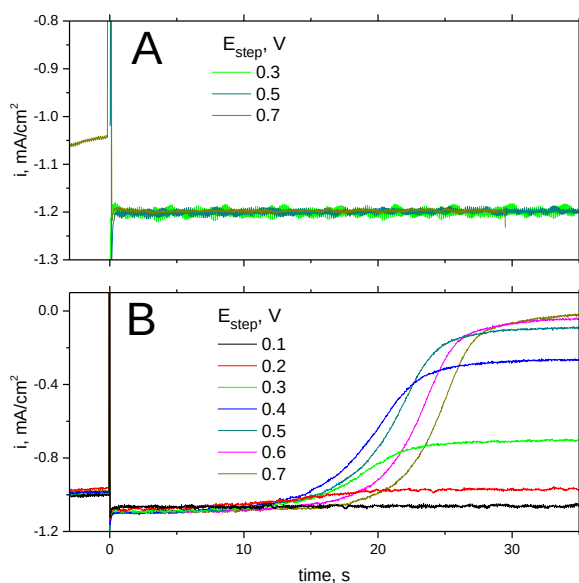


Figure 7.11. Chronoamperometric curves recorded at $\omega = 400$ rpm in 0.1 M HClO_4 containing 1 mM H_2O_2 in the absence (Panel A) and in the presence of 10 μM KBr at $\omega = 400$ rpm (Panel B) in solution starting from $E_0 = 0.05$ V (neat) or $E_0 = 0.03$ V (bromide) to various E_{step} values as indicated in the legends.

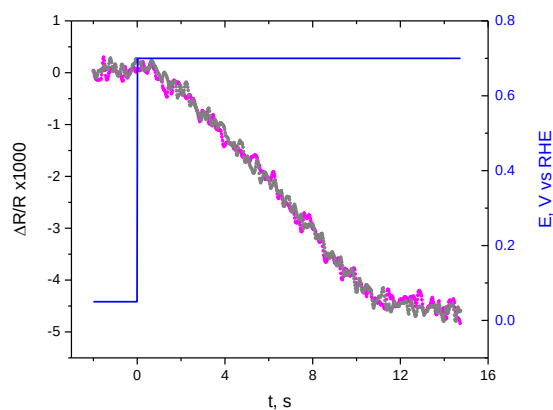


Figure 7.12. Comparison between $\Delta R/R = [R(0.05\text{V}) - R(0.7\text{ V})]/R(0.05\text{V})$ transients collected for the Pt disk of the Pt|Pt RRDE at $\omega = 900$ rpm in Ar-saturated (black) and in H_2O_2 -saturated (magenta) 10 μM KBr in 0.1 M HClO_4 following a potential step from $E_0 = 0.05$ to $E_{\text{step}} = 0.7$ V applied at $t = 0.0$ s (blue, right ordinate, see text for details).

The same type of analysis as that performed for the ORR in Figure 7.8 was used to construct plots of the $\text{H}_2\text{O}_2(\text{aq})$ reduction currents as a function of θ_{Br^-} , as well as of E_{step} (see Panels A and B, Figure 7.13, respectively). Although highly speculative, the peculiar hump found in Panel B in this figure may be due to the interplay between a possible peroxide dismutation (also known as disproportionation) and the reduction of dioxygen derived from it.

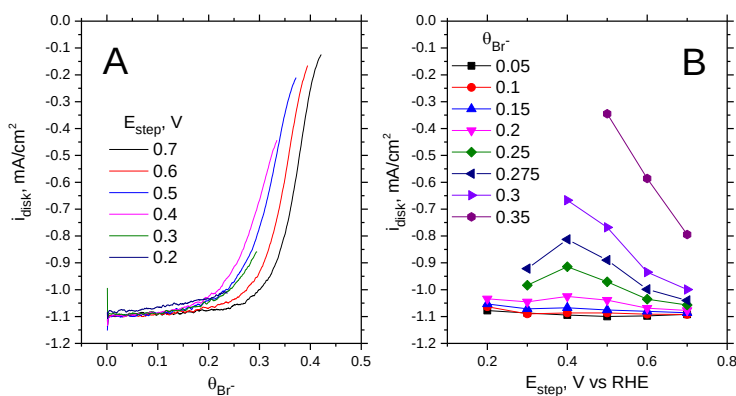


Figure 7.13. Panel A. Peroxide reduction currents, i_{disk} , as a function of θ_{Br^-} recorded in a 1 mM H_2O_2 solution containing 10 μM KBr in 0.1 M HClO_4 with the Pt disk of the Pt|Pt RRDE at $\omega = 200$ rpm, based on chronoamperometric data acquired following a potential step from $E_0 = 0.03$ V to various values of E_{step} , as specified Panel B. Plot of i_{disk} vs E_{step} for various values of θ_{Br^-} as indicated.

Theoretical

The primary aim of this section is to account for some of the effects induced by adsorbed bromide on the diffusion-limited currents for the reduction of $\text{H}_2\text{O}_2(\text{aq})$. The theoretical strategy employed relies on a model developed by Levart in his pioneering studies of partially blocked rotating disk electrodes, pb-RDE.¹⁶ His approach assumes the pb-RDE consists of a central active disk of radius, r_c , surrounded by a crown-type array of active disks of radii r_s embedded into an otherwise inactive disk surface of radius r_e (see Figure 7.14). As reported by this author, the diffusion-limited currents observed for this type of pb-RDE are very similar to those found

for a theoretically more tractable pb-RDE involving the same active central disk surrounded by alternating inactive (light grey) and active (dark grey) rings of thicknesses a and b , respectively, so as to preserve the total fraction of inactive area, θ . A schematic diagram of this arrangement for an electrode with $\theta = 0.6$ and a number of active rings, $N_A = 3$, is shown Figure 7.14. Further details regarding this system, including the mathematical relationships between the parameters involved, can be found in the original literature and will not be repeated here.

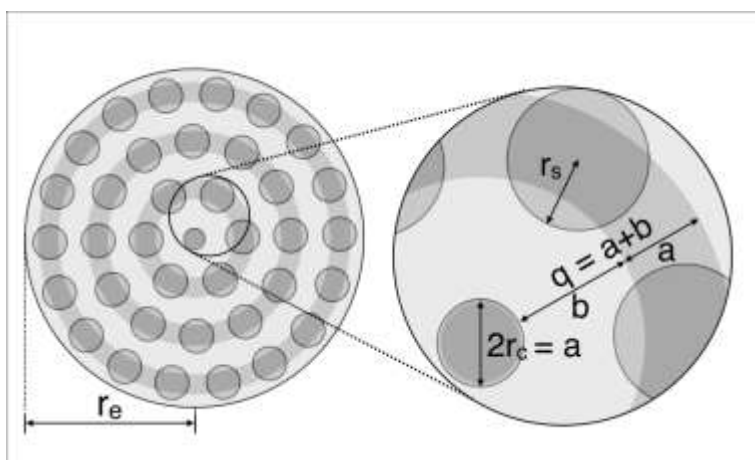


Figure 7.14. Schematic diagram of the equivalent array of active disks of radius r_s and alternating active and inactive rings of thickness a and b , respectively, surrounding a central disk of radius r_a described by Levart. In this figure, q is the radial period and b_{ext} is the width of the outermost inactive ring (see text and the original paper for details).

Within the Levart formalism, the problem so stated reduces to finding solutions to the steady state convective-diffusion equation in a 2D-axisymmetric system under conditions in which the reaction on the active surfaces of the electrode (dark grey regions) proceeds under strict diffusion control (see Table 7.1, where all parameters are defined in Table 7.2). These are precisely the same conditions under which the experimental data in Figure 7.11 were collected. The domain used by Levart is shown in Figure 7.15, where the depth (length of boundary 5) was ca. $7/3$ the thickness of the diffusion boundary layer, δ , and the corresponding boundaries and

boundary conditions are defined in Table 7.2. In our case, we set the depth and width of the domain to be large enough, ca. three times δ and r_e , respectively, for the results to become independent of any further increases in the actual size of the domain within the range of parameter values of interest.

Table 7.1. Governing Differential Equations and Boundary Conditions Involved in the Implementation of Levart's Formalism to Calculate Attenuation Factors for Partially Blocked Rotating Disk Electrodes.

	Dimensioned	Dimensionless
Governing Differential Equations	$D \left\{ \frac{\partial^2 c}{\partial z^2} + \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial c}{\partial r} \right) \right\} = u_r \frac{\partial c}{\partial r} + u_z \frac{\partial c}{\partial z}$ $u_r = 0.51023 \cdot (r\omega)(\omega/\nu)^{1/2} z$ $u_z = -0.51023 \cdot (\omega\nu)^{1/2} (\omega/\nu) z^2$	$\frac{1}{Sc} \left\{ \frac{\partial^2 C}{\partial Z^2} + \frac{1}{R} \frac{\partial}{\partial R} \left(R \frac{\partial C}{\partial R} \right) \right\} = U_r \frac{\partial C}{\partial R} + U_z \frac{\partial C}{\partial Z}$ $U_r = 0.51023 \cdot R \cdot Z$ $U_z = 0.51023 \cdot Z^2$
Boundary Conditions	$c_{z=0, \text{active rings}} = 0$ $c_{z \rightarrow 3r_e} = c_{r \rightarrow 5r_e} = c_{bulk}$	$C_{Z=0, \text{active rings}} = 0$ $C_{Z \rightarrow 3R_E} = C_{R \rightarrow 5R_E} = 1$
Characteristic Parameters And Results	$\delta = 1.612 \cdot (D^2 \omega^{-3} \nu)^{1/6}$ $i_{Lev}^{lim} = (\pi r_e^2) n F c_{bulk} D / \delta$ $Re = r_e^2 \omega / \nu$ $\log(r_e / \delta) = \log \left\{ r_e / 1.612 (D^2 \omega^{-3} \nu)^{1/6} \right\}$ $\rho = \frac{I_{lim}}{I_{Lev}^{lim}} = - \int \left(\frac{dc}{dz} \right)_{z=0} \frac{2\delta r}{r_e^2 c_{bulk}} dr$	$\Delta = 1.612 \cdot Sc^{-1/3}$ $\Psi_{Lev}^{lim} = 1 / (Sc \cdot \Delta) = 0.62 (\pi R_E^2) Sc^{-2/3}$ $Re = R_E^2$ $\log(R_E / \Delta) = \log \left\{ R_E / 1.612 \cdot Sc^{-1/3} \right\}$ $\rho = \frac{\Psi_{lim}}{\Psi_{Lev}^{lim}} = -3.22 \int \left(\frac{dC}{dZ} \right)_{Z=0} \frac{R}{Sc^{1/3} R_E^2} dR$
Dimensionless Quantities	$U = u / (\omega\nu)^{1/2}, \quad R = (\omega/\nu)^{1/2} r, \quad Z = (\omega/\nu)^{1/2} z, \quad C = c / c_{bulk}$ $Sc = \nu / D, \quad R_E = (\omega/\nu)^{1/2} r_e, \quad \Psi = i / (n F c_{bulk}),$	

Table 7.2. Parameters Employed in the Simulations

Parameter	Symbol	Value
Kinematic viscosity	ν	$0.010 \text{ cm}^2/\text{s}$
Diffusion Coefficient	D	$10^{-5} \text{ cm}^2/\text{s}$
Bulk Concentration	C_{bulk}	$500 \text{ mol}/\text{cm}^3$
Electrode Radius	r_e	0.35 mm
Angular Velocity of Electrode	ω	$94.24/\text{s}$ (900 rpm)
Inactive Coverage	θ	0.5

Table 7.3. Boundary Conditions Associated with the Domain Shown in Figure 7.11

Boundary	Transport of Diluted Species (chds)
1	Symmetry Axis
2	Bulk Concentration
3	Concentration at the Electrode Surface
4	No Flux
5	Outflow

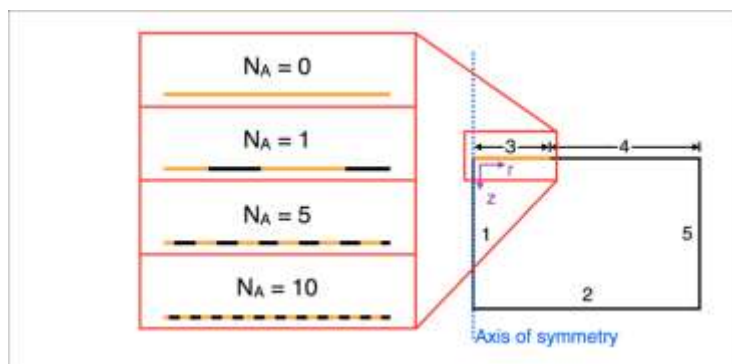


Figure 7.15. Schematic diagram of the domain used in COMSOL for implementing the multi-ring rotating electrode model described by Levart.

Calculations were performed for the parameters specified in Table 7.3 using an initial surface mesh with an adaptive mesh refinement (AMR) as implemented in COMSOL 4.3a, of maximum size $0.5 \mu\text{m}$, and an extremely fine domain mesh optimized for fluid dynamics. Prior to tackling the multi-ring problem, however, a number of tests were performed to validate Levart's formalism within the COMSOL platform using a rotating ring-disk electrode, RRDE, of

the same dimensions as those used in the experiments above. Excellent agreement, i.e. < 1%, was found for the limiting currents both at disk and ring electrodes, for the collection efficiency of the assembly, and for the shielding factor, when compared with the well-known analytical expressions.

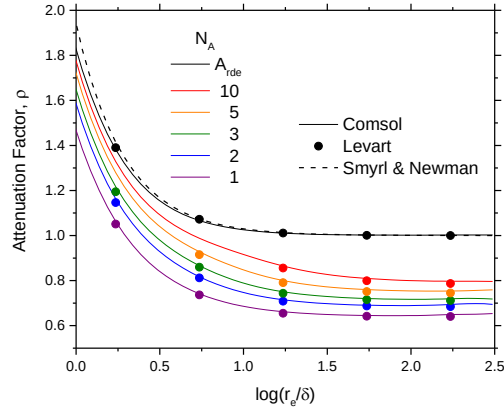


Figure 7.16. Comparison between plots of ρ vs $\log(r_e/\delta)$ reported by Levart (solid symbols) and obtained by solving the convective diffusion equation using COMSOL (solid line) for $\theta = 0.5$ and selected values of N_A assuming the same domain and boundary conditions as those employed by Levart. The black thick dotted line is based on ρ_{SN} values calculated from Eq. (7). The symbol A_{rde} refers to a completely unblocked disk electrode, i.e. $N_A = 0$.

Shown in solid thin lines in Figure 7.16, are plots of the attenuation factor, ρ , as defined in Table 1, as a function of $\log(r_e/\delta)$, for a few selected values of N_A , which match rather accurately the data reported by Levart (see solid circles in the figure). As clearly indicated, for large values of $\log(r_e/\delta)$, i.e. when the radius of the entire disk is much larger than the diffusion boundary thickness δ , or for very fast rotation rates, ω , for an electrode of fixed dimensions, ρ depends only on N_A . By contrast, as either r_e decreases, or the rotation slows down, ρ increases rapidly as δ surpasses r_e . Also shown in the dotted line in Figure 7.16 are values of ρ for a single disk electrode, A_{rde} , predicted by Smyrl and Newman,¹⁷ ρ_{SN} , based on their analysis for high Schmidt numbers, Sc , namely:

$$\rho_{SN} = 1 + 1.9193 \cdot Sc^{-1/2} Re^{-3/4} \quad [7.7]$$

where Re is the Reynolds number (see Table 7.1). Since bromide is a poisoning species the model above was modified by exchanging the active and inactive areas). As indicated, for very large N_A , hardly any attenuation is observed for small coverages of the blocking species, i.e. $\theta < 0.2$, in very good agreement with the results obtained experimentally for the ORR and HPRR. For higher coverages, i.e. $\theta > 0.2$, the experimental data show a rapid decrease in i_{disk} as θ increases (see Panel A, in Figure 7.8 and Figure 7.11). One possible explanation for this effect may be found in a decrease in the activity of the disk as N_A decreases, which would be consistent with a reorganization of the adsorbed bromide on the Pt substrate

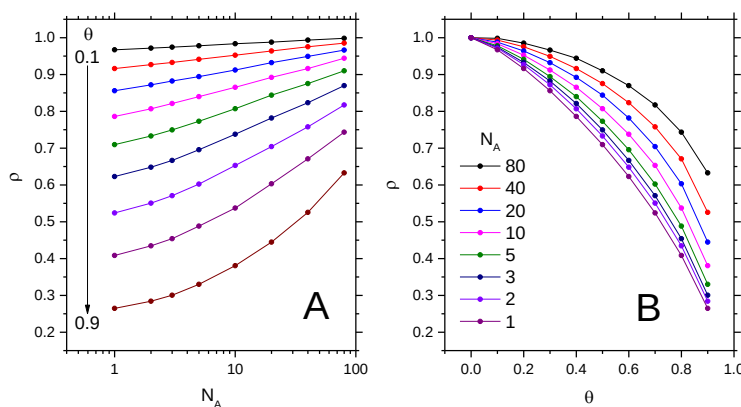


Figure 7.17. Plots of ρ vs N_A , for various values of θ (Panel A), and of ρ vs θ , for various values of N_A (Panel B), for $\omega = 900$ rpm.

Conclusions

A unique chronoamperometric method has been developed and implemented that allows correlations to be made between the current due to oxygen reduction and the extent of bromide adsorption on a rotating Pt(poly) disk of a Pt(poly)| Pt(poly) RRDE in acidic solutions using the method reported originally by Markovic *et al.*, to determine θ_{Br^-} quantitatively. This tactic also

made it possible to monitor changes in the reaction mechanism by detecting $\text{H}_2\text{O}_2(\text{aq})$ generated at the disk using the Pt ring polarized at a potential positive enough for the oxidation of $\text{H}_2\text{O}_2(\text{aq})$ to proceed under diffusion-limited conditions. The same strategy was employed for studies involving the reduction of $\text{H}_2\text{O}_2(\text{aq})$ in the same media. As evident from the results obtained, the ORR and HPRR currents do not seem to be affected by the presence of adsorbed Br^- up to θ_{Br^-} ca. 0.2, regardless of the applied potential, E_{step} . This behavior is in agreement with calculations performed based on the model proposed by Levart. However, in the case of oxygen reduction, a clear increase in the amount of $\text{H}_2\text{O}_2(\text{aq})$ was observed even for smaller θ_{Br^-} , in the low potential region.

Chapter 8 Electrocatalytic Properties of Adsorbed Halides: The Reduction of Hexaaquairon(III) on Polycrystalline Gold in Aqueous Electrolytes

Contributors

Adriel J.J. Jeberaj – Experimental Measurements

Nicholas S. Georgescu – All calculations from the experimental data

Daniel A. Scherson - Advisor

Abstract

Enhancements in the rates of reduction of hexaaquaairon(III), $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$, induced by the adsorption of bromide, Br^- , on a polycrystalline Au, Au(poly), electrode, have been examined in 0.1 M HClO_4 aqueous solutions containing KBr at concentrations in the μM (< 1 ppm) range. The tactic employed involved chronoamperometry at a Au(poly)|Au(poly) rotating ring|disk electrode (RRDE), whereby the shielding of the diffusion-controlled oxidation of Br^- detected by the ring in the absence of $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$, allowed for the amount of Br^- adsorbed on the disk to be determined with great accuracy. Evidence that Br^- adsorption is not affected by the reduction of $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ was obtained from in situ differential reflectance measurements performed in the absence and in the presence of the metal complex in solution. Information derived from these measurements was then used to establish quantitative relationships between the disk current, i_{disk} , due to the reduction of $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ and the normalized Br^- coverage, θ_{Br}^N , at constant potential, E , and between i_{disk} and E , at constant θ_{Br}^N . Rather surprisingly, no evidence for electrocatalytic activity was found for $\theta_{\text{Br}}^N < 0.15$. Possible explanations for this unexpected behavior are discussed. At higher coverages, the results obtained were in qualitative agreement with theories put forward in years past that correlate the activities of arrays of active microdisks arranged in a hexagonal pattern on otherwise inactive surfaces that do not account explicitly for the specific mode of convection associated with the rotating disk electrode.

Introduction

The influence of adsorbed species on the kinetics and mechanism of a number of redox reactions has been recognized for decades and continues to be the subject of numerous studies of both fundamental and technological importance. This contribution provides a quantitative measure of the extent to which the adsorption of a halide affects the rates of a simple redox process involving a solution-phase species, using the reduction of hexaaquairon(III), $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$, on a polycrystalline Au electrode, Au(poly), in 0.1 M HClO_4 solutions containing bromide, Br^- , as a model system. Perchlorate ion displays very weak affinity for Au surfaces and, as such, may be regarded as uniquely suited for these studies. The specific strategy employed in this investigation, described in detail in a recent communication, is based on a combination of chronoamperometric, optical and rotating ring-disk electrode, RRDE, techniques, that allows the time evolution of the Br^- coverage to be determined while the redox process is taking place. Although some aspects of this methodology share common features with tactics reported earlier in the literature, there are important differences that makes it particularly advantageous. Specifically,

- i. the disk of the RRDE electrode is first polarized ***under rotation*** at a potential, E_o , negative enough to induce the full desorption of Br^- , and then stepped to a potential E_{step} , at which adsorption of Br^- is expected to ensue under strict diffusion control for times on other order of tens of seconds.
- ii. The flux of bromide to the disk, and, thus, the absolute amount of Br^- adsorbed on the disk is determined from the shielding of the diffusion-limited ring current associated

with Br⁻ oxidation, and, as such, avoids uncertainties associated with the analysis of the disk current.

- iii. *In situ* optical differential reflectance is used to monitor adsorption of Br⁻ on the disk and confirm that the redox reaction does not interfere with the Br⁻ adsorption process.

An illustration of the power of this technique was provided recently by Jebaraj *et al.*,⁶⁸ who examined the effects of Br⁻ adsorption of Pt(poly) on the reduction of oxygen and hydrogen peroxide in aqueous 0.1 M HClO₄ solutions. As will be shown, correlations between i_{disk} and the normalized Br⁻ coverage, $\theta_{\text{Br}^-}^N$ for various values of E_{step} , revealed that significant enhancements in the electrocatalytic activity of Au(poly) for the reduction of [Fe(H₂O)₆]³⁺ could only be discerned for $\theta_{\text{Br}^-}^N > 0.15$.

Experimental

All measurements were performed in an all-glass, three-compartment cell, filled with deaerated (Ar, UHP, 99.999%, Airgas) 0.1 M HClO₄ solutions (75 mL) prepared from ultrapure perchloric acid (Ultrapure, EMD), and ultrahigh purity, UHP, water generated by a Barnstead system (18.3 MΩcm, EASYpureUV), either neat or containing 10 μM KBr (Fisher Scientific, IR grade, > 99%), and/or hexaaquairon(III) perchlorate hydrate (Fe(ClO₄)₃·xH₂O, Sigma Aldrich, < 0.1% chloride), as received. Experiments were carried out with a Au(poly)|Au(poly) RRDE working electrode (Pine Instruments, disk radius: 0.2285 cm; ring inner diameter 0.2465 cm; gap, G = 180 μm and ring outer diameter 0.2690 cm, or, correspondingly, disk and ring areas of 0.164 cm² and 0.037 cm², respectively, and a collection efficiency, N = 0.22). A carbon rod was used as a counter electrode, and a reversible hydrogen electrode (RHE) as a reference. The disk

and ring potentials, E_{disk} and E_{ring} , respectively, were controlled with a bipotentiostat (Pine Instruments, Model AFCBP1) and the rotation rate of the RRDE, ω , was adjusted with a commercial rotator (Pine Instruments, Model AFMSRX). All the data were recorded using an acquisition card (National Instruments, USB-6009) programmed in Labview. Cyclic voltammetric curves recorded in deaerated 0.1 M HClO₄ at $\omega = 400$ rpm, yielded features characteristic of clean Au(poly) surfaces.

Normalized optical differential reflectance measurements, $\Delta R/R = [R(E_{\text{ref}}) - R(E_{\text{sam}})]/R(E_{\text{ref}})$, where $R(E_{\text{ref}})$ and $R(E_{\text{sam}})$ represent the intensity of the light collected at the detector following reflection from the Au electrode polarized at a reference, E_{ref} , and an arbitrary sampling potential, E_{sam} , respectively, were performed using a blue diode laser (Polaris 700, $\lambda = 447$ nm, 900 mW, Laserglow), while collecting chronoamperometric data employing methods and instrumentation specified in detail elsewhere⁶⁸. Chronoamperometric experiments were carried out for fixed $[\text{Br}^-]$ and variable ω . In all instances, the disk current, i_{disk} , was monitored following application of a potential step from $E_0 = -0.15$ V, i.e. no Br⁻ adsorption, to values of E_{step} in the range $0.0 \leq E_{\text{step}} \leq 0.9$ V vs RHE, i.e. negative to the onset of oxide formation on Au(poly) in neat 0.1 M HClO₄, and to the onset of $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ reduction (vide infra)

Results and Discussion

Correlations between the coverage of bromide, θ_{Br} , on the Au(poly) RDE and its activity for the reduction of $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ were pursued by first using the method of Gasteiger *et al.* (see Section A. below) to determine θ_{Br} in the absence of the redox couple in solution. As mentioned briefly above, this tactic relies on measurements of the ring currents associated with bromide oxidation under diffusion limited conditions following a potential step applied to the disk from a

value negative enough to induce full desorption of Br^- to a higher value, E_{step} , at which adsorption of Br^- is expected to ensue. The results of these experiments are given in Section A below, which include in situ optical evidence that halide adsorption and redox activity do not affect one another, i.e. the θ_{Br} values obtained in the absence of $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ are applicable when analyzing its reduction, to be presented in Section B.

A. Bromide Adsorption Under Forced Convection: Chronoamperometric Studies

Shown in Figure 8.1 (see left ordinate), are plots of the ring currents, I_{ring} , collected with the ring polarized at $E_{\text{ring}} = 1.4$ V vs RHE, a value positive enough to oxidize Br^- under strict diffusion control, for a series of values in the range $0.0 \leq E_{\text{step}} \leq 0.9$ V. These transients can be used to predict the flux of Br^- associated with its adsorption (or consumption) on the disk, which can be expressed in terms of a current, $I_{\text{disk}}^{\text{pr}}$, as follows:

$$I_{\text{disk}}^{\text{pr}} = \frac{I_{\text{ring}}^{\circ} - I_{\text{ring}}}{N} \quad [8.1]$$

The symbols N and I_{ring}° in this equation represent, respectively, the collection efficiency, and the values of I_{ring} under conditions in which no Br^- is being consumed nor generated at the disk known as the *unshielded* ring current. The main advantage of this tactic is that uncertainties derived from the RC response of I_{disk} , due, for example, to leakage currents, and other adventitious effects.

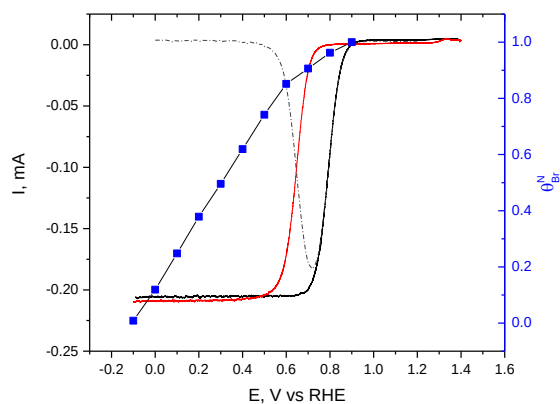
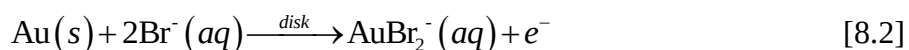


Figure 8.1. Dynamic polarization curves collected with the Au(poly) disk of the Au|Au RRDE in the absence (red) and presence (black) of 10 μM KBr in deaerated 0.1 M HClO_4 aqueous solutions containing 1 mM $\text{Fe}(\text{ClO}_4)_3$ at a scan rate $\nu = 10$ mV/s and a rotation rate $\omega = 400$ rpm, where the arrow specifies the direction of the scan. The blue scattered symbols in this figure (see right ordinate) represent values of the bromide coverage normalized by its value at $E = 0.9$ V, θ_{Br}^N . The difference between the red and black lines is represented by the dash-dotted gray line.

Careful inspection of these data reveals that at longer times and large values of E_{step} , the ring continues to be partially shielded. One possible explanation for this effect may be found in the dissolution of Au to form one or more soluble bromide complexes, as illustrated below for a Au(I) species, which, would be enhanced at positive potentials,



Support of this view has been obtained by O. M. Magnussen et al.,⁷⁰ who obtained in situ surface x-ray diffraction evidence for a bromide-induced step-flow etching of Au(111) in the high potential region. Such process will consume Br^- arriving at the disk, and thereby increase the observed shielded current. It should be emphasized that the magnitude of this residual current is small enough for its impact on the analysis to follow to be safely ignored.

Integration of the flux of Br^- to the disk predicted by the shielding as a function of time, with the background current at long times (namely $I_{\text{disk}}^{\text{pr},\infty}$) subtracted, provides a measure of the amount of Br^- adsorbed on the disk, as illustrated for various values of E_{step} in Figure 8.3, where

$$Q_{\text{Br}} = \int_0^t (I_{\text{disk}}^{\text{pr}} - I_{\text{disk}}^{\text{pr},\infty}) d\tau \quad [8.3]$$

These data can be divided by the corresponding value for Q_{Br} for $E_{\text{step}} = 0.9$ V over the entire time scale, Q^o , to yield a normalized coverage of bromide, θ_{Br}^N , namely

$$\theta_{\text{Br}}^N = Q_{\text{Br}} / Q^o \quad [8.4]$$

(see right ordinate in this figure), which can then be used to construct the adsorption isotherm shown in blue scattered symbols in Figure 8.1.

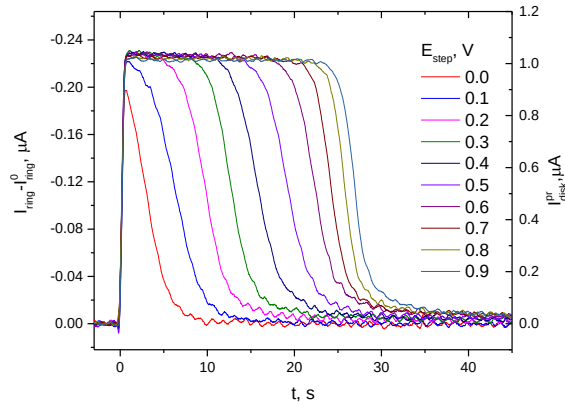


Figure 8.2. Chronoamperometric curves recorded with the Au ring of a Au|Au RRDE in 10 μM KBr in 0.1 M HClO_4 at $\omega = 400$ rpm, from $E_0 = -0.15$ V to E_{step} , where the thin lines are the predicted disk currents based upon the ring currents shown in the bottom panel in this figure.

Bottom Panel. Plots of $i_{\text{ring}} - i_{\text{ring}}^o$ vs time recorded with the ring polarized at $E_{\text{ring}} = 1.4$ V vs RHE, during the acquisition of the chronoamperometric disk data

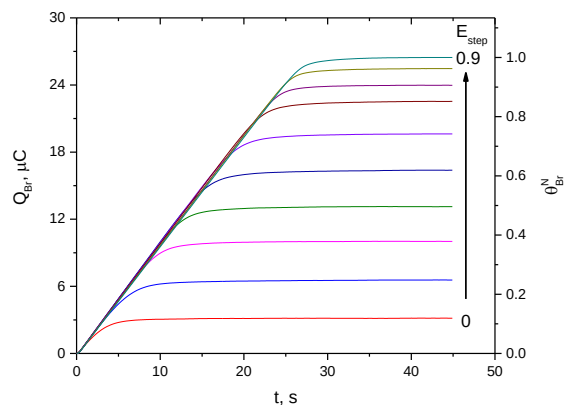


Figure 8.3. Plots of Q_{Br} vs t (left ordinate) and the corresponding θ_{Br}^N vs t (right ordinate) for various values of E_{step} , based on the data in the bottom panel, Figure 8.2 (see text for details).

B. The Reduction of Hexaaquairon(III): Effects of Adsorbed Bromide

Chronoamperometric experiments similar to those described in the previous section were performed in deaerated 0.1 M $HClO_4$ aqueous solutions containing both 10 μ M KBr and 1 mM $Fe(ClO_4)_3$ to examine the effects of adsorbed Br^- on the reduction of $[Fe(H_2O)_6]^{3+}$. As before, the potential was stepped from $E_0 = -0.15$ V, which according to the adsorption isotherm is negative enough for θ_{Br}^N on Au(poly) to be negligible (see blue symbols, Figure 8.1), to values of E_{step} , where adsorption of Br^- would be expected to ensue.

The results obtained for five ω in the range 200 to 1600 rpm are shown in Panels A through E in Figure 8.5, which include, in each case, data collected for different values of E_{step} (see legend in this figure). All the curves, except the overlaid jagged ones, which correspond to the raw data, have been smoothed either by an 11-, to 13-point moving average. As illustrated in Panels A, D and E, the current immediately following the step to $E_{step} = 0.4$ V, attained a

constant value for the entire duration of the run. This should not be surprising, as the effects of Br^- adsorption for E_{disk} in the range $0.15 < E \leq 0.45$ V, are hardly noticeable (see Figure 8.1, *vide supra*), i.e. the current is controlled by mass transport. A markedly different behavior was found for curves involving $E_{\text{step}} \leq 0.85$, for which the current jumped suddenly followed by an exponential-type decay, as would be expected for the response of an RC circuit, to reach, shortly thereafter, a relatively constant value. This plateau, which lasted for a time τ_{pl} (see Figure 8.4) on the order of a few seconds, was followed by a gradual increase in the current to yield a final steady state value, i_{ss} .

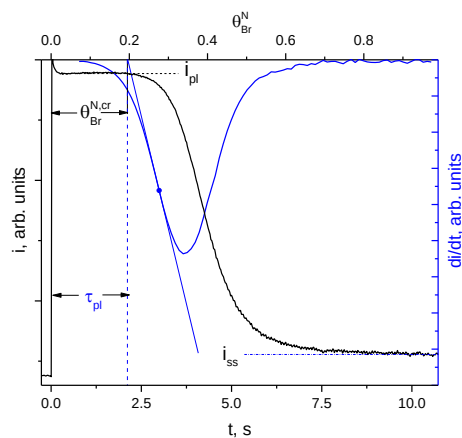


Figure 8.4. Schematic diagram describing the graphical method employed to extract values of τ_{pl} and $\theta_{\text{Br}}^{\text{N,cr}}$ from chronoamperograms recorded with the Au(111) electrode under the conditions specified in Figure 8.5 using one such curve as an example (see solid black line), where the blue line represents its derivative.

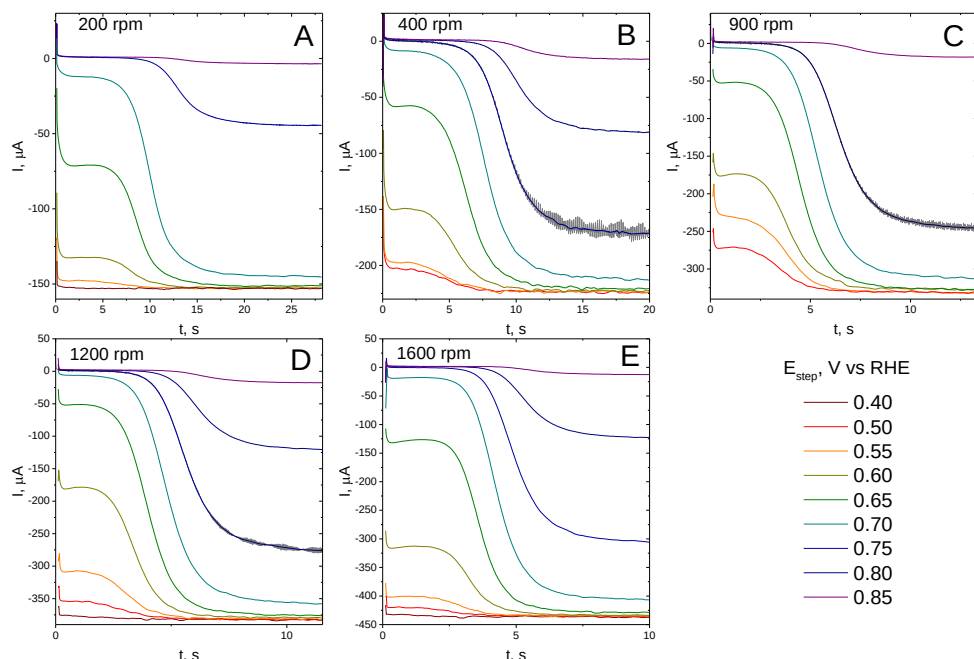


Figure 8.5. Chronoamperometric curves recorded with the Au disk of a Au|Au RRDE in deaerated 0.1 M HClO₄ aqueous solutions containing both 10 μM KBr and 1 mM Fe(ClO₄)₃ for various values of ω as specified in each of the panels. For each experiment, the potential was stepped from $E_0 = -0.15$ V to different E_{step} , as indicated in the legend. Except for the jagged curves (raw data), collected for $E_{\text{step}} = 0.75$ V, all others represent averages (see text for details).

Evidence that the reduction of $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ does not interfere with the Br^- adsorption process was obtained from in situ differential reflectance measurements, $\Delta R/R$, in deaerated 10 μM KBr in 0.1 M HClO₄, in which the potential was stepped from E_0 , and $E_{\text{step}} = 1.0$ V (see magenta line, right ordinate) both before and after adding 1 mM Fe(ClO₄)₃. The results obtained shown, respectively, in black and blue in Figure 8.6, yielded virtually no differences between the two sets of results. On this basis, it becomes possible to correlate the electrocatalytic activity of Au(poly) for the reduction of $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ and θ_{Br}^N , as illustrated in Figure 8.7 for $E_{\text{step}} = 0.70$ V. At this potential, (see Figure 8.1) Au(poly) displays almost no activity for the redox reaction in the absence of Br^- in solution, but elicits nearly diffusion-controlled currents in its presence. In

fact, the differences in activity between the two dynamic polarization curves (see dotted line in Figure 8.1), was largest over the range 0.65-0.8V with the maximum occurring at 0.715 V.

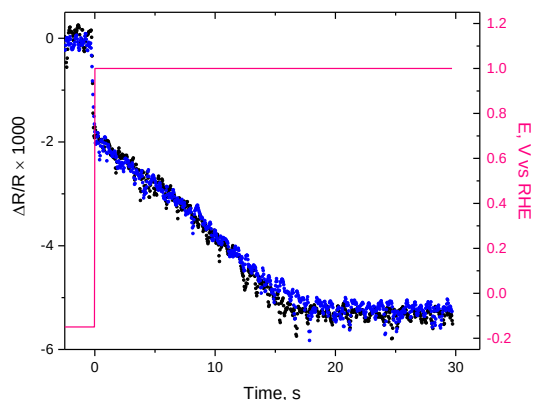


Figure 8.6. Plots of $\Delta R/R = [R(-0.15) - R(1.0)]/R(-0.15)$ vs time collected with the Au disk of the Au|Au RRDE at $\omega = 900$ rpm in Ar-saturated (black) 0.1 M HClO₄ containing 10 μ M KBr before (black) and after (blue) addition of 1 mM [Fe(H₂O)₆]³⁺ following a potential step from $E_o = -0.15$ V to $E_{\text{step}} = 1.0$ V applied at $t = 0.0$ s (magenta, right ordinate, see text for details).

Several interesting observations could be gleaned from a more detailed analysis of data of the type shown in Figure 8.5. Specifically, the presence of a current plateau region shortly after the step was applied, clearly indicates that for very small values of θ_{Br}^N , the activity of Au(poly) for the reduction of [Fe(H₂O)₆]³⁺ remains virtually unaffected. This finding may be regarded as very surprising, as adsorbed bromide (and a few other halides as well) are known to promote this redox process. Although somewhat speculative, this behavior may be related to the presence of “low energy” sites, i.e. first to fill, on the Au(poly) surface, the activity of which toward [Fe(H₂O)₆]³⁺ reduction remains unaltered following adsorption of Br⁻.

Further insight into this phenomenon was obtained from plots of τ_{pl} at constant ω , or, correspondingly, the coverage θ_{Br}^N required for the electrocatalytic activity to “turn on”, defined

as $\theta_{Br}^{N,cr}$ which were found to increase monotonically with E_{step} (see Figure 8.8). Shown in thick solid lines in Figure 8.4 is one of the transients following application of the potential step at $t = 0$ depicted in Figure 8.5 (black) and its corresponding time derivative (blue), where the thin blue line is the second derivative of the transient evaluated at the first inflection point (see blue solid circle). Extrapolation of this line down to the time axis, allows definition of a graphical parameter that represents the value of θ_{Br} required for the electrocatalytic activity to “turn on”, θ_{Br}^{cr} , or equivalently, the time during which the current attains a fairly constant value, τ_{pl} . As time elapses, the current slowly increases to achieve a final steady state value denoted as i_{ss} .

This behavior is equivalent to a decrease in τ_{pl} as the overpotential for the reduction of $[Fe(H_2O)_6]^{3+}$ is increased. These results suggest that the number of such “low energy” sites increases as the potential shifts in the positive direction. Furthermore, the fact that $\theta_{Br}^{N,cr}$ at constant E_{step} was found to be fairly independent of ω (see Figure 8.8), at least for $\omega < 1600$ rpm, suggests that the effect may be derived from thermodynamic rather than kinetic considerations. Some support for this view is provided by changes in the arrangement of bromide on the surface induced by the applied potential as inferred from in situ x-ray diffraction measurements for Br^- adsorption on Au(111) in neutral NaBr solutions. As reported by Magnussen *et al.*, the data collected are consistent with the existence of a hexagonal close-packed adlayer phase above a critical potential, E_{cr} , where the bromide-bromide spacing changes in a continuous fashion between 4.24 Å at the critical potential and 4.03 Å at a potential 300 mV more positive. On this basis, it seems conceivable that the adlayer formed at the lower potential may involve fewer number of bromide inactive sites than those in the adlayer formed at the higher potentials,

leading, under the same mass transport conditions, to shorter τ_{pl} , i.e. smaller $\theta_{Br}^{N,cr}$, as the experimental data show.

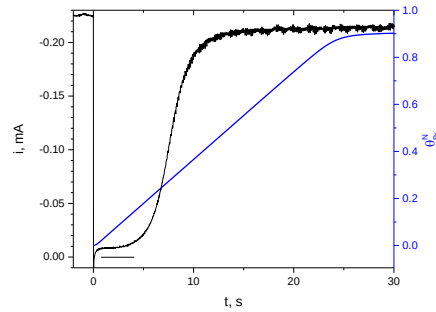


Figure 8.7. Plot of the Au(poly) disk current associated with the reduction of $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ (left ordinate; black) and θ_{Br}^N as a function of time for $E_{\text{step}} = 0.70$ V from Figure 8.5. The blue curve is the normalized coverage of bromide from Figure 8.3.

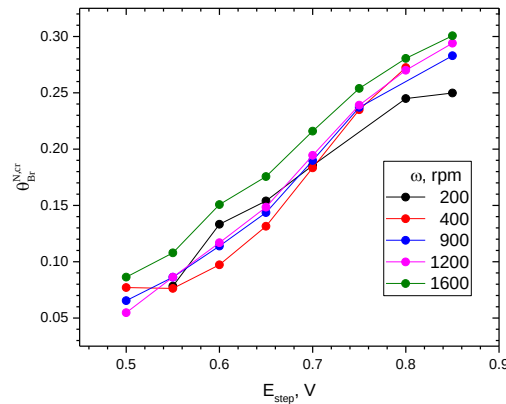


Figure 8.8. Plot of the normalized critical bromide coverage, $\theta_{Br}^{N,cr}$ for the reduction of $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ as a function of the step potential, E_{step} for different rotation rates, ω , calculated from fits of the data to Figure 8.5.

For $\theta_{Br}^N > \theta_{Br}^{N,cr}$ the current increased monotonically (see Figure 8.4) as would be expected for an increase in the numbers of bromide active sites to later reach a steady state value which corresponds to a situation where the entire surface is active, i_{ss} in that figure. This behavior has been discussed by a number of authors over the last three decades and will not be pursued here.

Particularly noteworthy is the pioneering work of Amatore, et al. who implemented a mathematical model based on the classical formalism of Smythe to predict the behavior of the current as a function of the size and number of active clusters distributed in a hexagonal pattern on top of an underlying inactive substrate. As predicted by that theory, the range of values of θ_{Br}^N over which the current changes is rather narrow, i.e. 0.2 to 0.5 (see blue line in Figure 8.4), A thorough analysis for the specific case of arrays of electrocatalytic nanoparticles dispersed on an inactive rotating disk electrode based strictly on numerical calculations using the COMSOL platform will be presented in a forthcoming communication. Further insight into this overall behavior was gained from plots of the normalized current, i.e. i/i_{lim} , as a function of the applied potential, E, for various values of θ_{Br}^N (see Figure 8.9). These curves display a sigmoidal type shape which shifts monotonically toward positive potentials for $\theta_{Br}^N > \theta_{Br}^{N,cr}$. as predicted by Amatore's formalism.

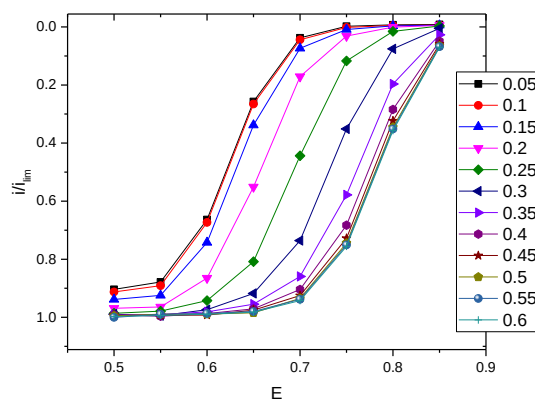


Figure 8.9. Left Panel: experimental curves for the current as a function of potential at 400 rpm. Slices of the current were taken at times corresponding to the coverages as indicated in the legend.

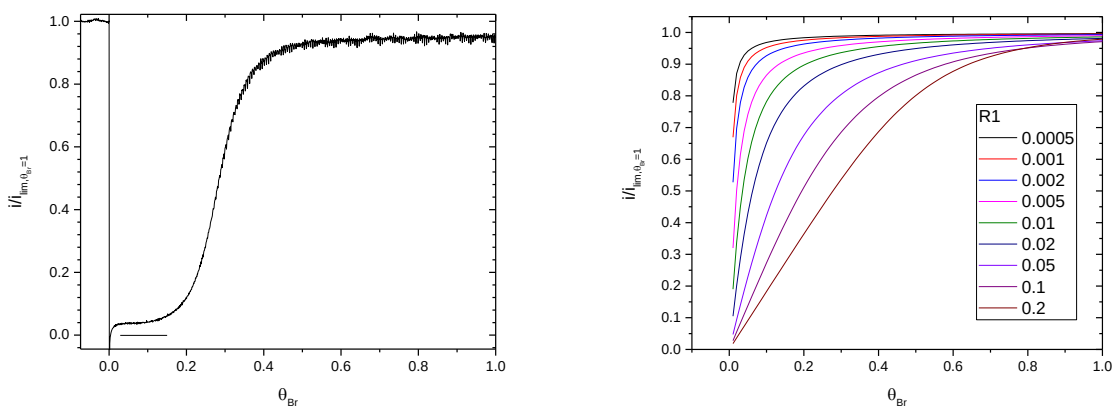


Figure 8.10. Left panel: Plot of current as a function of time, with the time axis normalized to the corresponding coverage of bromide. Right panel: simulations of various $R1$ as a function of coverage. The current axis is again normalized such that the limiting current is 1.

Conclusions

A chronoamperometric method was implemented to examine changes in the rates of 0.1 M $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ reduction in aqueous 0.1 M HClO_4 on a Au(poly) induced by the adsorption of bromide. This tactic, described in rather detail in an earlier communication, relies on measurements performed in the presence of the convective flow generated by a rotating disk electrode, RDE, whereby the coverage of bromide is determined independently using the method of Gasteiger, and the lack of interference between the adsorption and redox reaction is validated by differential reflectance spectroscopy. The results obtained showed, rather unexpectedly, that a critical coverage of bromide, $\theta_{\text{Br}}^{\text{N},cr}$, is required to elicit any detectable electrocatalytic effects, and ascribed, tentatively, to the presence of sites on the Au surface, which once filled by adsorbed bromide elicit no electrocatalytic activity for the redox process. This explanation was supported by the fact that $\theta_{\text{Br}}^{\text{N},cr}$ was found to be fairly independent of the rotation rate. In addition, $\theta_{\text{Br}}^{\text{N},cr}$ was found to decrease as the overpotential increased. This suggested that the

applied potential affects the distribution of bromide on the surface, so that the sites leading to catalysis are filled preferentially as the overpotential is increased.

Once $\theta_{Br}^{N,cr}$ is reached, however, the polarization curve shifted toward lower overpotentials as the coverage was increased as predicted by a strict application of Amatore's formalism.

**Chapter 9 Oxidation of Bromide on Platinum Electrodes in Aqueous Acidic
Solutions: Electrochemical and In Situ Spectroscopic Studies**

Publication Details

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Contributors

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Abstract

Certain aspects of the oxidation of bromide in acidic aqueous electrolytes have been investigated via a combination of rotating disk electrode (RDE) and in situ reflection absorption UV-visible spectroscopic techniques. Koutecky-Levich plots of polarization curves recorded with a Pt RDE in 5 mM KBr in 0.1M HClO₄ at rotation rates in the range 400 to 2000 rpm yielded for low overpotentials rates independent of the applied potential. This conclusion supports the reports of Conway et al. (*J. Chem. Soc., Faraday Trans.*, **1995**, 91, 283) which relied on the analysis of the results different type of experiments performed in more concentrated bromide solutions. Advantage was taken of the very high molar absorptivity of tribromide, i.e. $\epsilon(\lambda=266\text{nm}) = 4.09 \times 10^7 \text{ cm}^2\text{mol}^{-1}$, a solution phase species generated via the reaction of bromine and bromide, to monitor quantitatively its presence within the diffusion boundary layer and thus gain insights into the reaction mechanism. Numerical simulations of the electrochemical-chemical (EC) mechanism that take into account the dynamics of formation and

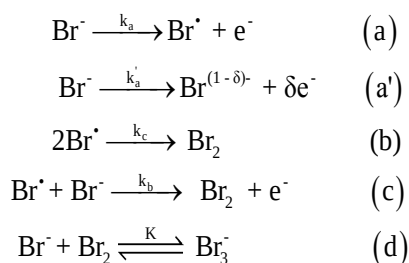
dissociation of tribromide, yielded good agreement with both the optical and electrochemical data collected in this work under strict diffusion control.

Introduction

The development and optimization of efficient and economical means of storing energy from renewable and intermittent sources, including wind and solar, will have a pronounced impact toward mitigating problems derived from the depletion of oil reserves, as well as improve the way in which the electrical grid is currently managed. Although their capital costs per cycle are higher than those associated with pumped-storage hydroelectricity, the versatility of electrochemical energy storage devices, such as batteries, fuel cells and double layer capacitors may indeed hold the key to accomplishing these goals. Capacitors display very high power densities and sub-second response times and, as such, are suitable for power quality management. Batteries, on the other hand, including those of the flow redox type, and fuel cells, exhibit much higher specific energy densities than capacitors and therefore can meet the requirements of large-scale electrical energy storage.

Whereas batteries store energy within, and, therefore, their energy and power densities are determined by the characteristics and size of its constituent electrodes, the redox active materials in a fuel cell (or redox flow battery) are stored externally; hence, their energy capacity is dictated by the size of the containers. The latter constitutes a great advantage, as in analogy to an internal combustion engine, the power and energy densities are governed, respectively, by the size of the engine and that of the fuel tank. In stark contrast, the energy of a battery of a specific chemistry can only be augmented by increasing the number of modules and thus of all its integral

components, including among others, separators and current collectors. Yet another complicating factor relates to changes in the intrinsic structure, including dimensionality, of the battery electrodes as they undergo charge and discharge, which ultimately will limit its lifetime and performance.



Scheme 9.1

A class of electrochemical devices that hold particular promise is the so-called regenerative fuel cells, whereby chemicals produced during the charging cycle are stored either as gases or liquids and subsequently used as fuels during discharge generating electricity as in conventional fuel cells. In particular, rather recent reports⁷² have indicated that the recombination of hydrogen, H₂, and bromine, in a fuel cell allows up to 90% of the chemical energy stored in the reactants to be converted to electricity. This value is much higher than that associated with state of art H₂/air fuel cells, for which the conversion can reach only about 50%. This translates into an electric-to-electric efficiency approaching 80% for H₂/Br₂ system compared to 40% for H₂ /air and 35% for most fossil-fuel-fired power generators. Not surprisingly, efforts are being made in various laboratories toward gaining a better understanding of the factors that control the overall performance of this type of devices.

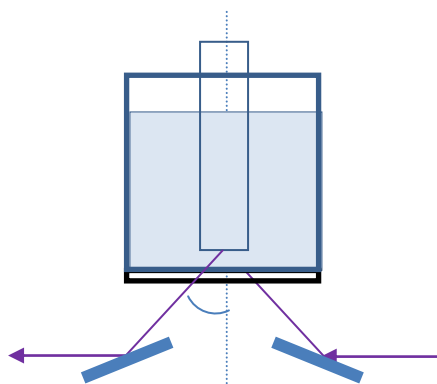
The present contribution illustrates the combined use of rotating disk electrode (RDE) and reflection absorption UV visible spectroscopic techniques for the quantitative detection of

tribromide in the diffusion boundary layer, to elucidate various aspects of the oxidation of bromide on polycrystalline Pt, Pt(poly), in 0.1 M HClO₄ aqueous solutions. Also reported herein, are mathematical simulations of the electrochemical-chemical (EC) mechanism associated with the overall oxidation of bromide to bromine and tribromide, Br₃⁻, to assess the effects of the chemical reaction kinetics under the conditions employed in the experiments (see Scheme I). Overall, the information provided in this work complements a large body of knowledge regarding the mechanism associated with the Br⁻|Br₂ redox couple in aqueous electrolytes on both Pt(poly), and carbon electrodes. In particular, Conway and coworkers⁷³, concluded that the oxidation of Br⁻ on Pt(poly) in aqueous perchloric acid solutions proceeds via a sequence of elementary steps shown in Scheme I. As specified therein, the initial step (a) involves oxidative adsorption of Br⁻ on a Pt surface site, to yield Br[·], which then generates bromine, Br₂, either via dimerization (b) and/or by reacting electrochemically with Br⁻ in solution (c). Also considered in this scheme is the possibility that Br⁻ undergoes a partial electron transfer (a'), which would then be followed at higher potentials by further oxidation, i.e. neutralization (not shown), to form Br[·]. More recent experiments involving the CO displacement technique have shown that for the three low index planes of Pt single crystal δ appears to be very close to 1.⁷⁴⁻⁷⁶ Based on a combination of electrochemical measurements Conway et al.⁷³ suggested that under all conditions, the kinetics of Br⁻ oxidation are governed by the recombination of adsorbed bromine atoms (b), in direct analogy with the chlorine evolution reaction⁷⁷. Much earlier, Mastragostino and Gramellini⁷⁸, concluded that in the case of glassy carbon (GC), the rate controlling step for Br⁻ oxidation was step (a). As is well known⁷⁹, at relatively high concentrations Br⁻ reacts with Br₂ to form solution phase tribromide anion, Br₃⁻ (step d in Scheme 9.1), a species that can undergo

reactions parallel to those of Br^- and, as such, should be included in the analysis of the $\text{Br}_2|\text{Br}^-$ electrode in a regenerative fuel cell environment.

Experimental

Electrochemistry - All experiments were carried out at room temperature (290 K) in Ar (UHP, 99.999%, Airgas)-purged aqueous 0.1 M HClO_4 (70%, Ultrapure, EMD) solutions in ultrapure water (18.3 $\text{M}\Omega\text{ cm}$, EASYpureUV system, Barnstead) either pure or containing KBr solutions (Fisher Scientific, IR grade) at concentrations ranging from 0.1 to 10 mM. The same all-Teflon cylindrical cell described in a previous communication⁸⁰ was employed for these studies. A Pt wire in the form of a coil inserted in a glass tube with a glass frit at its end, and a homemade reversible hydrogen in 0.1 M HClO_4 (RHE) were used as counter and reference electrodes respectively, and inserted into individual vertical holes drilled through the cap of the cell to connect to its main body. The actual volume of electrolyte used in these experiments was about 15 mL.



Scheme 9.2

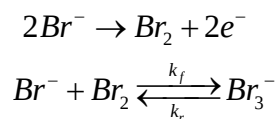
Rotating disk electrode (RDE) measurements were performed using the Pt disk of a Pt(ring)|Pt(disk) electrode (Area = 0.164 cm^2 , Pine Instruments, Model AFE7R8PTPT) and a

commercial rotator (Pine Instruments, Model MSRX). Except where otherwise noted, potential control of the RDE was achieved using an Autolab potentiostat (PGSTAT302N) connected to a personal computer.

Spectroscopy - In-situ reflection-absorption UV-visible spectra were acquired using a Cary 60 (Agilent Technologies) using a special optical attachment incorporating two UV flat mirrors to direct the light upward at an angle $\Theta = 45^\circ$ toward the surface of the RDE and then to the detector following reflection (see Scheme II). Experiments aimed at correlating bromide oxidation with Pt surface oxidation were performed under normal incidence using a laser employing an optical system and data acquisition strategies described in detail elsewhere⁸⁰. Data are displayed in the form $A(E) - A(E_{\text{ref}})$, where E and E_{ref} represent an arbitrary and reference potentials, respectively, where the latter was set at 0.05 V vs RHE, a value at which no tribromide is formed and also the amount of adsorbed bromine may be regarded as negligible.

Results and Discussion

Theoretical Considerations – According to models proposed in the literature⁸¹, the electrochemical oxidation of bromide to bromine in aqueous electrolytes proceeds via an electrochemical-chemical (EC) mechanism shown in Scheme 9.3 below.



Scheme 9.3

The primary objective of the work presented in this section was to determine:

- i. the extent to which the chemical reaction following the electron transfer process, i.e. formation of tribromide, will affect the values of the diffusion limited currents,

- ii. the solution phase absorbance associated with the presence of solution phase tribromide as measured with a RDE.

A quantitative analysis of this complex problem requires finding solutions of a system of three non-linear coupled differential equations that govern convective mass transport for each of the species involved, including the source/sink terms associated with the homogeneous chemical reaction. The specific system of equations and boundary conditions employed in this study are given in Table 9.1 both in dimensioned as well as in dimensionless form, where, for simplicity, the charge of the various species was omitted, and the superscripts 0 and ∞ for the concentration of the various species refer to values at the electrode surface and in the bulk of the solution, respectively. Also defined in that table are the dimensionless variables and dimensionless parameters selected for this study. All simulations were carried out within the Levich formalism which ignores, among other factors, components to the fluid velocity other than that along an axis parallel to main axis of rotation, using Comsol 4.3, assuming a one-dimensional domain to represent the solution phase and a no flux boundary condition for all species at the end opposite to the electrode surface. The domain length, intermediate between the thickness of the hydrodynamic and diffusion boundary layers, as well as the mesh was adjusted until no further changes in the flux at the surface and the shape of the concentration profiles were found. A few validation tests were performed to assess accuracy, including a different EC mechanism examined earlier in our laboratory involving nitrite oxidation in aqueous acidic electrolytes⁸², yielding virtually identical results.

Table 9.1. Steady State Governing Differential Equations and Boundary Conditions for the EC Mechanism in Scheme 2 in One-Dimension at the Diffusion Limited Current both in Dimensioned and Dimensionless Form.

	Dimensioned	Dimensionless
Governing Equations	$D_j \frac{d^2 c_j}{dx^2} = u_x \frac{dc_j}{dx} + r_j$ $u_x = 0.51023 \frac{\omega^{3/2} x^2}{\nu^{1/2}}$ $r_{Br} = r_{Br_2} = k_f c_{Br} c_{Br_2} - k_r c_{Br_3}$ $r_{Br_3} = -r_{Br}$	$\frac{1}{Sc_i} \frac{d^2 C_j}{d\gamma^2} = H(\gamma) \frac{dC_j}{d\gamma} + R_i$ $H(\gamma) = 0.51023 \gamma^2$ $R_{Br} = R_{Br_2} = \kappa (\xi C_{Br} C_{Br_2} - C_{Br_3})$ $R_{Br_3} = -R_{Br}$
Boundary Conditions	$c_{Br}^0 = 0$ $D_{Br_2} \left. \frac{dc_{Br_2}}{dx} \right _{x=0} = -\frac{1}{2} D_{Br_2} \left. \frac{dc_{Br}}{dx} \right _{x=0}$ $D_{Br_3} \left. \frac{dc_{Br_3}}{dx} \right _{x=0} = 0$ $c_{Br}^\infty = c_{bulk}; c_{Br_2}^\infty = c_{Br_3}^\infty = 0$ $D_j \left. \frac{\partial c_j}{\partial x} \right _{x=\infty} = 0 \text{ for all } j$	$C_{Br}^0 = 0$ $\left. \frac{1}{Sc_{Br_2}} \frac{dC_{Br_2}}{d\gamma} \right _{\gamma=0} = -\left. \frac{1}{2Sc_{Br}} \frac{dC_{Br}}{d\gamma} \right _{\gamma=0}$ $\left. \frac{1}{Sc_{Br_3}} \frac{dC_{Br_3}}{d\gamma} \right _{\gamma=0} = 0$ $C_{Br}^\infty = 1; C_{Br_2}^\infty = C_{Br_3}^\infty = 0$ $\left. \frac{\partial C_j}{\partial \gamma} \right _{\gamma=\infty} = 0$
Current Density	$i_{Lev} = 0.621 F c_{Br}^\infty D_{Br}^{2/3} \omega^{1/2} \nu^{-1/6}$ $i = F D_{Br} \left. \frac{dc_{Br}}{dx} \right _{x=0}$	$I_{Lev} = 0.621 S c_{Br}^{-2/3}$ $I = \frac{S c_{Br}^{1/3}}{0.621} \left. \frac{dC_{Br}}{d\gamma} \right _{\gamma=0}$
Absorbance	$A = \frac{2\epsilon_{Br_3}}{\cos \theta} \int_0^\infty c_{Br_3} dx$	$A = \frac{2E}{\cos \theta} \int_0^\infty C_{Br_3} d\gamma$
Dimensionless Variables and Parameters	$\gamma = (\omega / \nu)^{1/2} x; C_j = c_j / c_{Br}^\infty; \xi = c_{Br}^\infty K; \kappa = \frac{k_f}{K\omega}; Sc_i = \nu / D_i; E = \frac{\epsilon_{Br_3}}{K} (\nu / \omega)^{1/2}$	

Shown in Figure 9.1 is a plot of the dimensionless steady state limiting current,

$$I = \frac{S c_{Br}^{1/3}}{0.621} \left. \frac{dC_{Br}}{d\gamma} \right|_{\gamma=0}, \text{ as a function of } \log_{10} \xi, \text{ where } \xi = c_{Br}^\infty K \text{ is the dimensionless equilibrium}$$

constant for the homogeneous reaction associated with the EC mechanism in Scheme 9.3, for

which the reported value of $K = 1.7 \times 10^4 \text{ cm}^3 \text{ mol}^{-1}$.⁷⁹ As indicated, I is virtually constant up to $\log_{10}\xi$ ca. 2.5, i.e. within the range $0.1 \leq c_{Br}^\infty \leq 10 \text{ mM}$ and $400 \leq \omega \leq 2000 \text{ rpm}$ used in our experiments, which are specified by the shaded area in the figure. In other words, I becomes independent of the kinetics of the homogeneous reaction. It should be stressed that even at a concentration of ca. 0.5 M, for which no diffusion limited current could be observed experimentally⁸³, the drop in I due to the homogeneous reaction amounts to only ca. 4%. This implies that reliable values of the diffusion coefficient of bromide can indeed be obtained from a conventional Levich plot⁸⁴ under the conditions employed in our measurements.

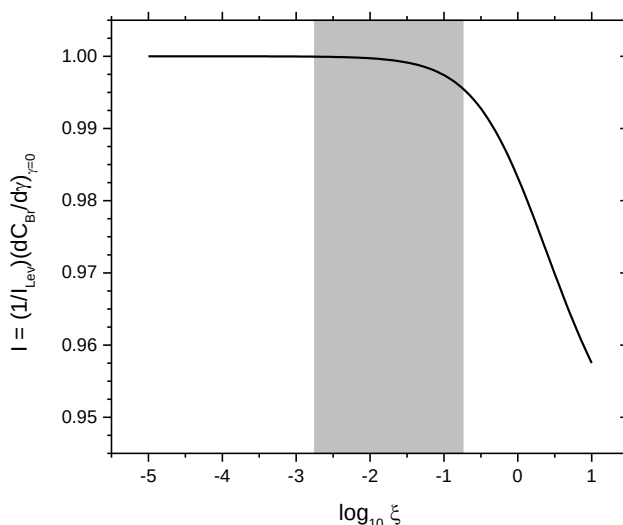


Figure 9.1. Plot of the dimensionless steady state limiting current, $I = \frac{Sc_{Br}^{1/3}}{0.621} \left. \frac{dC_{Br}}{d\gamma} \right|_{\gamma=0}$, as a function of $\log_{10}\xi$, where $\xi = c_{Br}^\infty K$ is the dimensionless equilibrium constant for the homogeneous reaction associated with the EC mechanism in Scheme 9.3. The shaded area represents the range of values of κ involved in the experimental measurements reported herein.

Additional calculations aimed at examining the effects of the dimensionless rate constant

$\kappa = \frac{k_f}{K\omega}$ on I (see Figure 9.2), where $k_f = 1.51 \times 10^{12} \text{ cm}^3 (\text{mol s})^{-1}$,⁸⁵ revealed the presence of a

rather small dip clearly observed for $\xi = c_{Br}^\infty K = 1.7 \times 10^{-2}$, 8.5×10^{-2} and 0.17 , or $c_{Br}^\infty = 1, 5$,

and 10 mM , respectively, which is outside the range of values accessed experimentally, shown

by the shaded area in this figure. This dip is most likely the result of the interplay between the

rate of the homogeneous reaction and convective mass transport.

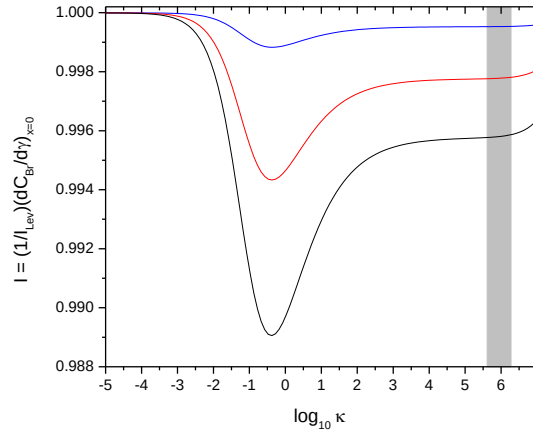


Figure 9.2. Plot of the dimensionless current density, $I = \frac{Sc_{Br}^{1/3}}{0.621} \frac{dC_{Br}}{d\gamma} \Big|_{\gamma=0}$ as a function of

the dimensionless rate constant $\kappa = \frac{k_f}{K\omega}$ for selected values of $\xi = c_{Br}^\infty K$, namely, 1.7×10^{-2} (blue), 8.5×10^{-2} (red) and 0.17 (black), which correspond to values of the bulk concentration of bromide, $c_{Br}^\infty = 1, 5$, and 10 mM . The shaded area represents the range of values of κ involved in the experimental measurements reported herein.

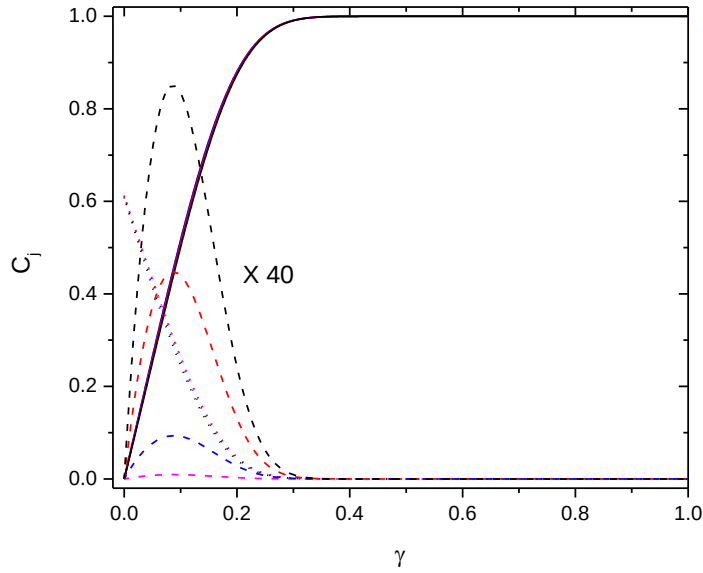


Figure 9.3. Plots for the dimensionless concentration C_j , $j = \text{Br}^-$ (solid line), Br_2 (dotted) Br_3^- (dashed, X 40) for various bulk concentrations of bulk bromide, $c_{\text{Br}}^\infty = 0.1$ (magenta), 1 (blue), 5 (red), and 10 mM (black) as a function of the dimensionless distance $\gamma = (\omega/\nu)^{1/2}x$.

The dimensionless concentration profiles, C_j vs $\gamma = (\omega/\nu)^{1/2}x$, for all the species involved, namely, $j = \text{Br}^-$ (solid line), Br_2 (dotted), and Br_3^- (dashed) are displayed in Figure 9.3, for the same values of c_{Br}^∞ as in Figure 9.2, i.e. 1 (blue), 5 (red), and 10 mM (black). As evident from these data, the profiles within the diffusion boundary layer both for Br^- and Br_2 vary monotonically, whereas that for Br_3^- displays a peak like feature. This information can be used to calculate the absorbance of the diffusion boundary layer by simply integrating the profiles along the beam trajectory (see Figure 9.4).

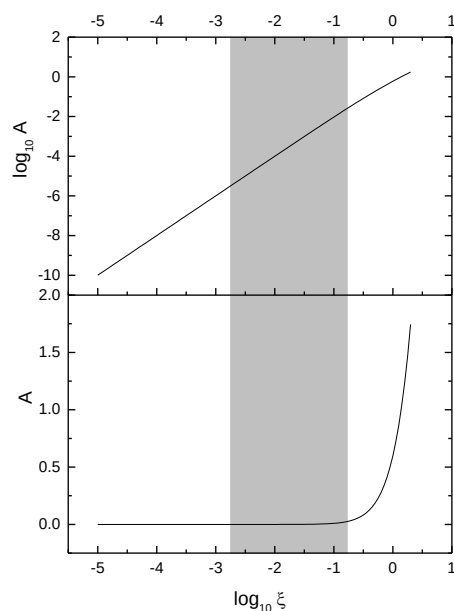


Figure 9.4. Plot of the absorbance, due to solution phase tribromide, $A = \frac{2\varepsilon_{Br_3}}{\sin \theta} \int_0^\infty c_{Br_3} dx$ (lower panel), and $\log_{10} A$ (upper panel), as a function of $\log_{10} \xi$ $\xi = c_{Br}^\infty K$.

Attempts were made to compare the results of these simulations with both theoretical and experimental work reported earlier work in the literature⁸¹. Unfortunately, in the first case, the concentrations of the various species in solutions were found to be inconsistent with the known values of the equilibrium constant for the chemical process that followed electron transfer, whereas in the second, the concentration of bromide was too high for the system to achieve a diffusion-controlled current for the oxidation⁸³. Lastly, the actual values of the rate constants for the chemical reaction involving tribromide were not known at the time.

Experimental Results - Shown in Figure 9.5 are a series of polarization curves recorded under steady state conditions for the oxidation of bromide on the Pt disk of a Pt|Pt rotating ring disk electrode, RRDE, collected in a solution 5 mM KBr in 0.1M HClO₄ at rotation rates, ω , in the range $400 \leq \omega \leq 2000$ rpm, as specified in the caption. Each of the scattered symbols

represents the disk current, i_{disk} , recorded following a potential step from $E_{\text{disk}} = 0.05$ V, at which no reaction ensues, to a specified value in the range $1.0 \leq E_{\text{disk}} \leq 1.5$ V vs RHE, and waiting for a very brief period of time, on the order of seconds, for i_{disk} to attain a constant value. As clearly seen, i_{disk} increases at about $E_{\text{disk}} = 1.15$ V to reach a limiting value, $i_{\text{disk}}^{\text{lim}}$, at E_{disk} ca. 1.4 V.

Evidence that $i_{\text{disk}}^{\text{lim}}$ is governed by mass transport to the disk was obtained based on the linear character of the $i_{\text{disk}}^{\text{lim}}$ vs. $\sqrt{\omega}$ extracted from these data (see Figure 9.6), as well as that recorded for three additional KBr solutions in 0.1M HClO₄ of concentrations, $[\text{Br}^-]$, in the range $0.1 \text{ mM} \leq [\text{Br}^-] \leq 0.01 \text{ M}$ with the same Pt RDE electrode, as specified in the caption Figure 9.6. The open and half open circles in this figure represent data collected in two independent experiments.

Diffusion coefficients for bromide, D_{Br^-} , extracted from $i_{\text{disk}}^{\text{lim}}$ vs $\omega^{1/2}$ at constant $[\text{Br}^-]$, and $i_{\text{disk}}^{\text{lim}}$ vs $[\text{Br}^-]$ at constant ω , yielded values within 10% on one another. For the theoretical simulations $D_{\text{Br}^-} = 1.58 \times 10^{-5} \text{ cm}^2/\text{s}$, $D_{\text{Br}_2} = 1.18 \times 10^{-5} \text{ cm}^2/\text{s}$, and $D_{\text{Br}_3^-} = 1.00 \times 10^{-5} \text{ cm}^2/\text{s}$.

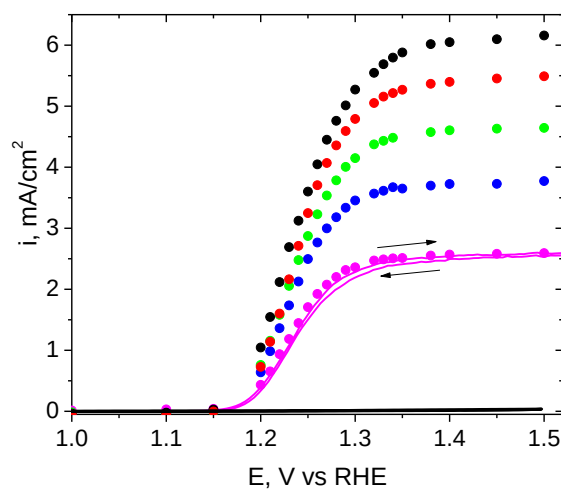


Figure 9.5. Steady state polarization curves (scattered symbols) for the oxidation of bromide collected with the Pt(poly) disk of a Pt|Pt rotating ring disk electrode, RRDE, in a solution 5 mM KBr in 0.1M HClO₄, at various rotation rates, ω , i.e. 400 (magenta), 800 (blue),

1200 (green), 1600 (red) and 2000 rpm (black). Corresponding dynamic polarization curves recorded for the same Pt electrode at a scan rate $v = 10$ mV/s and $\omega = 400$ rpm in the same electrolyte and in 0.1 M HClO₄ devoid of bromide are shown in magenta and black solid lines, respectively, where the arrows in the former indicate the direction of the scan.

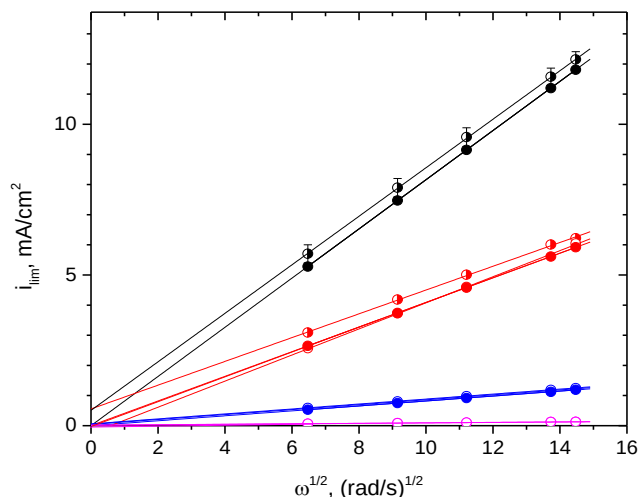


Figure 9.6. Plots of i_{lim} vs. $\sqrt{\omega}$ for the oxidation of bromide recorded with a Pt RDE electrode in 0.1M HClO₄ containing 10 mM (black), 5 mM (red), 1 mM (blue), and 0.1 mM KBr (magenta). The open and half open circles represent data collected in two independent experiments. The solid symbols are the results of the theoretical simulations assuming $D_{Br^-} = 1.58 \times 10^{-5}$ cm²/s (see text). The straight lines represent the best linear fits to the experimental data. The error bars represent the average of four independent runs.

Koutecky-Levich plots, i.e. $1/i_{disk}$ vs $1/\omega^{1/2}$, based on the results in Figure 9.5, shown in steps of 10 mV in Figure 9.7, were found to be linear with non-zero intercepts over the region $1.21 \leq E_{disk} \leq 1.3$ V. Kinetic currents, i_k , and rate constants, k_f , based on these data, yielded values virtually independent of the applied potential over the range $1.2 \leq E_{disk} \leq 1.24$ V (see Figure 9.8) affording strong evidence that the rate determining step for such low overpotentials is not potential dependent. This conclusion is in agreement with that reached by Conway et al.⁷³ who, as mentioned in the Introduction, suggested that the rate determining step was the recombination of adsorbed bromine on the surface.

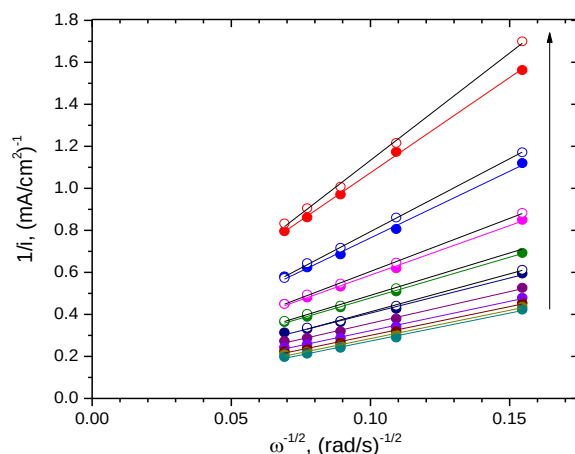


Figure 9.7. Koutecky-Levich plots for the oxidation of bromide at a rotating Pt disk electrode based on the results shown in Figure 9.5. The solid and open circles represent data collected in two fully independent runs and the arrow points in the direction of decreasing potentials from 1.3 to 1.21 V in steps of 0.01 V.

The unusually high molar absorptivity of tribromide, i.e. $\epsilon(\lambda=266\text{nm})^{86} = 4.09 \times 10^7 \text{ cm}^2\text{mol}^{-1}$, makes it possible to monitor its presence in the diffusion boundary layer of the RDE. Shown in Figure 9.9 are spectra collected in the region $230 \leq \lambda \leq 450 \text{ nm}$ with the system described in the experimental section with the RDE rotating at rates in the range, $400 \leq \omega \leq 2000 \text{ rpm}$. Based on the theory developed earlier in our research group⁸⁷, an increase in ω elicits an overall decrease in the absorbance. The time required for acquiring spectral data is rather long, on the order of 15 s, and several of such measurements were found to be required to obtain an acceptable signal to noise ratio. During this period, tribromide accumulates in the bulk solution and, thus, somewhat involved corrections must be introduced to the absorbance to correlate quantitatively the optical response and ω . A better approach, implemented in our work, involves stepping the potential while monitoring the absorbance at 266 nm and extrapolating the optical signal to the time immediately following application of the step. The results obtained (see

open circles in Figure 9.10) yielded a linear dependence between the potential difference absorbance, $A(1.4\text{V}) - A(0.05\text{ V})$, and $\omega^{-1/2}$ for the same values of $[\text{Br}^-]$ used in the experiments in Figure 9.6, where $A(1.4\text{ V})$ represents the absorbance while the electrode was polarized at 1.4 V, a potential at which the oxidation of bromide proceed under diffusion limited conditions (see Figure 9.10), and $A(0.05\text{ V})$ the absorbance at 0.05 V, where the contribution to the spectra is derived mostly from bromide accumulated in solution while the electrode was poised at 1.4 V. In this fashion, the difference in the absorbance can be attributed to tribromide present in the diffusion boundary layer. The straight lines in this figure are the best fits to the experimental data and the errors bars for data collected in a solution 10 mM KBr represent the average of four totally independent measurements. Also provided in this figure in solid symbols are the results of the theoretical simulations to be presented in the previous section. As clearly indicated, the intercepts for all three values of $[\text{Br}^-]$ appear to cluster at a value close to $A(1.4\text{V}) - A(0.05\text{ V})$ ca. 0.01. This phenomenon is caused by the change in the reflectance of the surface induced both by bromide adsorption and by the formation of Pt oxide. In fact, $A(1.4\text{V}) - A(0.05\text{ V})$ data collected in the absence of bromide in solution yielded a value of ca. 0.015 (see Figure 9.11). Additional evidence for this explanation is afforded by the results of similar experiments performed with a Pt RDE in 5 mM KBr in 0.1M HClO_4 (see Figure 9.11, open circles) at $\lambda = 266$ (blue), 390 (red), 633 nm (black) . In particular, for $\lambda = 390$ nm, the wavelength at which Br_2 exhibits maximum absorption in this spectral region and thus, one would expect for Br_2 to be detected, the magnitude of $A(1.4\text{V}) - A(0.05\text{ V})$ decreases because its corresponding molar absorptivity, $\epsilon(390\text{nm})^{86} = 1.75 \times 10^5 \text{ cm}^2 \text{ mol}^{-1}$ is far smaller than that of tribromide at 266 nm. Lastly, at 633 nm where none of species in solution absorb, $A(1.4\text{V}) - A(0.05\text{ V})$ becomes independent of ω . The dependence of the intercept on λ reflects differences in the absorbance

due to the oxide, which is larger as the energy of the incident beam increases, as clearly evident by data collected in neat 0.1M HClO₄ (solid circles in Figure 9.11).

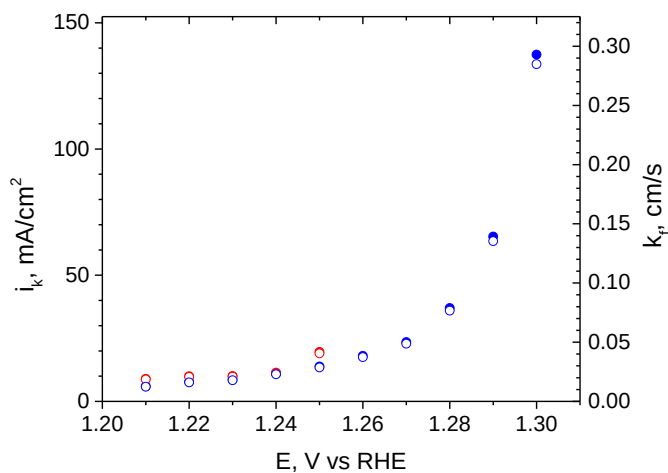


Figure 9.8. Plot of the kinetic currents, i_k , (left ordinate, solid symbols) and kinetic rate constants, k_f , (right ordinate, open symbols) based on the data in Figure 9.7.

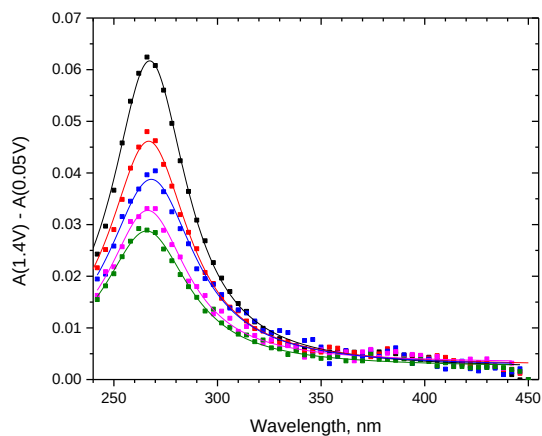


Figure 9.9. Potential difference reflection absorption spectra, $A(1.40\text{ V}) - A(0.05\text{ V})$, recorded at an angle of incidence of ca. 45° from a rotating Pt disk electrode in a 0.01M KBr solution in 0.1M HClO₄ at various values of $\omega = 400$ (black), 800 (red), 1200 (blue), 1600 (magenta) and 2000 rpm (olive). The prominent peak centered at $\lambda = 266\text{ nm}$ is characteristic of tribromide ion. Each of the spectra represents the average of 15 acquisitions (see text). The smooth line is the best fitting of the data.

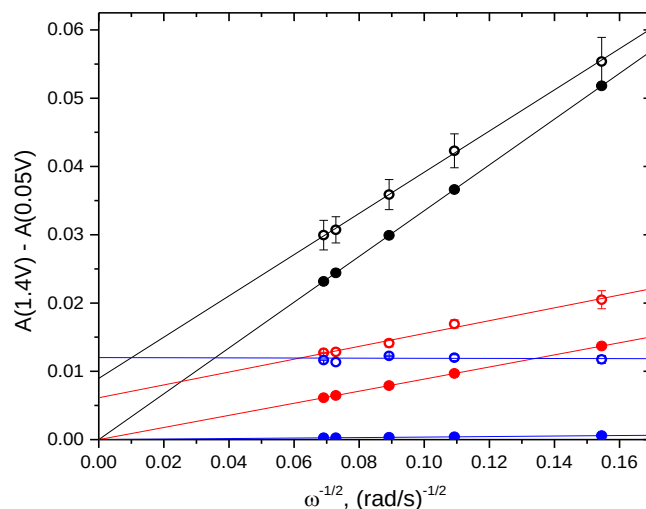


Figure 9.10. Potential difference absorbance, $A(1.4V) - A(0.05V)$ vs $\omega^{-1/2}$, collected at $\lambda = 266$ nm from a Pt RDE in solutions 1 (blue), 5 (red) and 10 mM (black) KBr in 0.1M $HClO_4$ and at an angle of incidence, θ ca. 45° (open symbols) during acquisition of the electrochemical data in Figure 9.6. The solid symbols are the results of the theoretical simulations (see text). The straight lines represent the best fits to the experimental data. The error bars represent the average of four or two independent measurements.

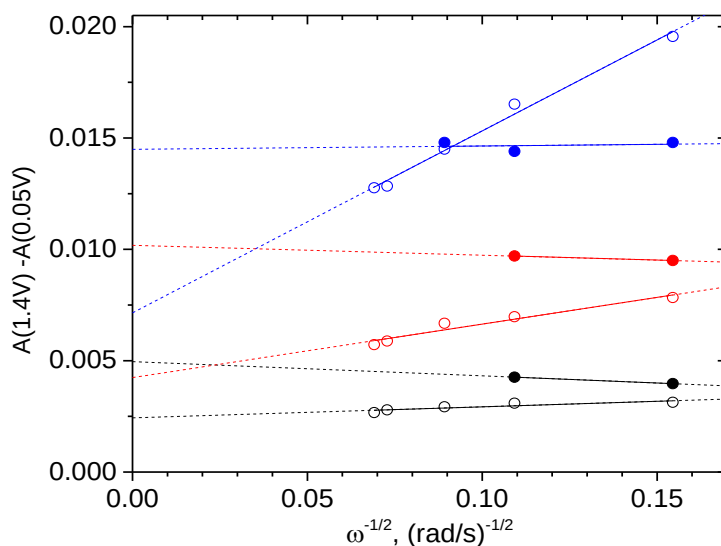


Figure 9.11. Potential difference absorbance, $A(1.4V) - A(0.05V)$ vs $\omega^{-1/2}$ collected from a Pt RDE in a 5 mM KBr in 0.1M $HClO_4$ (open circles) at $\lambda = 266$ (blue), 390 (red), 633 nm (black). The data in solid circles was acquired in neat 0.1M $HClO_4$ under otherwise identical conditions. The smooth lines are the best fit of the data.

The concerted use of electrochemistry under forced convection and differential reflectance makes it possible to monitor the state of oxidation of the Pt surface during the oxidation of bromide under quasi steady state conditions. This is illustrated in Figure 9.12 which compares dynamic polarization curves recorded in this case using instrumentation specified in an earlier communication⁸⁰, while scanning the potential toward positive values ($v = 10$ mV/s, upper panel) with $-\Delta R/R$ (lower panel) curves collected simultaneously in solutions of KBr in 0.1 M $HClO_4$ at concentrations of 0.1 mM (magenta), 1 mM (blue), 5 mM (red) and 10 mM KBr (black) at $\omega = 900$ rpm. Also shown therein are the results of the same experiments in the neat electrolyte (olive), where the smooth solid lines represent best fits to the data in the range $0.6 \leq V \leq 1.4$ V.

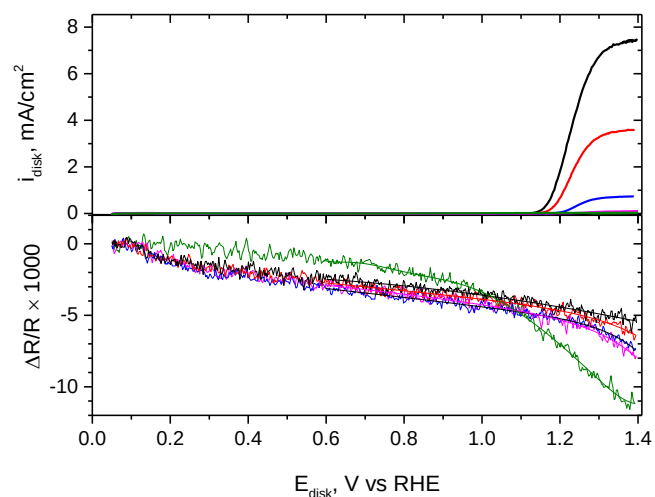


Figure 9.12. Upper panel: Dynamic polarization ($v = 10$ mV/s) curve collected with the Pt disk of the Pt | Pt RRDE in 0.1M HClO_4 containing 0 mM KBr(olive), 0.1 mM KBr(magenta), 1 mM KBr(blue), 5 mM KBr(red) and 10mM KBr(black) , $\omega = 900$ rpm. Lower panel: Corresponding differential reflectance collected simultaneously while scanning the potential toward positive values, where the smooth solid lines represent best fits to the data in the range $0.6 \leq V \leq 1.4$ V.

Conclusions

The oxidation of solution phase bromide in strongly acidic aqueous solutions was examined via a combination of rotating disk electrode and reflection absorption UV visible spectroscopic techniques under conditions that yielded well defined s-shaped polarization curves. Koutecky-Levich analyses of the data collected in 5 mM KBr in 0.1 M HClO_4 provided evidence that for very low overpotentials, the rates of the heterogeneous process are independent of the applied potential which is in agreement with conclusions made by other workers using different

methodologies.⁴ Numerical simulations were performed to account for the dynamics of formation and dissociation of tribromide, a species generated by the reaction between bromine formed electrochemically and solution phase bromide, using values for the corresponding kinetics rate constants reported in the literature.¹⁵ The results obtained assuming $[\text{Br}^-] = 0$ at the electrode surface, indicated that:

- i. the effects associated with the following chemical reaction are too small to be detected under the conditions employed in our measurements, as was verified by the dependence of the disk limiting currents on the rotation rate observed experimentally, and
- ii. the absorbance due to tribromide present in the diffusion boundary layer is in good agreement with values measured experimentally provided proper account is made of the effect of the oxide layer on the optical measurements.

Chapter 10 New Advances in Ohmic Microscopy

Contributors

Zhange Feng – Experimental work

Nicholas S. Georgescu – Calculations and simulation

Daniel A. Scherson – Advisor

Abstract

A series of electrochemical measurements involving metal disks, ring-disks and faceted single crystals in aqueous 0.1 M H_2SO_4 have been performed to monitor differences in the electrostatic potential in the electrolyte, $\Delta\phi_{\text{sol}}$, induced by the passage of current, and thus assess the prospects of ohmic microscopy as an in situ imaging tool of electrodes in solution. Excellent quantitative agreement was found between the experimental values of $\Delta\phi_{\text{sol}}$ and those predicted by theory for current pulses several mA in magnitude and tens of μs in duration, applied to a Pt disk electrode embedded in a coplanar insulating surface, assuming a strict primary current distribution. Cyclic voltammetry measurements involving a gapless Pt-Ir ring|Au disk electrode yielded $\Delta\phi_{\text{sol}}$ versus potential, E , curves, consistent with contributions derived from each of the two electrodes assuming a uniform current distribution. Also explored were extensions of ohmic microscopy to the study of faceted Pt single crystals using microreference electrodes housed in a double barrel capillary. Data collected in voltammetric experiments in which the tip of the capillary was placed directly above and at very close distance from one of the (111) facets recorded with the entire single crystal immersed in the electrolyte, yielded $\Delta\phi_{\text{sol}}$ vs E curves displaying pronounced features believed to be characteristic of that surface. Possible strategies toward improving the spatial resolution of this emerging technique are also discussed.

Introduction

The advent of novel techniques for the fabrication of microreference, μ -ref, electrodes, and the availability of fast, highly sensitive commercial voltmeters have opened new prospects for the further development of ohmic microscopy as a highly sensitive, and time-, and spatially-resolved probe of electrostatic potential differences in electrolytic solutions, $\Delta\phi_{\text{sol}}$.⁸⁸⁻⁹² As illustrated in previous communications,⁸⁸ ohmic microscopy can provide spatially-resolved maps of individual or composite electrodes in the electrolyte, by taking advantage of the unique character of potential-dependent capacitances and pseudocapacitances of different metal-electrolyte interfaces. Moreover, analyses of data collected during voltammetric cycles of a flat Pt disk embedded in an insulating coplanar surface in aqueous H_2SO_4 solutions have shown that the magnitude of $\Delta\phi_{\text{sol}}$ can be accounted for quantitatively,⁸⁹ based on the primary current distribution model introduced by Newman a few decades ago.⁹³ The present contribution extends ohmic microscopy to controlled current applications of relevance to neural stimulation, and the imaging of both composite and faceted single crystal electrodes.

Experimental

All measurements were performed in non-deaerated 0.1 M H_2SO_4 (Fisher Scientific, 98%) prepared with ultrapure water (UPW, 18.3 Ω cm, Barnstead UV pure system). The conductivity, κ , of this solution at $25^\circ\text{C} \pm 1$, as measured with a commercial instrument (Mettler, Toledo), was found to be 47.8 mS /cm, which compares well with that reported in the literature, i.e. 47.0 mS/cm.⁹⁴ Details regarding the preparation of microreference Ag|AgCl electrodes, μ -ref, have been provided in a previous communication.⁹² Double-barrel capillaries (Sutter instrument, O.D.1 mm, I.D. 0.75 mm, 2BF100-75-10 were pulled with a Sutter instrument (Model P80-PC)

yielding, generally, tips ca. 80 μm outer and ca. 60 μm inner diameters, respectively. Also described in prior work⁹² is the all-Teflon cell employed for experiments not involving single crystals (see Panel A, Figure 10.1). Experiments involving single crystals were performed in a conventional three compartment glass cell, using a Pt wire (0.5 mm, diameter, Alfa Aesar, 99.9%) as a counter electrode, and a reversible hydrogen electrode (RHE) in the same solution as a reference.⁹⁵

Table 10.1. Experimental and theoretical values $\Delta\phi_{\text{sol}}$ obtained in two different trials following application of two current pulses of different amplitude to a Pt disk electrode embedded in an insulating surface assuming a primary and uniform current distribution.

Current, mA			$\Delta\phi_{\text{sol}}$, mV (1,3 mm)	Error, %	$\Delta\phi_{\text{sol}}$, mV (1,2 mm)	Error, %
5	Trial	1	84.9	1.5	63.3	5.0
		2	83.6		60.2	
	Theory	Primary	83.4	1.0*	60.0	2.9
		Uniform	88.6	4.9*	64.5	7.4
2.4	Trial	1	39.7	2.3	30.9	5.7
		2	38.8		29.2	
	Theory	Primary	40.0	1.9*	28.9	4.0
		Uniform	42.5	7.6*	31.0	2.4

* All errors were calculated using the expression $\left\{ \left(\Delta\phi_{\text{sol}}^1 + \Delta\phi_{\text{sol}}^2 \right) / 2 - \Delta\phi_{\text{sol}}^{\text{th}} \right\} / \Delta\phi_{\text{sol}}^{\text{th}}$, where $\Delta\phi_{\text{sol}}^{\text{th}}$ refers to either the primary, or a *fictitious* uniform current distribution determined theoretically.

A commercial Pt disk electrode (Basi MF-2013, 0.16 cm in diameter) was employed for measurements involving current pulses. The Au-disk Pt-Ir ring electrode was constructed by filling the orifice of a Pt:Ir (95:5) electron microscope aperture (Ted Pella, Inc.) with molten Au. The resulting piece was then cast in epoxy and subsequently polished to expose a Au disk 1.2 mm in diameter surrounded by a Pt-Ir ring 2 mm outside diameter (see Panel B, Figure 10.1). Cyclic voltammetry curves were recorded in the range -0.2 to 1.4V at a scan rate of 10 V/s, while

acquiring $\Delta\phi_{\text{sol}}$ vs t data from either two single barrel, or a double barrel (175 μm edge to edge distance μ -ref electrodes via the 10X or the 100X differential amplifier (vide infra).

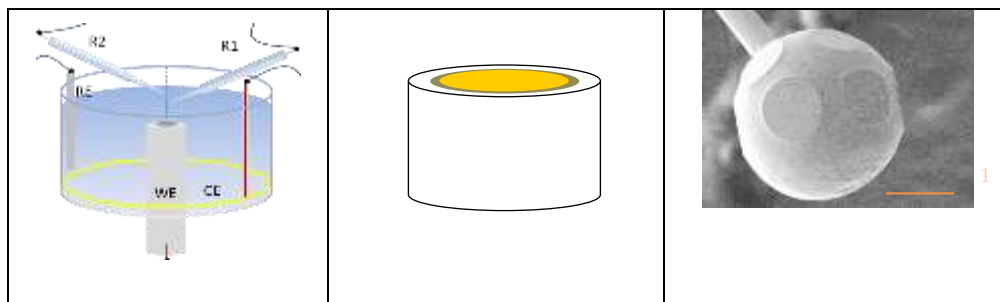


Figure 10.1. **Panel A.** Schematic diagram of the cell in which the ohmic microscopy experiments involving for current pulse and ring-disk electrodes were carried out. WE, working electrode; CE, counter electrode; RE, commercial reference electrode; R1 and R2, μ -ref electrodes. The vertical red line represents a glass insulated Cu wire that connects the Au ring counter electrode to the outside. **Panel B.** Schematic diagram of the Au (yellow) disk|Pt-Ir ring (grey) electrode. **Panel C.** Scanning electron micrograph (5.0 kV, $\times 250$) of a faceted Pt single crystal (see text for details).

Facetted single crystal Pt electrodes were prepared by melting the tip of a thin Pt wire (Alfa Aesar, 0.0508 mm diameter, 99.99%) using a hydrogen flame, and then annealed in a methane flame continuously for 24 to 48 h. Scanning electron micrographs of these electrodes were obtained with an FEI Helios Nanolab 650 after the measurements were completed (see Panel C, Figure 10.1). For these experiments, the tip of the two-barrel capillary was first carefully placed in contact with Pt facet surface, and then pulled back ca. 10 μm , at which point a series of cyclic voltammetric curves were recorded at 10 V/s while acquiring $\Delta\phi_{\text{sol}}$ vs t data with a digital multimeter (Keithley 3706 System Switch/Multimeter) through a 100X battery powered differential amplifier (Texas Instrument, INA131) using the Labview platform.⁹²

Potential and current protocols were applied to the working electrodes using a potentiostat (Autolab PGSTAT302N, Metrohm). In the case of current pulses, a voltage function generator (Agilent 33220A) was used to input the desired waveform to the potentiostat operating

under the ultrahigh speed chronopotentiometry mode. A high quality oscilloscope (Agilent, DSO-X 2014A) was used to record $\Delta\phi_{\text{sol}}$ vs t data through a 10X battery powered differential amplifier (Texas Instrument, Ina141, $G = 10$), as well as other inputs and outputs into and from the potentiostat, which required no additional amplification. It is important to note that small differences were found between the potentials of any two μ -ref electrodes. These, however, were virtually unchanged during the experiments, and their values were subtracted from the reported $\Delta\phi_{\text{sol}}$.

Theoretical

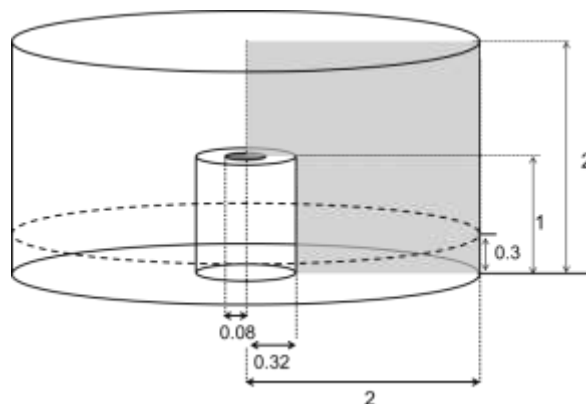


Figure 10.2. Schematic diagram and associated dimensions of the cell (in cm) used for the current pulse experiments, where the shadowed area represents the domain involved in the theoretical simulations, and the dotted line, the counter electrode loop.

Numerical solutions of Laplace's equation, $\nabla^2\Phi = 0$, within the axisymmetric domain specified by the shaded area in Figure 10.2, were obtained using COMSOL Multiphysics v5.2. The dimensions specified therein are those of the actual electrochemical cell in which the pulse experiments were performed. Except for the working electrode and the point corresponding to the location of the counter electrode, which was set at a potential of zero, all boundaries were

assumed to be insulating. Hardly any changes in the equipotential profiles in areas of relevance to our work were found by replacing the single point counter electrode by a more realistic circle.

Results

A. Current Pulses

We seek to demonstrate in this section, that the application of current pulses of a few mA lasting few tens of μs to a disk electrode in the cell specified in Figure 10.2, elicits $\Delta\phi_{\text{sol}}$ values in quantitative agreement with those predicted by Newman's theory. This type of pulses is commonly used in neural stimulation⁹⁶ to depolarize the membrane and thus induce an action potential that propagates along the nerve. For simplicity, the notation $(X,Y)_z$ will denote the position of the tips of the R1(X) and R2(Y), μ -ref electrodes in mm, and z the position of the axis normal to the surface, $z = c$ (center of the disk), and ie and oe , the inner and outer edges of the ring), respectively.

Shown in blue in the top panel, Figure 10.3 (right ordinate) is a plot of the temporal response of $\Delta\phi_{\text{sol}}$ measured across $(1, 3)_c$, i.e. (see insert in this panel) along an axis normal to the surface passing through the center of the disk, upon application of a current pulse 5 mA in amplitude and 50 μs in duration (left ordinate, black curve). As indicated, $\Delta\phi_{\text{sol}}$ observed during the pulse was 84.9 mV (see Trial 1, Table I). A subsequent experiments in which the pulse amplitude was reduced to 2.4 mA yielded $\Delta\phi_{\text{sol}} = 39.7$ mV. The tip of R2 was then placed at 2 mm, i.e. $(1,2)_c$, and an identical set of measurements performed, the results of which are listed in Table I.

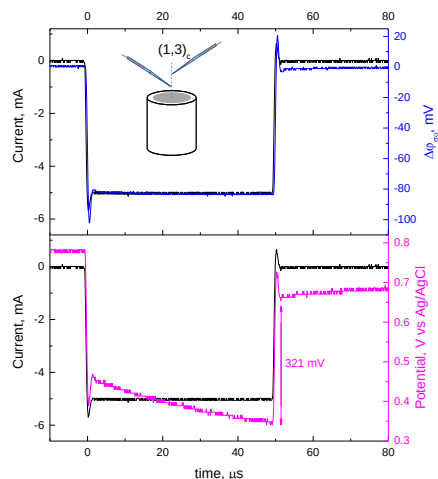


Figure 10.3. **Top Panel.** Plot of the $\Delta\phi_{sol}$ vs t (right ordinate, blue curve) recorded while applying a current pulse of 5.0 mA for 50 μ s to the Pt disk (left ordinate, black curve) with the μ -Ag/AgCl electrodes placed at (1,3)_c as shown schematically in the insert. The spikes observed immediately before and after the pulse was applied are instrumental artifacts. **Bottom Panel.** Plot of the potential of the Pt disk working electrode vs a conventional Ag|AgCl reference electrode (right ordinate, magenta curve) during application of a similar current step (left ordinate, black curve).

In a second fully independent trial involving freshly prepared μ -ref electrodes and the same precise set of measurements (Trial 2 in Table 1), the data collected for (1,3)_c yielded for currents pulses of 2.4 and 5 mA, $\Delta\phi_{sol} = 38.8$ and 83.6 mV, respectively, which are within ca. 1.5 and 2.3 % of those obtained in Trial 1 (see Table 1). For (1,2)_c, however, the corresponding errors increased to 5.0 and 5.7%, due, in all likelihood, to the “crowding” of the equipotential lines close to the electrode surface, where slight differences in the position of the tips would lead to large changes in $\Delta\phi_{sol}$.

Shown in the right panel, Figure 10.4 are equipotential lines calculated using COMSOL assuming a primary current distribution, where the potential of the disk electrode vs that of the counter electrode was adjusted so as to match the magnitude of the applied current pulse, i.e. 5 mA. An expanded view of the area neighboring the electrode is displayed in the left panel in this

figure. Based on the entries in Table I, the theoretical values of $\Delta\phi_{\text{sol}}$ were within 1% and 2.9% of the mean experimental values found experimentally for the two positions of the tips, and thus a bit lower than the corresponding values for a current of 2.4 mA, 1.9 and 4.0%, respectively (see entries in Table I).

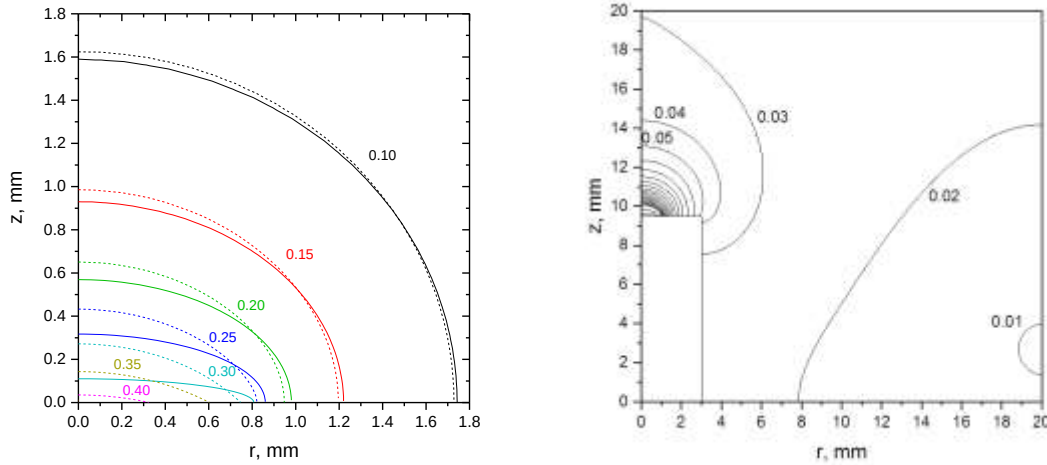


Figure 10.4. **Right Panel.** Equipotential lines obtained from COMSOL simulations involving the shaded domain in Figure 10.2 (see text for details) assuming a current of 5 mA under primary current distribution. **Left Panel.** Expanded region of the area just above the working electrode in the right panel in this figure assuming a primary (solid lines) and a *fictitious* uniform current distribution (dotted lines).

It is interesting to compare these results with those of a *fictitious* uniform current distribution obtained by solving Laplace's equation assuming a current density, i.e. $I/\pi a^2$, where a is the disk radius, to be the same across the entire electrode. The equipotentials determined for this model for $I = 5$ mA are given in dotted lines in Panel C, Figure 10.4, and the predicted values of $\Delta\phi_{\text{sol}}$ for the two positions of the tips are listed in Table I. As evidenced from the data therein, the non-uniform current distribution provides far better agreement with the experimental values than derived from the uniform current model. Unfortunately, in the case of the tips at

(1,2), the error was much higher than the predicted differences between $\Delta\phi_{\text{sol}}$ values calculated from the two models.

Lastly, we also determined, for completeness, the ohmic drop by the conventional current interruption method using the commercial Ag|AgCl reference electrode yielding for $I = 5 \text{ mA}$, a value of ca. 320 mV (see magenta line in the bottom panel, Figure 10.3), which compares very well with that predicted from Newman's theory, i.e. $\Phi_0 = \frac{I}{4\kappa a} = 327 \text{ mV}$ ($\kappa = 0.0478 \text{ S}\cdot\text{cm}^{-1}$; and $a = 0.08 \text{ cm}$). It should be stressed that Newman's model assumes the composition of the electrolyte to be invariant; hence, contrary to the claims made by Ignatowitz and Oesterschulze,⁹¹ $\Delta\phi_{\text{sol}}$ in the experiments of the type described therein, is not caused by the concentration gradient of the redox active species in solution.

B. Ring-Disk Electrode

A better assessment of the spatial resolution that can be attained with ohmic microscopy in its present stage of development was pursued using the Pt-Ir ring|Au disk assembly in Panel B, Figure 10.1 (see Experimental Section). To this end, a series of $\Delta\phi_{\text{sol}}$ vs E were recorded during voltammetric cycles using individual μ -ref electrodes placed at (0.5,1) (Left Panels, Figure 10.5) along the normal to the center, c (upper panels), to the inner edge, ie, (middle panels), and to the outer edge of the ring, oe, respectively (lower panel) . (see black curves in Figure 10.5), and the results obtained are given by the red traces in these panels. Immediately thereafter, the tip of R2 was lifted to a distance 1.5 mm from the plane of the electrode and a set of identical data were collected yielding the curves in the right panels in this figure. As evident from the data in the upper panels in this figure, a larger separation distance between the tips of the two μ -ref electrodes causes an increase not only in the overall magnitude of $\Delta\phi_{\text{sol}}$, but also in

the relative amplitudes of the voltammetric features characteristic of Pt-Ir with respect to those of Au. In other words, an increase in sensitivity is accompanied by a loss in spatial resolution. This trend was also found for the tips placed along the other two axes normal to the surface.

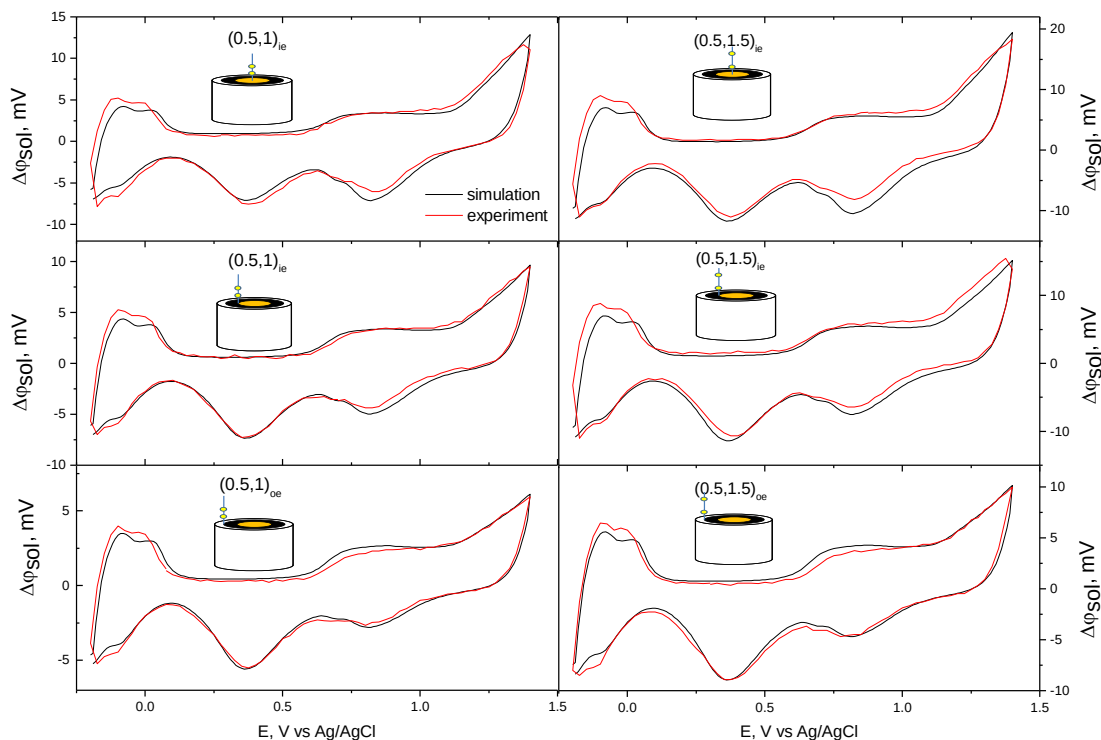


Figure 10.5. Plots of $\Delta\phi_{\text{sol}}$ vs E for the Pt-Ir ring|Au disk electrode collected with individual microreference electrodes, along three different axis normal to the electrode surface as shown where the dots represent the position of the tips (see also text for details). The red curves represent the actual recorded data and the black curves are the results of the numerical simulations.

A more quantitative measure of this effect was obtained from numerical simulations in which, for simplicity, the current density flowing through each of the electrodes was assumed to be independent of the radius, i.e. effects due to the edges were fully neglected. On this basis, ϕ_{sol} at every point in the solution was regarded as the sum of contributions derived from each electrode. Specifically, the voltammetric response of the ring was obtained by encasing the entire

Pt-Ir cylinder in epoxy, including the hollow cavity, which was later polished to expose the ring surface (see red line, Left Panel, Figure 10.6), whereas that ascribed to the Au disk in the assembly was obtained from a commercial Au disk. Statistical techniques were applied to determine the contributions of each of the electrodes to the measured voltammogram (see black curve in the left panel, Figure 10.6).

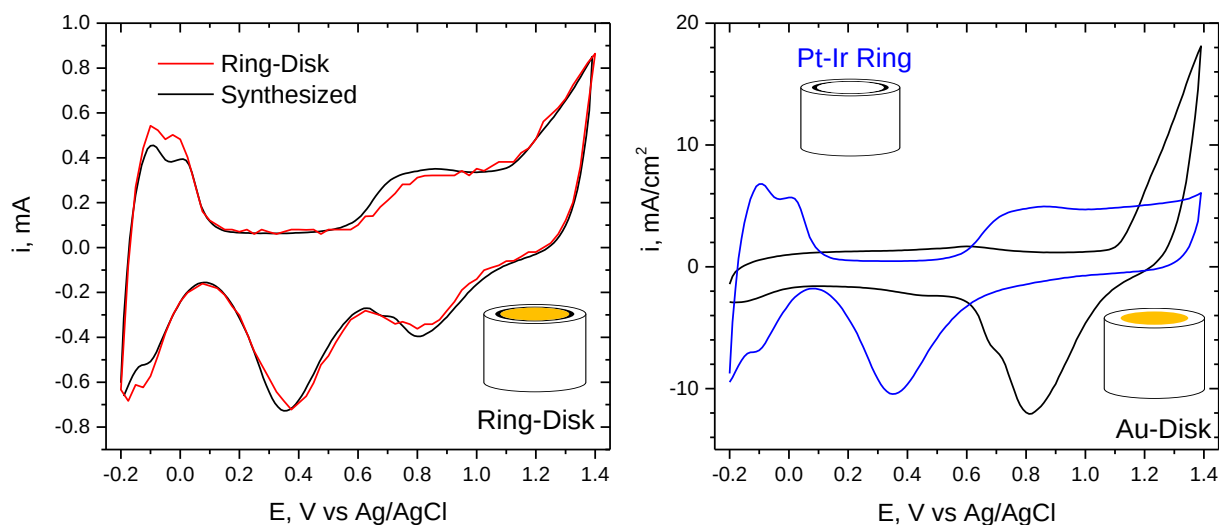


Figure 10.6. **Left Panel.** Cyclic voltammogram of the Pt-Ir|Au ring-disk assembly (red curve, $\nu = 10$ V/s), and of a best fit analysis using individual voltammetric curves recorded independently shown in the Right Panel in this figure. **Right Panel.** Individual voltammetric curves recorded at 10V/s for the Pt-Ir ring with the center cavity filled with epoxy (blue curve), and for a commercial Au disk encased in Teflon (black curve, see text for details).

An analysis of these curves was then carried out assuming a uniform current distribution yielding the curves in in Figure 10.6. As evidenced from these results the agreement between theory and experiment was indeed quite good.

In a second set of experiments, $\Delta\phi_{\text{sol}}$ vs E measurements were performed with the tip of a double-barrel capillary housing the μ -ref electrodes (see Experimental Section) placed at a

fixed distance of 50 μm from the surface while scanning the potential of the Pt-Ir ring Au disk electrode at 10 V/s for various points along its radius. The results obtained are shown in colored curves in Figure 10.7. As a cursory examination of these curves indicates, the features observed for the tip placed along the center line of the Au disk displays mostly, but not exclusively, Au-like voltammetric characteristics, which then decrease in intensity as the tip is displaced away from the center, while at the same time, the curves attain features characteristic of Pt. In fact, no Au features could be gleaned for the data collected at $X = 1.1$ mm.

These data were then analyzed assuming, as above, a uniform current through each of the electrodes, and an average position of the tips at 93.75 and 181.25 μm , which correspond to one fourth above and below the dividing wall in the dual capillary with respect to the electrode surface. Based on the results obtained (see black curves in Figure 10.7), the agreement between theory and experiment up to $X = 0.9$ is quite good. In particular, the relative heights of the oxide reduction peak at either 0.8 for the Au disk or 0.4 for the Pt-Ir ring are rather accurately predicted lending credence to the overall procedure involved in the simulation. Lastly, the lack of a quantitative agreement for the curves at $X = 1.1$, i.e. beyond the outer edge of the ring can be ascribed to inability of this simple model to account for effects due to the Au ring-insulator edge.

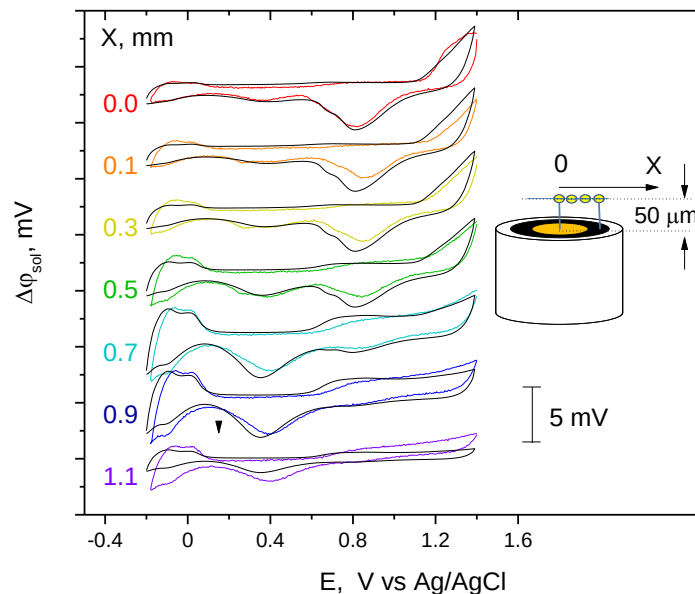


Figure 10.7. Plots of $\Delta\phi_{\text{sol}}$ vs E acquired with the double barrel capillary while the working electrode was scanned at 10 V/s, where each of the curves represents data collected with the tip placed along a line parallel to the surface of the ring disk intersecting the center of the disk ($X = 0$), at a fixed distance from the surface, i.e. 50 μm .

C. Single-Crystal Experiments

Shown in the upper panels, Figure 10.8, is the cyclic voltammogram of the faceted Pt single crystal in 0.1M H_2SO_4 recorded at a scan rate $v = 10 \text{ V/s}$, displaying a hydrogen adsorption|desorption region characteristic of polycrystalline Pt, Pt(poly) in this electrolyte. As has been thoroughly discussed in the literature, the two highly symmetric peaks at 80 mV and 220 mV are ascribed to the weak and strong hydrogen adsorption on Pt (110) and Pt (100), respectively, and the two small peaks centered at about 0.45 V vs RHE, are unique to the Pt(111) surface in this electrolyte, as shown by the red curve in this same panel for a genuine Pt(111) electrode.⁹⁷ These features became significantly more enhanced in the $\Delta\phi_{\text{sol}}$ vs E curves recorded

simultaneously (see black curve in lower left panel, Figure 10.8) when the barrel-type capillary was placed in the center of the Pt(111) facet (see inset in this figure). Similar conclusions could be drawn from data collected with the capillary placed in the center of the Pt(100) facet as shown in the Right Panel, Figure 10.8. In that case, however, the $\Delta\phi_{\text{sol}}$ vs E curves displayed in addition to a peak unique to Pt(100) at 0.37 V, a prominent feature associated with Pt(100) steps.⁹⁸ As the capillary was moved away from the single crystal the $\Delta\phi_{\text{sol}}$ vs E curves attained characteristics similar to those found in the conventional voltammetry (see Figure 10.9).

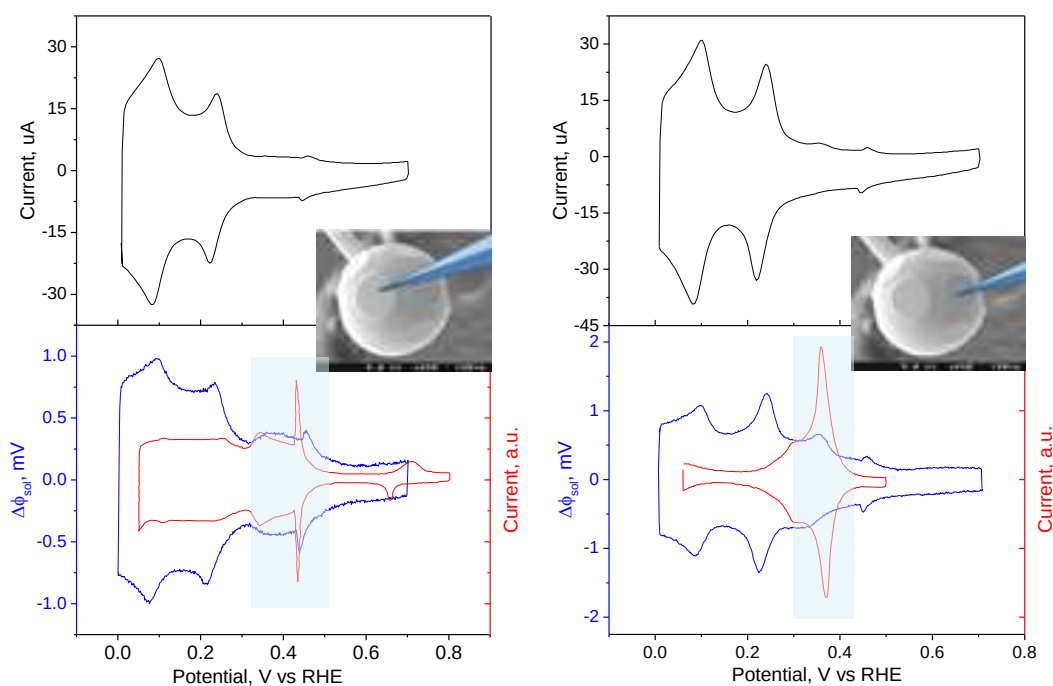


Figure 10.8. Cyclic voltammetry for the single crystal Pt faceted microsphere recorded at 10V/s in 0.1M H₂SO₄ solutions (Upper Panels), and $\Delta\phi_{\text{sol}}$ vs E (black curve lower Panels) acquired simultaneously with the tip of the double barrel capillary placed very close to the center of the Pt(111) (Left Panel) and Pt(100) microfacets (Right Panel). The red curves are the corresponding voltammetric curves for Pt(111) and Pt(100) faces collected using conventional single crystal electrodes reported in the literature in deareated 0.5 M H₂SO₄ solution at 50 mV/s.⁹⁷

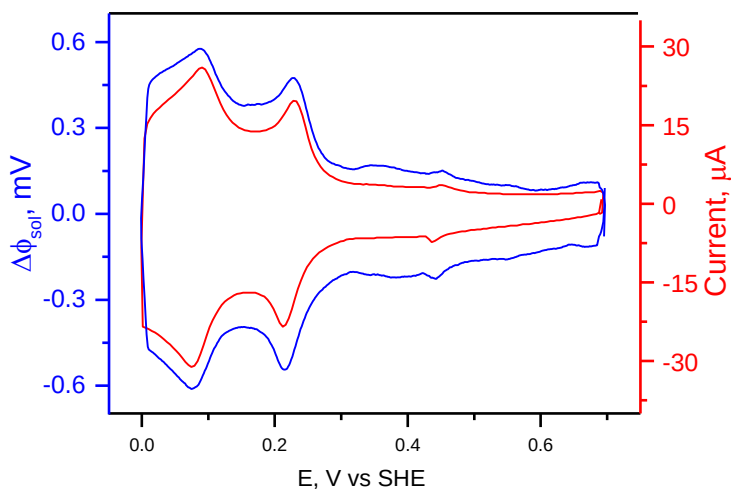


Figure 10.9. Cyclic voltammogram (right ordinate, red curve) and $\Delta\phi_{\text{sol}}$ vs E recorded simultaneously (left ordinate, blue curve) under the same conditions as specified in the caption Figure 10.8 with the tip of the capillary 0.3 mm away from the microfaceted single crystal sphere.

Concluding Remarks

The results presented herein support the view that for capacitive and/or pseudocapacitive processes, differences in the electrostatic potential in the electrolyte, $\Delta\phi_{\text{sol}}$, can be rather accurately accounted for by the primary current distribution, as predicted by Newman,⁹³ provided the changes in the composition of the solution are negligible. However, very reasonable agreement could be obtained between theory and experiment for data collected for a gapless Pt-Ir ring Au disk electrode assuming uniform currents both through the disk and the ring. Analysis of these results provided evidence that keeping the tip of the μ -ref electrode closer to the surface fixed and bringing the tip of the second μ -ref electrode further away elicits an increase in the sensitivity, i.e. larger $\Delta\phi_{\text{sol}}$, but also a loss in resolution, higher contribution of the electrode farther away from the tip to the signal. On this basis, the possibility of using much smaller reference electrodes, currently being explored in our laboratories, is expected to improve

both the sensitivity and resolution opening encouraging prospects for ohmic microscopy to become a viable tool for in situ imaging of electrode surfaces in the sub μm domain. Efforts in this direction are also being pursued by the group of Wittstock^{89, 90}, who implemented a modulation scheme to determine the $\Delta\phi_{\text{sol}}$ response as a function of position. The results so far published by these authors, however, do not seem to offer much advantage over our dual electrode strategy in terms of the quality of the data obtained and the acquisition time.

**Chapter 11 The Electrocatalytic Properties of Adsorbed Hemin and its
Nitroxyl Adduct on Glassy Carbon Surfaces Toward Hydroxylamine
Reduction in Aqueous Neutral Electrolytes**

Contributors

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Doe Kumsa – Preliminary Calculations

Nicholas S Georgescu – Theoretical Section below

Daniel Scherson – Advisor

Abstract

Cyclic voltammetry and dynamic and steady state polarization measurements have been used to examine the electrocatalytic activity of glassy carbon, GC, bearing a layer of adsorbed hemin, Hm, Hm|GC, toward hydroxylamine, NH_2OH , in neutral aqueous solutions. The results obtained have provided ample evidence that Hm|GC promotes the reduction and oxidation of NH_2OH , as well as its disproportionation. Also presented is a mechanistic model that accounts quantitatively for steady state polarization data for the reduction of NH_2OH recorded with a Hm|GC rotating disk electrode, RDE. This model predicts that a very large fraction of the current is attributed to the reduction of a dimeric Hm species axially coordinated by a NH_2OH molecule on either side of the porphyrin ring.

Introduction

Chloro[3,7,12,17-tetramethyl-8,13-divinylporphyrin-2,18-dipropionato(2-)]iron(III), known as hemin (Hm), is an iron macrocycle derived from biological sources, often regarded as a model electrocatalyst for the reduction of a number of simple molecules, including nitric oxide, NO ,⁹⁹ and dioxygen.¹⁰⁰ The vast majority of the studies published so far have involved Hm adsorbed at coverages on the order of a single monolayer on carbonaceous surfaces, including the elegant work of Schmickler and Tao,¹⁰¹ who examined the electronic and redox properties of

a highly ordered Hm monolayer spontaneously assembled on the basal plane of highly oriented pyrolytic graphite using *in situ* scanning tunneling spectroscopy.

This contribution presents the results of cyclic voltammetry and rotating disk electrode, RDE experiments in buffered neutral aqueous solutions aimed at exploring the electrocatalytic activity of Hm adsorbed on glassy carbon (GC) surfaces, Hm|GC, toward hydroxylamine, NH_2OH . This species is not only of technological importance, but has been identified as the only significant product of the reduction of nitric oxide (NO) in aqueous solutions of relevance to biological systems.⁹⁹ As will be shown, Hm|GC electrodes promote the reduction and oxidation of NH_2OH , as well as its disproportionation, a behavior analogous to that found for certain solution phase iron porphyrins in anaerobic, neutral aqueous media.¹⁰² Also included in this work is a mathematical model for the Hm|GC-mediated reduction of NH_2OH found to be consistent with experimental data collected under strict steady state conditions with a RDE.

Experimental

Electrochemistry - All measurements were performed either with a rotating GC RDE (Pine Instruments, area, $A_{\text{disk}} = 0.164 \text{ cm}^2$) from the same manufacturer (Disk diameter: 0.5 cm; Ring outer diameter: 0.6 cm; Ring inner diameter 0.5982 cm; Gap = 180 μm ; $A_{\text{ring}} = 0.037 \text{ cm}^2$; collection efficiency, $N = 0.22$) using a commercial rotator (Pine Instruments, Model AFMSRX), and either a potentiostat (Autolab, Metrohm, The Netherlands) for RDE or a bipotentiostat (Pine Instruments, Model AFCBP1) for RRDE measurements, respectively.

Hemin (Hm, Porphyrin Products, Inc., now Frontier Scientific, Inc.), sodium nitrite (Fisher Chemicals), acetone (Fisher Scientific) and hydroxylamine (NH_2OH , Aldrich, 99.999%

pure) were used as received. Phosphate buffer solutions (PB, 0.5 M, pH 7) were prepared from monosodium dihydrogen phosphate (Ultrapure, J.T. Baker), disodium hydrogen phosphate (Ultrapure, J. T. Baker) and ultrapure water (UPW, 18.3 MΩcm, EASYpureUV system, Barnstead). Experiments were carried out in a conventional all-glass, three-compartment cell with a carbon rod and a Ag/AgCl (BAS, RE-5B, West Lafayette, IN) as counter and reference electrodes, respectively.

Preparation of Hemin-modified Glassy Carbon Electrodes (Hm|GC). Prior to chemical functionalization, the GC disk electrode was first polished using 1 μm alumina powder and then sonicated for 15 min to remove particles adhering to the surface. Subsequently, it was immersed in deaerated 0.5 M PB and a voltammogram recorded over the potential range $-0.9 < E < 0.7$ V vs Ag/AgCl yielding a small peak at about 0.0 V associated with the presence of redox active functional groups on the GC surface. The electrode was then removed from that solution, rinsed thoroughly with UPW and then immersed in a deaerated 25 μM Hm solution in 0.1 M borate buffer (BB, pH 9). Adsorption of Hm was achieved under rotation at a rate $\omega = 400$ rpm. At specific time intervals, the rotation was stopped and a voltammogram recorded. The electrode was then removed from the solution, rinsed with UPW, immersed in deaerated neat PBS and a new voltammogram recorded. Subsequently, it was removed from the solution and, following rinsing, reintroduced in the Hm-containing BB electrolyte and the rotation was then resumed to allow further adsorption. About 15 to 20 such measurements were collected over a period of about three hours. Coulometric analyses of these data indicated that a time of ca. 70 min was sufficient for the GC surface to undergo saturation. All studies reported herein were performed with Hm-modified GC surfaces (Hm|GC) at full saturation in PB.

Preparation of the nitrosyl adduct of Hm adsorbed on GC. The NO adduct of adsorbed Hm (NOHm) was formed by placing the Hm|GC electrode in a 10 mM NaNO₂ in PB. A series of cyclic voltammetric curves were then recorded in the potential range 0 V to -0.6 V. Over time, the characteristic redox peak of adsorbed Hm centered about -0.38 V vs Ag/AgCl (*vide infra*) gradually diminished, signaling formation of NOHm, yielding, eventually, a featureless curve reminiscent of bare GC(*vide infra*) over a wide potential range. Additional experiments showed that conversion of Hm to NOHm could be readily accomplished by holding the potential at -0.6 V for times as short as ca. 30 s in a quiescent 0.5 M PB solution containing 10 mM NaNO₂. That was not the case when the electrode was left at open circuit, for which the rates of NOHm formation were considerably lower. All NOHm|GC surfaces were rinsed with UPW followed by PB and transferred quickly to the solution of interest for further measurements.

Results and Discussion

The cyclic voltammogram of a Hm|GC electrode (see Experimental Section) recorded in a quiescent 0.5 M PB (pH 7) solution at a scan rate, $\nu = 0.5$ V/s, displayed a prominent peak centered at -0.37 V vs Ag/AgCl characteristic of adsorbed Hm (see thick black curve in Figure 11.1). This feature has been attributed to an iron-centered one-electron transfer process. Also shown in this figure, is the corresponding featureless voltammogram of the bare GC electrode collected under identical conditions (grey line). The interfacial capacity of bare GC (C_{GC}) was found to be about 250 $\mu\text{F}/\text{cm}^2$ of cross-sectional area. This value is more than an order of magnitude larger than those reported earlier by other workers,¹⁰³ but less than twice that determined from data provided by the electrode manufacturer in sulfuric acid, i.e. 150 $\mu\text{F}/\text{cm}^2$.

Regardless of the possible reasons behind such differences, which might include the polishing procedure as well as variations in the quality of the material, C_{GC} does not seem to be affected by the presence of adsorbed Hm for potentials outside the range associated with the Hm redox peaks.

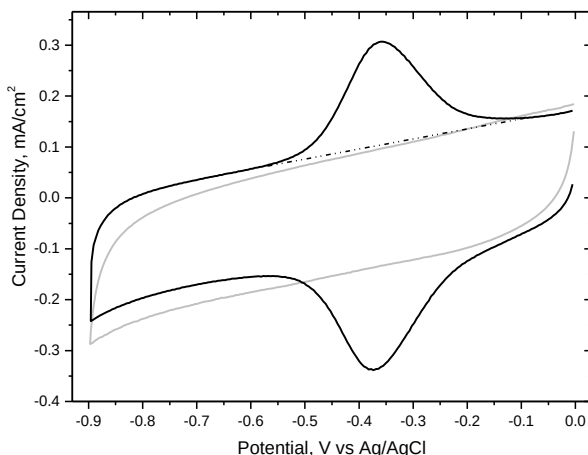


Figure 11.1. Cyclic voltammograms of a bare GC (grey line) and a Hm|GC electrodes (black line) recorded in deaerated 0.5 M PB (pH 7) at a scan rate, $v = 0.5$ V/s under quiescent conditions. The dotted line was used as a background for the coulometric analysis.

A coulometric analysis based on data of the type shown in Figure 11.1, using the dotted line therein as a baseline, yielded, for various trials, Hm surface concentrations in the range 1.6 - 1.8 nmol per cm^2 of cross sectional surface. The area occupied by a Hm molecule flat on a surface is on the order of $2 \times 10^{-14} \text{ cm}^2$; hence, a single monolayer packed in a tile-like fashion would yield a surface concentration of ca. 0.08 nmol/cm^2 .¹⁰⁴ This value is much smaller than that found in this work, and suggests that the topography of the GC is highly corrugated and/or Hm is bound to the surface forming a rather large angle, an arrangement which would increase the packing density. Unfortunately, no methods have yet been devised to determine the true area

available for adsorption; hence, conclusions based on these arguments are subject to large uncertainties.

Electrochemical Properties of Adsorbed NOHm on GC Electrodes

A. Reduction - Shown in Figure 11.2 (see black line) is a cyclic voltammogram of a freshly prepared NOHm|GC electrode (see Experimental Section) in neat, deaerated PB initiated at 0.0 V vs Ag/AgCl toward negative potentials. As has been discussed in the literature,⁹⁹ this first linear scan is virtually featureless down to the onset of the prominent peak centered at about -0.77 V, a feature ascribed to the irreversible reduction of adsorbed NOHm. This process renders the macrocycle in its reduced form, as indicated by the appearance of the characteristic Hm oxidation peak centered at -0.35 V upon reversing the scan at -0.9 V. Further evidence that the reduction of NOHm leaves the macrocycle intact is provided by the close overlap of the third cycle (see magenta line) with that of the neat Hm|GC obtained before the NOHm was formed (see dotted line). Based on a coulometric analysis using the third cycle after full reduction of the adduct as a baseline, the number of electrons involved in the reduction of the NOHm is three times larger than the one-electron redox couple characteristics of adsorbed Hm. This value is in line with that reported elsewhere,⁹⁹ and supports the view that the product of this reaction is predominantly hydroxylamine, NH_2OH .

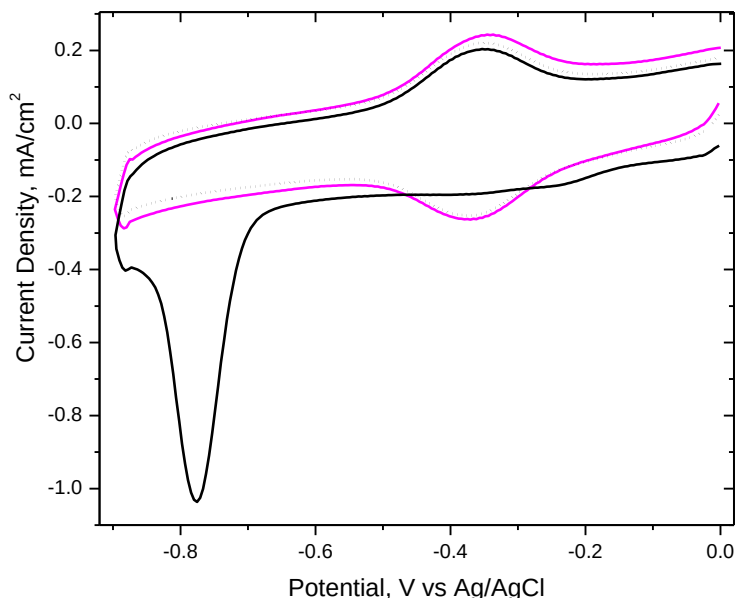


Figure 11.2. Cyclic voltammograms ($v = 0.5$ V/s) of NOHm|GC (black line) recorded in deaerated PB under quiescent conditions. The scan was initiated at 0.0 V vs Ag/AgCl in the negative direction. The magenta line was obtained during the subsequent third cycle. Also shown for comparison is the voltammogram of Hm|GC in neat PB (dotted curve).

B. Oxidation. For these experiments, individually prepared NOHm|GC surfaces were transferred to neat 0.5 M PB (pH 7) for further electrochemical characterization. The latter involved linear scans at a rate $v = 0.05$ V/s initiated at 0.0 V in the positive direction, which were reversed at various potentials E_+ in 0.1 V increments up to 0.7 V, and subsequently reversed once again at a common negative value, i.e. -0.9 V, to end at 0.0 V. As evidenced by the results obtained (see Figure 11.3), no changes in the magnitude of the NOHm reduction peak at ca. -0.7 V were discerned for values of E_+ up to ca. 0.4 V. The same behavior was found upon holding the potential at 0.3 V for times up to 6 min. and then scanned in the negative direction down to -0.9 V. On this basis, it can be concluded that NOHm is stable in the potential range $-0.65 \text{ V} < E < 0.25 \text{ V}$. A clear increase in the current was observed, however, by extending the scan beyond

0.35 V which was followed by a small plateau in the range around 0.55 - 0.65 V, and yet another rise at more positive potentials. Close inspection of these data revealed a gradual decrease in the magnitude of the NOHm reduction peak as E_+ was increased, accompanied by an increase in the magnitude of the redox peak attributed to adsorbed Hm. Also included in this figure (see dashed-dot curve) are data recorded after scanning the potential three times over the range $-0.9 \text{ V} < E < +0.7 \text{ V}$ under otherwise identical conditions for which the feature associated with the reduction of NOHm was very much diminished.

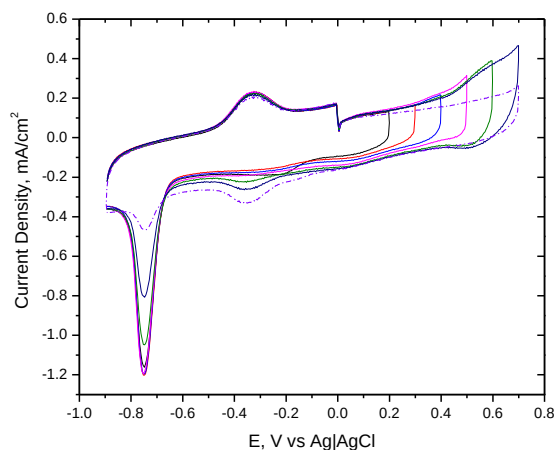


Figure 11.3. Series of cyclic voltammograms of NOHm/GC ($v = 0.05 \text{ V/s}$) in 0.5 M PB (pH 7). The scan was initiated at 0.0 V in the positive direction reversed at the indicated limit, and then reversed once again at $-0.9 \text{ V vs Ag/AgCl}$ to end at 0.0 V. The curve in dashed-dot was recorded after scanning the potential three times over the entire potential range under otherwise identical conditions.

These observations are consistent with the oxidation of adsorbed NOHm to yield Hm in its oxidized form. Insight into the nature of the products of this reaction was gained from a series of sequential voltammograms for a NOHm/GC surface in the range $-0.6 < E < 0.7 \text{ V}$, shown in Figure 11.4, where the grey and dash-dot lines represent the second and fifth cycles. A coulometric analysis of the peak associated with the oxidation of NOHm using as a background

the curve obtained after multiple oxidation scans up to 0.7 V (dot dashed), yielded a charge of about 2.6 electrons per adsorbed Hm molecule. Since, the oxidation of NOHm renders Hm in a formal ferric state, NO in the adduct would undergo a transfer of ca. 1.6 electrons per molecule, which is consistent with the formation of nitrite and its partial further oxidation to nitrate, where the latter process is known to proceed on bare GC.¹⁰⁵

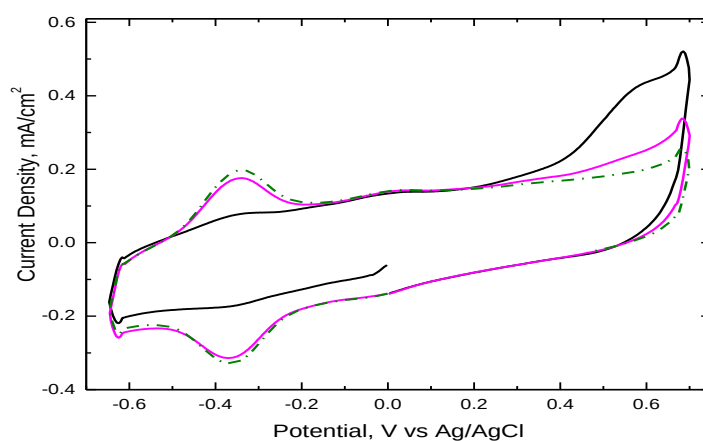


Figure 11.4. Cyclic voltammograms of NOHm|GC ($v = 0.5$ V/s) (first cycle, black curve) initiated at 0.0 V in the negative direction in 0.5 M PB (pH 7) recorded under the same conditions as those specified in the caption of Figure 11.2. The magenta and dash-dot line represent the second and fifth subsequent cycles.

Catalytic and Electrocatalytic Properties of Hm|GC Toward Hydroxylamine

Catalysis - For these measurements, a freshly prepared Hm|GC electrode was immersed in deaerated 1 mM NH_2OH in PB and rotated at $\omega = 400$ rpm for ca. 60 minutes at open circuit. Subsequently, it was removed from the solution, rinsed copiously with neat PB, and introduced into a separate cell filled with deaerated PB and a voltammogram recorded (see black curve, Figure 11.5). As shown therein, this electrode displayed a prominent peak centered at ca. -0.77 V

characteristic of the reduction of the NOHm adduct. In fact, longer exposures led to an increase in the magnitude of the NOHm reduction peak and a corresponding decrease in the amplitude of the Hm redox peaks. This finding provides rather unambiguous evidence that adsorbed Hm is capable of catalyzing the disproportionation of NH_2OH to yield NOHm as one of the products, although the rates are significantly lower than those observed in nitrite-containing solutions presented earlier in this work. The same reaction was studied recently by Bari et al.¹⁰² in neutral aqueous solutions containing a variety of water-soluble iron macrocycles, including iron(III) meso-tetrakis-(N-ethylpyridinium-2yl)-porphyrinate ($[\text{Fe}^{\text{III}}(\text{TEPyP})]^{5+}$) and iron(III) meso-tetrakis-(4-sulphonatophenyl)-porphyrinate ($[\text{Fe}^{\text{III}}(\text{TPPS})]^{3-}$). According to these authors, the NH_2OH disproportionates to yield ammonia as the sole reduction product and both nitrous oxide, N_2O , and N_2 , as its oxidation counterparts.

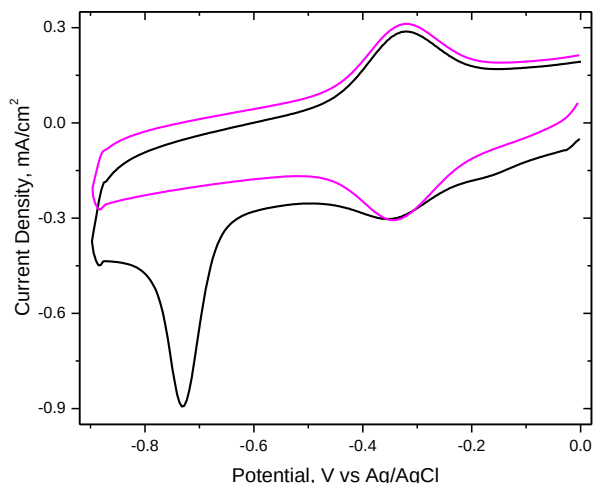


Figure 11.5. Cyclic voltammetry ($v = 0.5 \text{ V/s}$) for a Hm/GC electrode in neat PBS which had been previously exposed to a 1mM NH_2OH in PBS at open circuit, ca. 0 V for 60 min. under rotation ($\omega = 400 \text{ rpm}$), and subsequently rinsed with neat PB (black curve). The magenta curve was obtained without exposure to the NH_2OH solution. Unlike the data collected in nitrite solutions the nitrosyl adduct does not form readily on Hm under these conditions.

Electrocatalysis

a. Bare Glassy Carbon - The cyclic voltammograms of a bare GC electrode in PB both in the absence and in the presence of 1 mM NH_2OH are shown in the grey and black curves in Figure 11.6, respectively. Whereas bare GC exhibits no activity for NH_2OH reduction, as evidenced by the lack of any differences between the curves recorded in the two media, the increase in the current at potentials $E > 0.2$ V is clearly indicative that bare GC can promote NH_2OH oxidation. This last finding seems at variance with the reports of Piela and Wrona,¹⁰⁶ who claimed no detectable activity of GC for this process in acid media up to pH ca. 5.6.

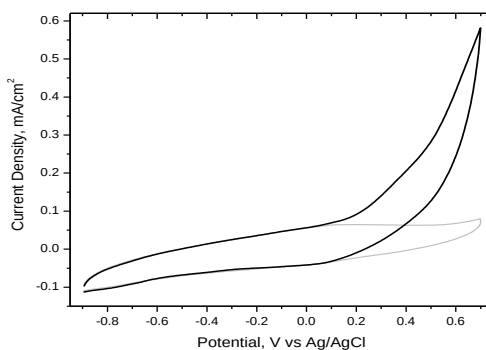


Figure 11.6. Cyclic voltammograms ($v = 0.05$ V/s) of a GC RDE electrode recorded under quiescent conditions deaerated PB before (grey) and after addition of 1 mM NH_2OH (black). See text for details.

b. Hemin-modified GC (Hm|GC)

In stark contrast with the behavior found for bare GC (see grey curve, Figure 11.7), negative currents were clearly observed in the cyclic voltammograms of Hm|GC in PB for potentials $E < -0.3$ V in the presence of NH_2OH in solution (see black curve in the same figure), affording strong evidence that Hm|GC displays electrocatalytic properties toward the reduction

of NH_2OH . In addition, the corresponding oxidation region also showed some differences compared to that found for the bare GC (see black trace in Figure 11.6), which suggest possible modifications in the mechanism of NH_2OH oxidation induced by the adsorbed Hm including its axial coordination. More detailed aspects of the reduction NH_2OH on Hm|GC are discussed below.

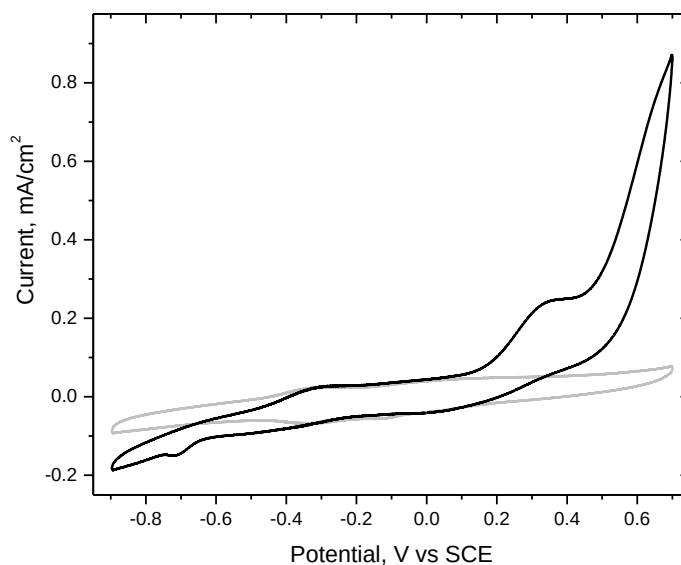


Figure 11.7. Cyclic voltammogram ($v = 0.05$ V/s) of a GC (grey curve) and a HM|GC electrode recorded under quiescent conditions in deaerated PB (pH 7) after addition of 1 mM NH_2OH (black curve). See text for details.

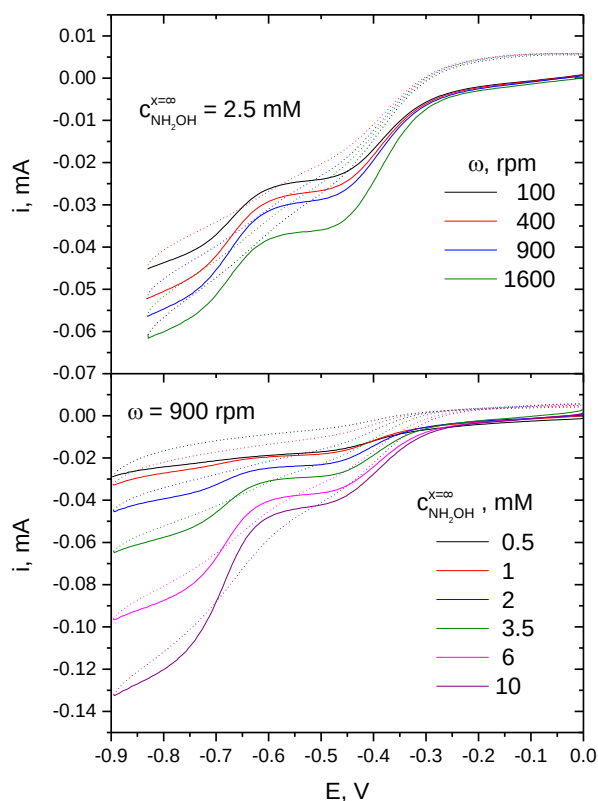


Figure 11.8. Dynamic polarization curves recorded at $v = 10$ mV/s with a Hm|GC RDE in PB buffer pH 7 as a function of ω as indicated in the legend for $c_{\text{NH}_2\text{OH}}^{x=\infty} = 2.5$ mM (Top Panel), and, as a function of $c_{\text{NH}_2\text{OH}}^{x=\infty}$ as indicated in the legend, at $\omega = 900$ rpm (Bottom Panel). The solid and dotted curves for each set of data correspond to scans toward negative and subsequent positive potentials, respectively.

A. Hydroxylamine Reduction on Hm|GC – Shown in Figure 11.8 are dynamic polarization curves recorded at $v = 10$ mV/s with a Hm|GC RDE at a fixed bulk concentration of hydroxylamine, $c_{\text{NH}_2\text{OH}}^{x=\infty} = 2.5$ mM, for different ω (Top Panel, Figure 11.8), and for rotation rate, i.e. $\omega = 900$ rpm for a range of $c_{\text{NH}_2\text{OH}}^{x=\infty}$ (Bottom Panel in the same figure), using Hm|GC surfaces prepared independently. Close inspection of these results revealed a number of distinctive features:

1. For scans toward negative potentials (see solid lines), NH_2OH reduction commences at ca. -0.35 V, reaching a limiting value at ca. -0.5 V, which extends down to about -0.65 V. At more negative potentials, the current markedly increases and begins to bend over at ca. -0.75 V.
2. The onset of the process at more positive potentials occurs at ca. -0.35 V, a value that is ca. 0.2 V more negative than the onset of Hm reduction in neat PB (see Figure 11.1), and very close to the peak maximum for that redox process. In contrast, the onset of the more negative process coincides almost precisely with the onset of the reduction of ONHm (see grey curve in Panel A) for data collected in PB devoid of NH_2OH in solution.
3. Upon reversing the scan at the negative limit, i.e. -0.9 V (see dotted lines), the currents for the highest $c_{\text{NH}_2\text{OH}}^{x=\infty}$ examined were found to increase especially in the plateau region, 0.5 - 0.6 V. This behavior may be caused by the full reduction of a Hm adduct, yielding an iron center in the macrocycle displaying much higher reactivity toward NH_2OH reduction than the adduct. As discussed above, the rate of NH_2OH disproportionation to yield the adduct is rather slow compared to the scan rate; hence, the interfacial conditions are different during the scans in the negative and positive directions, a factor that complicates the overall analysis of the data.
4. As $c_{\text{NH}_2\text{OH}}^{x=\infty}$ is increased keeping ω constant, or ω is increased for a fixed $c_{\text{NH}_2\text{OH}}^{x=\infty}$, the current increases, while the overall shape of the curves remains virtually unchanged
5. Except for a very narrow range of negative potentials, the overall currents recorded for $c_{\text{NH}_2\text{OH}}^{x=\infty} \geq 2 \text{ mM}$ during the scan in the positive direction are higher than the negative-going counterparts, whereas, for $c_{\text{NH}_2\text{OH}}^{x=\infty} \leq 1 \text{ mM}$, portions of the reverse scan have a

larger current than at the corresponding potential in the forward scan. Furthermore, this trend is maintained regardless of the value of ω , as illustrated by the data in the bottom panel in this figure collected for $c_{\text{NH}_2\text{OH}}^{x=\infty} = 1 \text{ mM}$ over the range $100 \leq \omega \leq 1600 \text{ rpm}$.

In view of the differences found in data collected in scans toward negative and positive potentials, a single polarization curve was recorded under strict steady state conditions (see scattered symbols in the top panel, Figure 11.9), and used as a basis to explore mechanisms that could reproduce the experimentally observed behavior. Considerable insight into possible reaction pathways was gained from the work of Bari *et al.*, who studied various aspects of the homogeneous disproportionation of NH_2OH catalyzed by a series of solution-phase iron porphyrins in aqueous electrolytes.¹⁰²

The symbol Hm in Scheme A represents hemin devoid of any nitrogen-containing axial ligands, a species that can bind one (R_1), and subsequently a second (R_2) NH_2OH molecule to yield a monomeric or dimeric adduct, HOH_2NHm and $(\text{HOH}_2\text{N})_2\text{Hm}$, respectively. As indicated, HOH_2NHm can then react with yet another NH_2OH molecule to form the nitroxyl adduct, OHNHm (R_3), which can either decompose, to yield nitrous oxide, N_2O , or undergo a four-electron reduction to ammonium ion, NH_4^+ (R_4 and R_6 , respectively). Also included in this pathway is the electrochemical reduction of the $(\text{HOH}_2\text{N})_2\text{Hm}$ dimer to produce NH_4^+ , and regenerate HOH_2NHm , (R_5).

Theoretical

Shown in Table 11.1 are the mass balance equations for the major species involved in the mechanism specified in Table 11.2 and Scheme 11.1. This set of non-linear expressions leads to

a second order algebraic equation in $c_{NH_2OH}^{x=0}$, i.e. the concentration of NH_2OH at the electrode surface (see Eq. 11.1), with only a single physically significant root, and to explicit expressions for the surface concentrations of all relevant adsorbed species j , denoted as γ_j .

$$\begin{aligned} & \left(c_{NH_2OH}^{x=0} \right)^2 \left(k_3 / (k_4 + k_6) + k_2 / k_5 \right) - c_{NH_2OH}^{x=\infty} \left(1 + k_3 / k_1 \right) + \\ & c_{NH_2OH}^{x=0} \left\{ \left(\left(1 + k_3 / k_1 \right) + \gamma_s \left(k_2 + 2k_3 \right) / (D / \delta)_{NH_2OH} \right) \right. \\ & \quad \left. - c_{NH_2OH}^{x=\infty} \left(k_3 / (k_4 + k_6) + k_2 / k_5 \right) \right\} = 0 \end{aligned} \quad [11.1]$$

$$\gamma_{HOH_2NHm} = \gamma_s / \left\{ \left(1 + k_3 / k_1 \right) + c_{NH_2OH}^{x=0} \left(k_3 / (k_4 + k_6) + k_2 / k_5 \right) \right\} \quad [11.2]$$

$$\gamma_{(NH_2OH)_2Hm} = \gamma_s / \left\{ \left(1 + k_3 / k_1 \right) k_5 / \left(k_2 c_{NH_2OH}^{x=0} \right) + \left(k_3 k_5 / \left(k_2 (k_4 + k_6) \right) + 1 \right) \right\} \quad [11.3]$$

$$\gamma_{OHNHm} = \gamma_s / \left\{ \left(1 / k_1 + 1 / k_3 \right) (k_4 + k_6) / \left(c_{NH_2OH}^{x=0} \right) + \left(1 + k_2 (k_4 + k_6) / k_3 k_5 \right) \right\} \quad [11.4]$$

Where $(D / \delta)_{NH_2OH}$ represents the diffusion coefficient of hydroxylamine, 3.2×10^{-6} cm²/s, divided by the diffusion boundary layer thickness, and γ_s is the surface concentration of adsorbed hemin, which was assumed to be 1.8 nmol/cm².

Table 11.1. Mass Balance Equations for the Major Species Involved

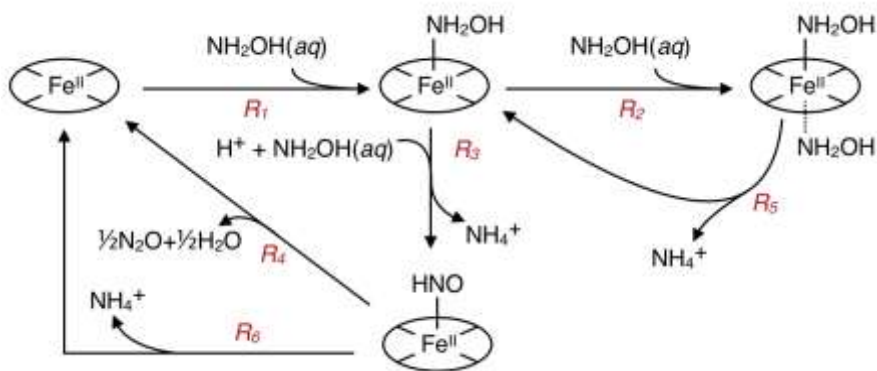
Species	Mass Balance Equation
$NH_2OH(aq)$	$(D / \delta)_{NH_2OH} (c_{NH_2OH}^{x=\infty} - c_{NH_2OH}^{x=0}) = c_{NH_2OH}^{x=0} (k_1 \gamma_{Hm} + (k_2 + k_3) \gamma_{HOH_2NHm})$
H_2OHNHm	$k_1 c_{NH_2OH}^{x=0} \gamma_{Hm} + k_5 \gamma_{(NH_2OH)_2Hm} = (k_2 + k_3) c_{NH_2OH}^{x=0} \gamma_{HOH_2NHm}$
$(H_2OHN)_2Hm$	$k_2 c_{NH_2OH}^{x=0} \gamma_{HOH_2NHm} = k_5 \gamma_{(NH_2OH)_2Hm}$
$OHNHm$	$k_3 c_{NH_2OH}^{x=0} \gamma_{HOH_2NHm} = (k_4 + k_6) \gamma_{OHNHm}$

Table 11.2 Proposed Mechanism for the Electrocatalytic Reduction of NH_2OH Mediated by Hm|GC

$\text{Hm} + \text{NH}_2\text{OH}(\text{aq}) \leftrightarrow \text{HOH}_2\text{NHm}$	R_1
$\text{HOH}_2\text{NHm} + \text{NH}_2\text{OH}(\text{aq}) \leftrightarrow (\text{HOH}_2\text{N})_2\text{Hm}$	R_2
$\text{HOH}_2\text{NHm} + \text{NH}_2\text{OH}(\text{aq}) + \text{H}^+ \rightarrow \text{HONHm} + \text{NH}_4^+ + \text{H}_2\text{O}$	R_3
$\text{HONHm} \rightarrow \text{Hm} + \frac{1}{2} \text{H}_2\text{O} + \frac{1}{2} \text{N}_2\text{O}$	R_4
$(\text{HOH}_2\text{N})_2\text{Hm} + 3\text{H}^+ + 2\text{e}^- \rightarrow \text{NH}_4^+ + \text{H}_2\text{O} + \text{HOH}_2\text{NHm}$	R_5
$\text{HONHm} + 5\text{H}^+ + 4\text{e}^- \rightarrow \text{NH}_4^+ + \text{Hm} + \text{H}_2\text{O}$	R_6

Table 11.3. Best Fit Parameters Based on the Model in & Scheme 11.1

Symbol	Magnitude
k_1	$10^8 \text{ cm}^3/\text{mol s}$
k_2	$2.3 \times 10^6 \text{ cm}^3/\text{mol s}$
k_3	$300 \text{ cm}^3/\text{mol s}$
k_4	$7 \times 10^{-5} / \text{s}$
k_5	$2 \times 10^{-4} e^{-20(E+0.030)} / \text{s}$
k_6	$2 \times 10^{-4} e^{-16(E+0.740)} / \text{s}$



Scheme 11.1

On this basis, the current density at the electrode can be written as

$$i = F \left(2k_5 \gamma_{(NH_2OH)_2 Hm} + 4k_6 \gamma_{OHNHm} \right) \quad [11.5]$$

or, equivalently, from Eqs. 11.3 and 11.4,

$$i = \frac{\gamma_s F (2k_2 + 4 / (k_4 / k_6 + 1))}{(1 / k_1 + 1 / k_3) / (c_{NH_2OH}^{x=0}) + (1 / (k_4 + k_6) + k_2 / k_5)} \quad [11.6]$$

These expressions were then used to fit the true steady state data yielding, for the set of parameters specified in Table 11.3, excellent agreement between theory and experiment (see solid line in the Top Panel, Figure 11.9). It should be stressed that this set of parameters is by no means unique, but their magnitude is within the range of those found experimentally for other chemical and electrochemical systems. It should be noted that the rates correspond to a transfer coefficient for k_5 of ~ 0.25 and for k_6 of ~ 0.10 for the two- and four-electron processes, respectively.

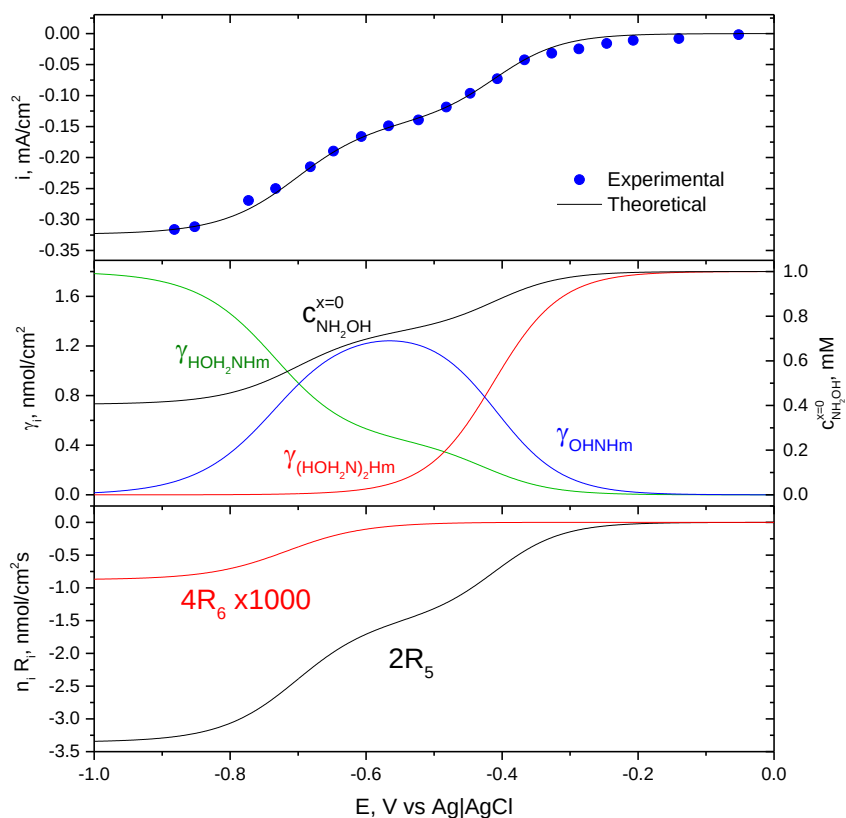


Figure 11.9. **Top Panel.** Plot of the experimental polarization curve collected at true steady state at $\omega = 900$ rpm (scattered symbols) and the simulation (solid line) based on Model in Table I and Scheme A and the parameters specified in Table III. **Middle Panel.** Plots of the concentrations of the four relevant species as a function of the applied potential based on the best fit set of parameters. **Lower Panel.** Contributions to the current from the two electrochemical processes, where the contribution from R_6 , i.e. the reduction of the OHNHm adduct, has been multiplied by 1000.

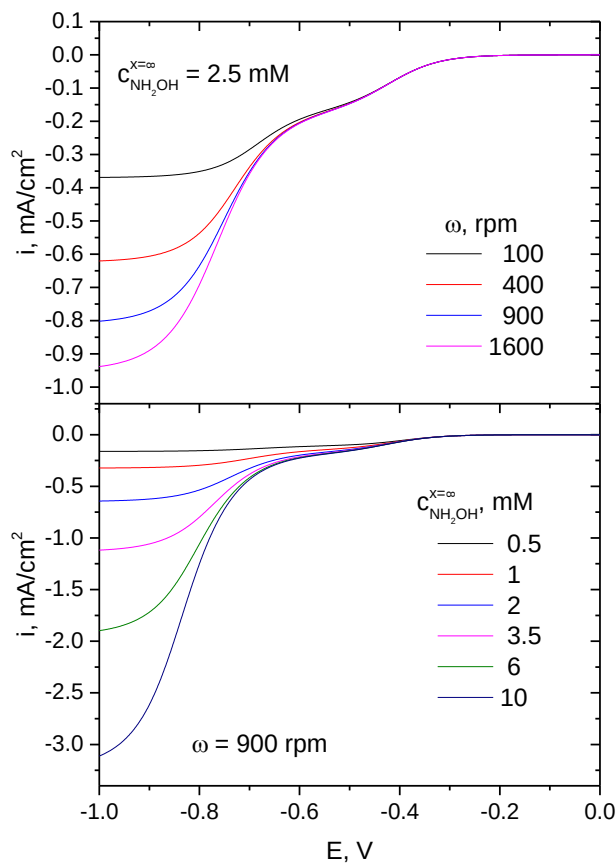


Figure 11.10. Simulated polarization data based on the parameters used to fit the steady state data in Figure 11.9 (see legends for details).

Cursory inspection of the results of the simulation shown in Figure 11.9 reveals a number of interesting observations:

- $(\text{H}_2\text{OHN})_2\text{Hm}$ is by far the dominant species at low overpotentials.
- The magnitude of the current is about half of the diffusion-limited current associated with the reduction of NH_2OH via a one-electron process.
- The potentials at which the surface concentration of $(\text{H}_2\text{OHN})_2\text{Hm}$ begins to decrease corresponds to the onset of the reduction of Hm.

- d. As the potential is further decreased, the surface concentration of H_2OHNHm and OHNHm increases. In fact, the highest concentration for the OHNHm occurs between -0.55 and -0.6 V which happens to coincide with the brief plateau in the top panel.
- e. According to our model, the rate of formation of $(\text{H}_2\text{OHN})_2\text{Hm}$ is independent of the potential, whereas the rate of its reduction increases with higher overpotentials. Hence, and not surprisingly, the surface concentration of $(\text{H}_2\text{OHN})_2\text{Hm}$ at even further negative potentials drops to negligible values.
- f. At the very negative limit the only species that is present on the surface is H_2OHNHm .

As indicated by the results in the bottom panel, Figure 11.9, the only significant contribution to the current at all potentials is made by the reduction of $(\text{H}_2\text{OHN})_2\text{Hm}$. Nevertheless, it is the comparatively very small current due to the reduction of OHNHm that creates sites for the main reaction to occur. Also noteworthy is the fact that the onset for reduction of the HNO adduct is very similar to that for the reduction of the NO analogue, which suggests that the electronic structure, and, thus, the redox properties of these two species are indeed very similar. Furthermore, and in line with suggestions put forward in the literature, R_3 is the slow process i.e. four orders of magnitude slower than the formation of the mono- or di-adduct, which are all second-order reactions.

Despite the good agreement found between the fits and the experimental data collected under strict steady state conditions, the behavior predicted by our model for dynamic polarization measurements as a function of ω and $c_{\text{NH}_2\text{OH}}^{x=\infty}$, shown in Figure 11.10, accounted for only a few of the features observed in the actual measurements, including the overall shapes of the curves as well as the increase in the magnitude of the current with ω and $c_{\text{NH}_2\text{OH}}^{x=\infty}$ while keeping the other

variable constant. Several factors may be responsible for the lack of a better quantitative agreement, such as, the rate at which the potential was scanned, which may have been too high for the system to have achieved conditions approaching steady state. Although it may be argued that a more thorough statistical analysis may have led to improvements in the fits, the inherent complexities of the system involving several adduct-type species would have rendered any conclusions largely speculative. Efforts are currently being made in our laboratory to enhance the sensitivity and time resolution of *in situ* differential reflectance spectroscopy in order to detect and identify the various adsorbed intermediates involved, in order to elucidate the actual mechanism of this interesting electrocatalytic process.

Conclusions

The reduction of hydroxylamine in aqueous solutions mediated by irreversibly adsorbed hemin on a glassy carbon electrode was examined using rotating disk techniques yielding dynamic polarization curves characterized by two rather well-defined plateau features. A mechanistic model was presented which accounted quantitatively for polarization data collected under strict steady state conditions for a fixed rotation rate and hydroxylamine concentration. According to this model, the plateau observed in the intermediate potential range can be attributed to the formation of a nitroxyl Hm adduct a reaction with a much slower rate constant than that associated with the formation of other adducts. Validation of the proposed mechanism will require the use of *in situ* spectroscopic tools with sufficient sensitivity and specificity to unveil the nature of the species involved. Lastly and in analogy with the behavior reported for a number of solution phase iron porphyrins, adsorbed hemin was found to catalyze the disproportionation of hydroxylamine.

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